DEEP-DIVE RESEARCH REPORT • INDIA • OCTOBER 2026

From Seed to Cup

and a Café in Ahmedabad

How coffee is grown, processed, traded, roasted, ground and brewed, how the market works, and a three-statement + DCF model of a specialty café across four sites, two premises options, three roasting routes and four bean mixes.

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2 Growing coffee
3 Harvesting
4 Processing
5 Milling, grading and quality
6 Trade, prices and the money
7 Roasting
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20 Recommendation and roadmap
A Appendix



178.8 m
bags: record world crop 2025/26

0.04 kg
coffee per Indian per year

5,339
branded cafés in India, +12.7%

₹207 lakh
NPV, best café model (Base)

Desk research from public sources available to 10 Oct 2026. Model figures in ₹ lakh (1 lakh = ₹1,00,000); see Cafe_Financial_Model.xlsx.

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Executive summary

The answers in two pages

Part 1: coffee from seed to cup

- **Coffee is a fruit seed with a long chain.** From research station and nursery to cup takes 3-5 years for a new plant and 6-18 months for each crop, passing through grower, curing works, trader or exporter, roaster, distributor and café.
- **Quality is made upstream and lost downstream.** Variety, altitude, ripe-only picking and careful processing set the ceiling; stale roasts, bad grinding, wrong water and sloppy recipes decide how much of it reaches the cup.
- **Processing is a flavour choice.** Washed coffees are clean and bright; naturals and honeys fruity and full; India's monsooned coffee is mellow and heavy; new anaerobic and carbonic methods are loud and expensive.
- **Every brewing method trades clarity for body.** Pressure (espresso, moka), percolation (pour-over, kaapi), immersion (French press, cold brew) and boiling (Turkish) exist because they produce different cups from the same bean, each suited to a culture and a moment.
- **Grinding and water are under-rated.** Uniform particles and SCA-standard water improve every cup more than a pricier machine.
- **The market is volatile and growing.** World prices roughly tripled between 2019 and 2025; India grows about 6.4 million bags, exports 70% and drinks just 0.04 kg per person, but its café market (5,339 branded outlets) is growing about 13% a year.
- **In a café, the coffee itself is ~6-10% of the price.** The grower receives roughly ₹13-20 of a ₹220 cappuccino; labour, rent, milk and service make up most of the rest.

Part 2: the Ahmedabad café

Model (toll roasting, House blend)	Capital ₹ L	NPV ₹ L	Proj. IRR	Eq. IRR	Payback	NPV Worst
A. Sindhu Bhavan Road flagship	129.9	207.2	36.3%	42.2%	3.9 yrs	(166.2)
B. GIFT City captive café	97.6	159.5	35.3%	40.6%	4.2 yrs	(96.7)
C. Prahladnagar neighbourhood café	101.2	99.8	29.8%	34.4%	4.6 yrs	(180.9)
D. Sindhu Bhavan Road, owned unit	413.3	240.8	22.2%	29.2%	6.6 yrs	(187.5)

- **Go to Sindhu Bhavan Road, rented, toll-roasted House blend (Model A)**: ₹130 lakh capital, NPV ₹207 lakh, project IRR 36.3%, equity IRR 42.2%, payback 3.9 yrs.
- **GIFT City (Model B)** is the capital-light, most resilient alternative (₹98 lakh, NPV ₹159 lakh) and the ideal second outlet.
- **Prahladnagar (Model C)** works (NPV ₹100 lakh, IRR 29.8%) but has thinner margins and more chain competition. **Sanand** is NPV-negative today.
- **Rent, don't buy.** Owning at SBR (Model D) raises NPV to ₹241 lakh but needs ₹413 lakh and cuts the IRR to 22.2%.
- **Toll-roast from day one**: it beats buying roasted beans everywhere. Build an in-house roastery only once wholesale sales reach roughly 450-500 kg a month.
- **Demand is the risk.** Bills per day and average bill move value most; in the Worst case every site loses money, and shutting after year 3 would cost close to the whole investment. Validate footfall before signing a lease.

Deliverables

Coffee_Deep_Dive_and_Cafe_Model.pdf: this report.

Cafe_Financial_Model.xlsx: the full working (Cover with checks, Dashboard, Inputs, Locations, Sourcing, Capex, Model, DCF, Scenarios, Configs, Sources). Change any blue cell to test your own numbers.

PART 1

Coffee, from seed to cup

How coffee is bred, grown, harvested, processed, graded, traded, roasted, ground and brewed, and how the global and Indian markets work.

CHAPTERS

1 The big picture
2 Growing coffee
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1 The big picture

What coffee is, how it travels from seed to cup, and where India fits

Coffee is the roasted seed of a cherry. Between the research station that breeds the plant and the cup in your hand sit a dozen steps, five or six owners, and between two and six years. Each step can add flavour, destroy it, or simply add cost. This part of the report walks that whole chain, then looks at the market it feeds.

1.1 What coffee actually is

Coffee shrubs belong to the genus *Coffea* in the Rubiaceae family (the gardenia family). Two species matter commercially. **Arabica** (*Coffea arabica*), originally from the highlands of Ethiopia and South Sudan, gives roughly 55-60% of the world's coffee and nearly all specialty coffee. **Robusta** (*Coffea canephora*), from the lowland forests of central and west Africa, gives the other 40-45%. It is hardier, higher-yielding and carries about twice the caffeine. India grows both, and robusta makes up about three quarters of its crop.

The fruit is a **coffee cherry**. Inside it, two seeds lie face to face: these are the coffee beans. In about 5-10% of cherries only one seed develops; it grows round and is sold separately as **peaberry**. A raw ("green") bean is hard, smells of peas and grass, and makes an unpleasant drink. Roasting creates more than 800 aroma compounds and turns the seed into what we recognise as coffee. Coffee is grown in about 70 countries in the tropical "bean belt" between the Tropics of Cancer and Capricorn, on roughly 12.5 million farms, most of them smaller than five hectares.

1.2 The value chain at a glance

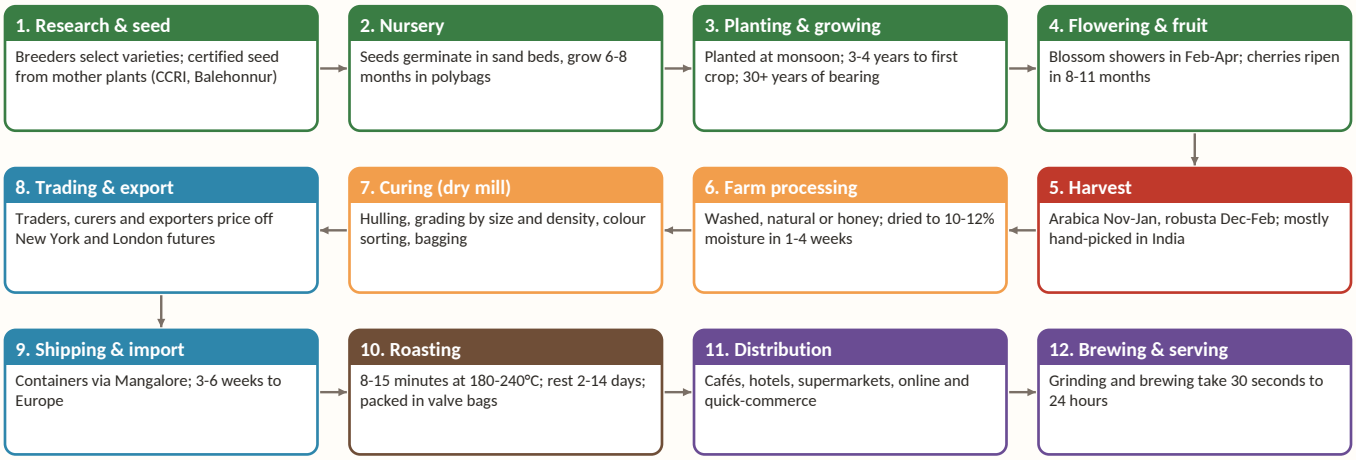


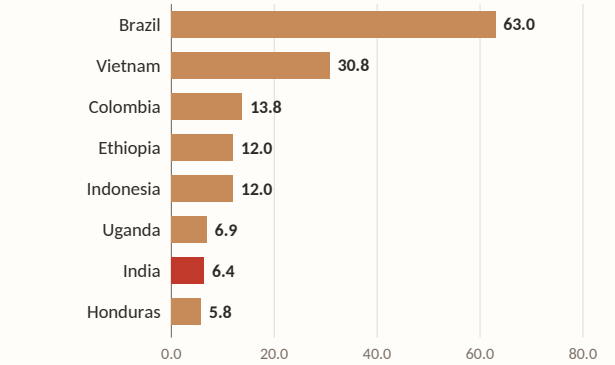
Figure 1.1. Twelve steps from seed to cup. Green = farm, orange = processing, blue = trade, brown = roasting, purple = selling and brewing.

Owner	What changes hands	Typical price (India, Oct 2026)	What they add
Grower	Farm-gate parchment or cherry	₹480-500/kg arabica parchment (₹24,100-24,800 per 50 kg, Oct 2026)	Years of cultivation, picking, processing
Curer / trader	Clean green coffee, graded	₹600-950/kg (commodity to estate specialty)	Hulling, grading, sorting, logistics, risk
Roaster	Roasted beans, wholesale	₹1,050-1,700/kg; micro-lots ₹3,000+	16% weight loss, roasting skill, packing, brand
Retailer	Retail bag	₹1,600-3,600/kg (₹400-900 per 250 g)	Shelf, marketing, freshness
Café	Beverages	~₹12,000 of drinks per kg of coffee (55 doubles at ₹220)	Labour, rent, milk, skill, experience

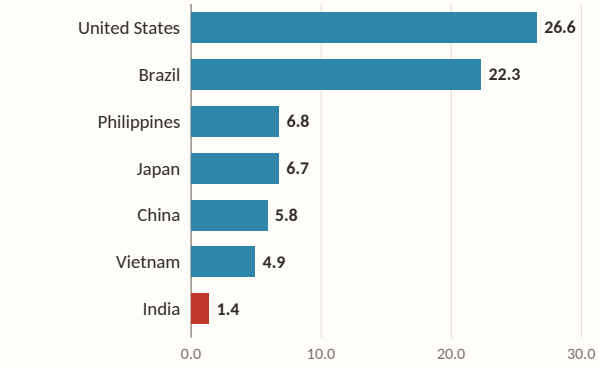
Source: Coffee Board indicative prices; Part 2 sourcing assumptions; Indian roaster price lists.

1.3 The world of coffee in numbers

Production 2025/26, million 60-kg bags



Consumption 2025/26, million bags (selected)



Source: USDA FAS, Coffee: World Markets and Trade (Dec 2025, Jun 2026); India figure from USDA Post (May 2026).

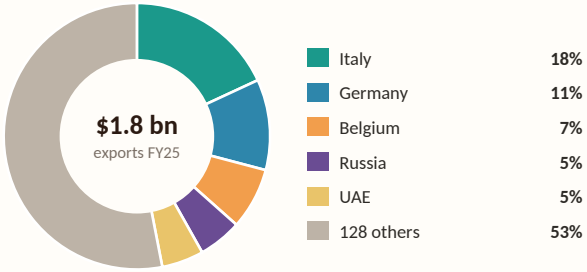
The 2025/26 world crop was a record **178.8 million bags** (about 10.7 million tonnes), against consumption of 173.9 million bags. Even so, ending stocks fell for a fifth straight year to 20.1 million bags, which is why prices stayed high. Brazil alone grows more than a third of the world's coffee, and Vietnam dominates robusta. For 2026/27 USDA expects another record crop, with Brazil at 71.9 million bags, while consumption also reaches a record 179.7 million bags.

1.4 India's place

India is the world's **7th-largest producer** (about 3.5% of output) and **5th-largest exporter** (about 5% of trade). Coffee covers about 4.9 lakh hectares in the Western and Eastern Ghats, supports more than two million livelihoods, and about 70% of the crop is exported to 128 countries. Around 99% of holdings are smallholdings, and they produce roughly 70% of the coffee.

Indian coffee is unusual in one respect: almost all of it is **shade-grown** under a two-tier canopy of about 50 native and planted tree species, intercropped with pepper, cardamom, vanilla, oranges and bananas. Exports set a record of **US$1.8 billion in FY2024-25** and about **US$2.06 billion in calendar 2025**. Italy is the largest buyer, using Indian robusta in espresso blends.

Where India's coffee goes (share of FY25 value)



Source: PIB backgrounder on India's coffee sector (29 Nov 2025); Coffee Board; export press reports Jan 2026.

1.5 A short history of coffee

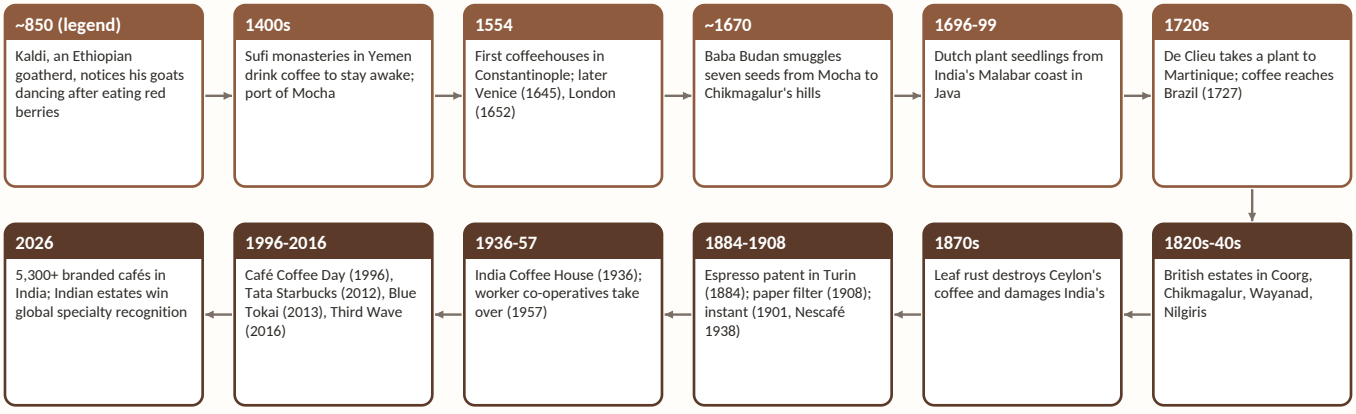


Figure 1.2. Twelve moments in coffee's history; India appears earlier than most people think.

Coffee's spread followed trade and empire. Arab traders guarded the plant until pilgrims and colonial powers carried seeds out of Yemen. Indian coffee entered the story early: the Dutch planted Java with seedlings from the Malabar coast, and Baba Budan's seeds, planted in the hills that now bear his name, are among the oldest coffee plantings outside Arabia and Africa. British planters turned the Western Ghats into commercial estates in the 19th century; after rust struck, many switched to tea and to the hardier robusta, which still dominates India's crop.

The 'waves' of coffee

Wave	Era	What it means	India's version
First wave	1800s-1960s	Coffee as a mass commodity: vacuum tins, instant, percolators	Instant and coffee-chicory blends; India Coffee House
Second wave	1970s-2000s	Espresso-based café chains, flavoured milk drinks, branded experiences	Café Coffee Day (1996), Barista (2000), Starbucks (2012)
Third wave	2000s-	Coffee as craft: origin, traceability, lighter roasts, manual brewing	Blue Tokai, Subko, KC Roasters, Black Baza and other estate-focused roasters
'Fourth wave'	2015-	Science, experimental processing, sustainability, ready-to-drink specialty	Anaerobic Indian micro-lots, cold-brew cans, specialty instant

1.6 How to read a coffee bag

Label item	Example	Why it matters
Origin / estate	Country, region and farm, e.g. 'Ratnagiri Estate, Chikmagalur'	The more specific, the more traceable
Altitude	Metres above sea level	Higher usually means denser beans and brighter acidity
Variety	S.795, SL9, Chandragiri, Catuai, Gesha…	Shapes the flavour potential
Process	Washed, natural, honey, anaerobic, monsooned	Clean versus fruity (Chapter 4)
Grade	Plantation A/AA, PB, Robusta Parchment AB…	Bean size and defect limits (Chapter 5)
Roast date and level	e.g. 'Roasted 2 Oct 2026, medium'	Look for a roast date, not just 'best before'
Tasting notes	e.g. 'dark chocolate, orange, jaggery'	Aromas the coffee resembles; nothing is added
Score / award	e.g. 'SCA 84', 'Fine Cup Award'	80+ = specialty
Blend ratio	e.g. '80:20 coffee-chicory'	Chicory is cheaper than coffee; check the share

1.7 The coffee economy at a glance

178.8 m bags
world crop 2025/26 (10.7 m tonnes)

~12.5 m
coffee farms worldwide, mostly < 5 ha

~70
producing countries in the bean belt

2 bn+
cups drunk around the world every day

6.4 m bags
India's crop; 7th-largest producer

US$2.06 bn
India's coffee exports, 2025

0.04 kg
coffee per Indian per year (world 1.3 kg)

5,339
branded cafés in India, +12.7% in a year

Source: USDA FAS; ICO; PIB; World Coffee Portal. Cups per day is a widely cited industry estimate.

1.8 India's coffee houses

The Coffee Board opened its first India Coffee House in Bombay in 1936 to promote coffee drinking. When the Board closed them in the 1950s, laid-off workers, backed by the communist leader A. K. Gopalan, took them over as co-operatives (from 1957). The Indian Coffee House chain, run by workers' co-operatives to this day, became a meeting place for writers, students and politicians, with College Street in Kolkata and Connaught Place in Delhi among the most famous branches. It shows that cafés as social spaces are not new in India; what is new is the scale, the specialty focus and the prices.

2 Growing coffee

The plant, the two species, varieties, terroir, India's regions, nurseries and the farming year

2.1 The plant

Left alone, an arabica plant grows into a small tree of 5-10 metres; on estates it is pruned to 1.5-2 metres so pickers can reach the fruit. Glossy, dark-green leaves grow in opposite pairs. White, jasmine-scented flowers appear in clusters where leaves meet the branch, open 8-10 days after a good shower of rain, and wilt within two or three days. The pollinated flowers become green berries that swell, then ripen from green to yellow to red (some varieties stay yellow). Arabica takes 8-9 months from flower to ripe cherry; robusta takes 10-11 months.

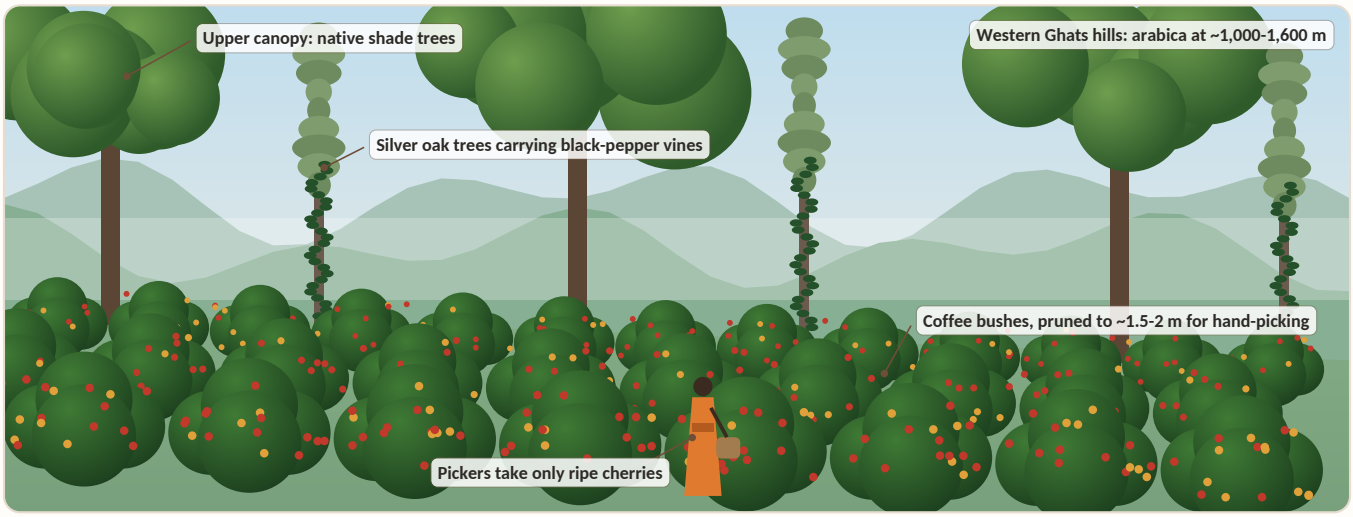


Figure 2.1. A shade-grown Indian estate: coffee bushes under silver oak and native trees, with pepper vines climbing the shade trees.

The two species differ in their genetics, and that shapes everything downstream. **Arabica** has four sets of chromosomes (44) and pollinates itself, so its varieties breed true from seed. **Robusta** has two sets (22) and cannot pollinate itself; bees and wind must carry pollen between plants, so every robusta plant is genetically a little different. Robusta is often multiplied by rooting cuttings (clones) of the best bushes.

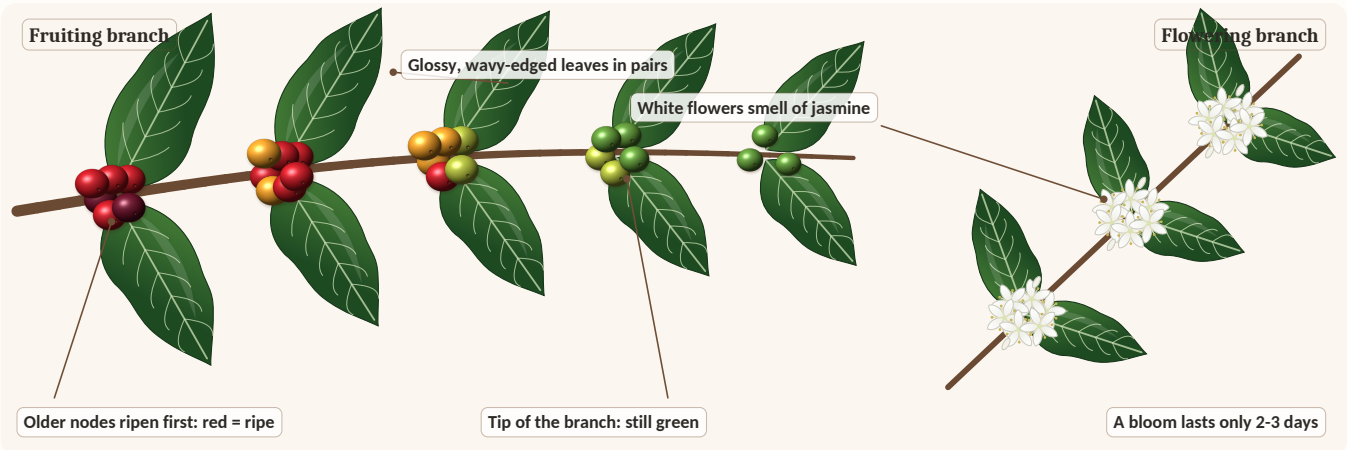


Figure 2.2. A fruiting branch: paired leaves, jasmine-like flowers and cherries ripening unevenly along the same branch, which is why selective picking takes several passes.

2.2 Arabica versus robusta

	Arabica	Robusta
Share of world output	~55-60%	~40-45%
Altitude in India	1,000-1,500 m (up to 2,000 m in the Pulneys)	500-1,000 m
Comfortable temperature	15-25°C	20-30°C
Chromosomes / pollination	44, self-pollinating	22, needs cross-pollination
Caffeine (dry weight)	1.2-1.5%	2.2-2.7%
Sugars (sucrose)	6-9%	3-7%
Lipids (oils)	15-17%	10-11.5%
Chlorogenic acids (bitterness precursors)	5.5-8%	7-10%
Bean shape	Oval, S-shaped centre cut	Rounder, straight centre cut
Yield in India (2026/27)	452 kg/ha	1,239 kg/ha
Main threats	Leaf rust, white stem borer	Berry borer, drought
Cup character	Sweeter, aromatic, brighter acidity	Heavy body, cocoa-woody, bitter, thick crema
Price signal (ICO, Apr 2025, US¢/lb)	Colombian milds 394	Robustas 246

Source: USDA FAS India Coffee Annual (May 2026); ICO Coffee Market Report (Apr 2025); composition ranges from coffee chemistry literature.

2.3 Varieties: the cultivars behind the names

Within each species there are many varieties (strictly, "cultivars"). They differ in plant height, yield, disease resistance and, above all, cup quality. Most of India's arabica comes from local selections bred for resistance to leaf rust, the fungus that wiped out coffee in Ceylon (Sri Lanka) in the 1870s and pushed many Indian planters towards robusta and tea.

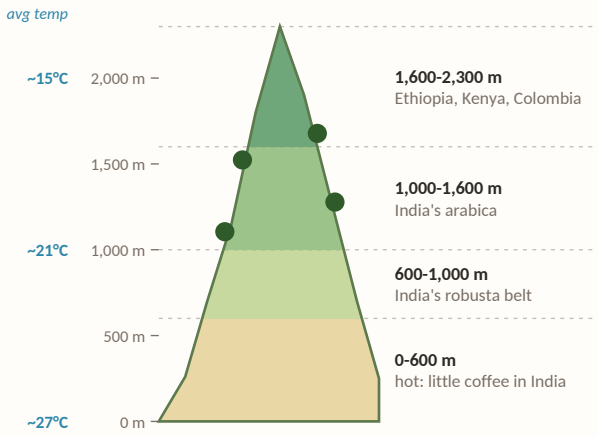
Variety	Origin	Why it matters
Typica, Bourbon	Global heirlooms	Tall, low-yield; sweet, complex cup; rust-susceptible
Kent's	India, 1920s	Selected on Kent's estate (Doddengooda); early rust tolerance
S.795 (Sln 795)	India, 1940s	Kent's x S.288; India's most planted arabica; balanced, mild acidity, chocolate and spice
Sln 9, Sln 5	India (CCRI)	Rust-tolerant selections involving Hibrido de Timor; good cup
Chandragiri	India (CCRI)	Semi-dwarf, bold beans, rust-tolerant, high-yielding
Cauvery (Catimor)	India	Caturra x Hibrido de Timor; dwarf, very high yield; plainer cup
SL28 / SL34	Kenya	Blackcurrant, vivid acidity; drought-tolerant
Gesha (Geisha)	Ethiopia via Panama	Jasmine, bergamot, tea; record auction prices
S.274 (Sln 274)	India (robusta)	Most popular Indian robusta; vigorous, bold beans
CxR	India (robusta)	C. congensis x robusta; compact bush, smoother cup

Source: Central Coffee Research Institute releases; TNAU Agritech portal; Counter Culture Coffee, India's arabica varieties.

2.4 Terroir: why place shows up in the cup

Terroir is the sum of place: altitude, latitude, soil, rainfall, sunshine and shade. The single biggest lever is **temperature**, and altitude is mostly a proxy for it: the air cools by roughly 0.6°C for every 100 m of height. In cooler air the cherry ripens slowly, the seed packs in more sugars and acids, and the bean grows denser. Dense, slow-grown beans give brighter acidity and more complex aromas, and they stand up to lighter roasts.

Closer to the equator, coffee must grow higher to find the same coolness, so 1,200 m in Coorg (12°N) is roughly equivalent to 1,600-1,800 m in Colombia. Soils matter too: coffee likes deep, well-drained, slightly acidic (pH 4.5-6.5), humus-rich forest soils. **Shade** slows ripening, protects soil and biodiversity, and is one reason Indian coffees are mellow and low in harshness.



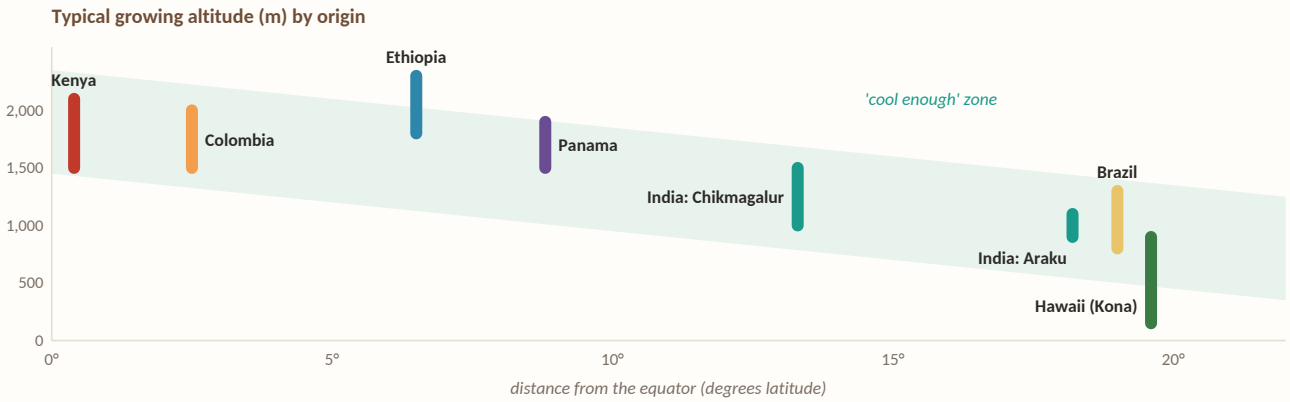


Figure 2.3. Latitude and altitude together set temperature: Indian estates at 11-14°N reach specialty-grade coolness at lower altitudes than equatorial origins.

2.5 India's growing regions

Region	State	Altitude	Species	Notes
Chikmagalur & Bababudangiri	Karnataka	1,000-1,500 m	Arabica, robusta	Birthplace of Indian coffee; GI Chikmagalur & Bababudangiri arabica; chocolate, spice
Coorg (Kodagu)	Karnataka	800-1,100 m	Mostly robusta	Largest district; GI Coorg arabica; robusta for espresso blends
Biligiris (BR Hills)	Karnataka	1,000-1,500 m	Arabica	Source of Mysore Nuggets EB
Wayanad	Kerala	700-1,200 m	Robusta	GI Wayanad robusta; smallholder belt
Travancore, Nelliampathy	Kerala	500-1,600 m	Both	Mixed estates with spices
Pulneys	Tamil Nadu	1,000-2,000 m	Arabica	India's highest coffee; bright, fruity
Nilgiris, Shevaroys, Anamalais	Tamil Nadu	900-1,800 m	Mostly arabica	Shevaroys (Yercaud) mild and sweet
Araku Valley	Andhra Pradesh	900-1,100 m	Arabica	Tribal, organic-leaning smallholders; GI; award-winning
Koraput & North-East	Odisha, NE states	600-1,200 m	Arabica	Non-traditional areas; small but growing (Koraput award at KYK 2024)

Karnataka produces about 70% of India's coffee (2.80 lakh tonnes in 2025-26), Kerala about a fifth and Tamil Nadu about 5-6%; together they make up 96%. The Coffee Board recognises 13 agro-climatic zones. Six names carry Geographical Indication (GI) tags: Coorg Arabica, Wayanad Robusta, Chikmagalur Arabica, Araku Valley Arabica, Bababudangiris Arabica and Monsooned Malabar (arabica and robusta).

Source: PIB (Nov 2025); Coffee Board post-blossom estimate 2025-26; Coffee Board GI list.

2.6 Before the estate: research, seed and the nursery

The chain starts in a research station. India's **Central Coffee Research Institute** (CCRI), founded in 1925 at Balehonnur in Chikmagalur and backed by regional stations, breeds and releases varieties and supplies certified seed. Growers can also select seed from their own healthiest, highest-yielding plants.

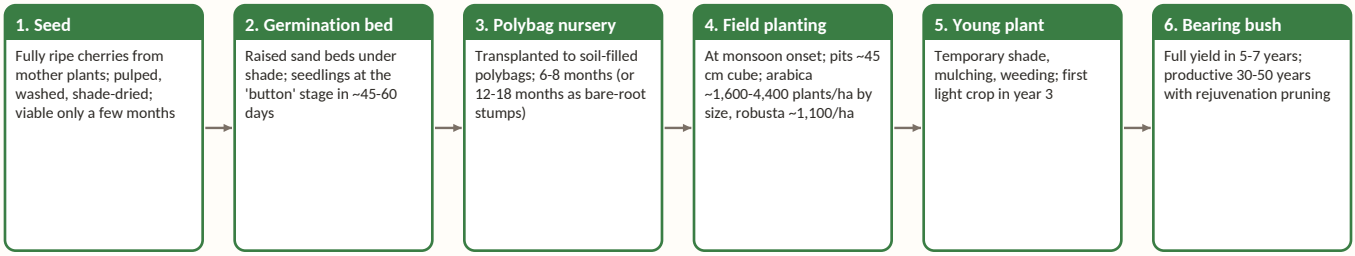


Figure 2.4. From seed to bearing bush takes three to five years, which is why replanting decisions are slow and costly.

2.7 The farming year in India

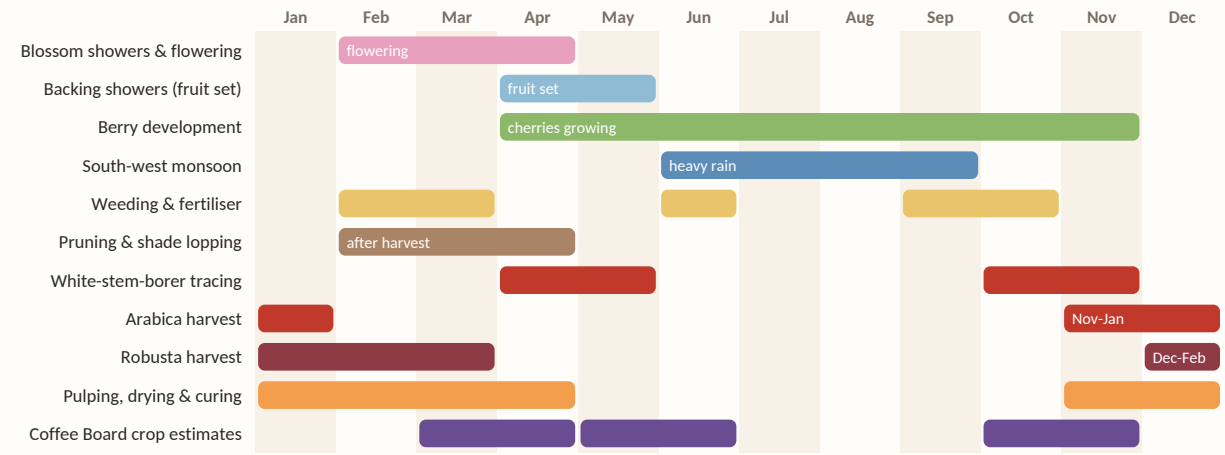


Figure 2.5. The Indian coffee calendar. A good blossom shower in February-March, followed by a 'backing' shower two to three weeks later, sets the size of the whole crop.

The year turns on rain. After the dry winter, a soaking **blossom shower** (or sprinkler irrigation that imitates it) triggers simultaneous flowering across an estate; a **backing shower** a few weeks later helps the fruit set. A failed blossom shower, or unseasonal rain that knocks flowers off, can cut a crop by a fifth. The south-west monsoon (June-September) brings most of the year's rain; too much of it causes black rot and berry drop, as in 2025. The Coffee Board publishes three crop estimates a season: post-blossom (May-June), post-monsoon (October-November) and final (March-April).

2.8 Estate economics and risks

Indian yields are modest by world standards: about 450-475 kg of clean coffee per hectare for arabica and 1,150-1,250 kg for robusta. Coffee is labour-intensive: **about 70% of the cost of production is labour**, most of it at harvest, and estates increasingly rely on migrant workers. Farm-gate prices are volatile: after two very profitable years, arabica farm-gate prices fell 16% and robusta 11% between October 2025 and May 2026 as the world crop recovered.

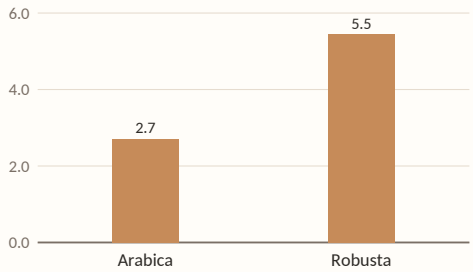
Risk	What it is	Impact and response
White stem borer	Beetle larvae tunnel through arabica stems	The main arabica pest in India; managed by shade, tracing and uprooting infested plants
Coffee berry borer	Tiny beetle bores into the berry	Damaged beans taste dirty; traps, field sanitation, clean picking
Leaf rust	Fungus (Hemileia vastatrix) strips leaves	Reason for India's rust-tolerant selections; fungicide sprays
Black rot, berry drop	Fungal rot in prolonged monsoon rain	Hit 2025-26 crop estimates
Climate change	Hotter, more erratic rainfall	Studies warn that up to half of today's arabica land may become unsuitable by 2050

Source: USDA FAS India Coffee Annual (May 2026); CCRI; Deccan Herald and StoneX reports on the 2025-26 crop.

2.9 Why India grows mostly robusta: the economics per hectare

Gross revenue per hectare (₹ lakh)

2026/27 yields x Oct-2026 green prices



Arabica sells for more per kilo, but robusta yields almost three times as much per hectare in India (1,239 against 452 kg). At October 2026 prices (about ₹600/kg for commodity arabica and ₹440/kg for robusta), a hectare of robusta grosses about **₹5.5 lakh** against **₹2.7 lakh** for arabica, and robusta needs less labour for pest control and is less exposed to white stem borer and rust. Arabica makes sense at altitude, where quality premiums (estate specialty lots fetch ₹900-1,000/kg and micro-lots far more) close the gap.

This is why robusta makes up roughly three quarters of the crop, and why India's specialty future depends on both better arabica and **fine (washed, selectively picked) robusta**, a category where India is a world leader.

2.10 Shade, biodiversity and sustainability

Most Indian coffee grows under a two-tier canopy: tall native forest trees above and planted shade trees such as silver oak and dadap below, often with pepper vines climbing them. The Western Ghats are a global biodiversity hotspot, and shaded estates act as wildlife corridors for birds, elephants and other animals. Shade lowers yields slightly but slows ripening (better flavour), protects soil, reduces the need for chemicals and gives growers second incomes from pepper, cardamom, oranges and timber.

Scheme	What it certifies	Relevance in India
Rainforest Alliance	Environmental and social farm standard	Widely used by Indian estates selling to Europe
Fairtrade	Minimum price + premium for smallholder co-operatives	Limited in India
Organic (NPOP / EU / USDA)	No synthetic fertilisers or pesticides	Araku, some Kerala and Karnataka growers
Bird-friendly / shade	Shade-cover standards (e.g. Smithsonian)	A natural fit for Indian estates; used by brands such as Black Baza
C.A.F.E. Practices, 4C	Buyer-led (Starbucks) and baseline sustainability codes	Large estates and exporters
EU deforestation rule (EUDR)	Proof that coffee was not grown on land deforested after 2020	Geolocation of every plot; costly for smallholders

2.11 Estate names worth knowing

India's specialty reputation rests on a set of estates and growing communities. Examples that appear on Indian roasters' menus include **Ratnagiri, Baarbara, Kelagur and Thogarihunkal** (Chikmagalur), **Attikan** (Biligiris), **Balanoor, Watapi, Palthope and Yellikodige** (Karnataka's coffee belt), and the tribal farmer co-operatives of **Araku Valley**. Many sell micro-lots by variety and process, for example a natural S.795 or an anaerobic Chandragiri, which is exactly what a specialty café's pour-over menu and retail shelf need.

2.12 The world's origins, and how India compares

Origin	Typical coffees	Cup profile	Note
Ethiopia	Heirloom landraces; washed and natural; 1,800-2,200 m	Jasmine, bergamot, lemon (washed); blueberry (natural)	Birthplace of arabica; huge genetic diversity
Kenya	SL28, SL34, Ruiru 11, Batian; double-fermented washed	Blackcurrant, grapefruit, tomato; vivid acidity	Weekly auctions in Nairobi
Colombia	Castillo, Caturra; two harvests a year	Caramel, red fruit, balanced	Strong grower federation (FNC)
Brazil	Natural and pulped natural; flat, mechanised farms	Nuts, chocolate, low acidity	A third of world supply; base of most espresso blends
Central America	Guatemala, Costa Rica, Honduras, Panama; mostly washed	Chocolate, citrus, stone fruit	Panama Gesha sets auction records
Vietnam	Robusta, dry-processed	Heavy, bitter, woody	No.1 robusta producer; instant and blends
Indonesia	Sumatra, Java, Sulawesi; wet-hulled	Earthy, herbal, cedar, syrupy	Distinctive 'giling basah' process
Yemen	Ancient landraces, natural	Winey, spicy, dried fruit	Where the coffee trade began
India	Shade-grown arabica and robusta; washed, natural, monsooned	Chocolate, spice, mild acidity, full body	Leader in fine robusta; Monsooned Malabar

Indian arabica is rarely as acidic or floral as East African coffee; its strengths are **balance, chocolate and spice, low bitterness and body**, which suit espresso and milk drinks. A café can still offer variety by rotating a guest East African or Latin American coffee on pour-over, but the commercial heart of an Indian specialty menu can be Indian.

2.13 Climate change and adaptation

Coffee is one of the crops most exposed to climate change. Warmer temperatures push arabica uphill, erratic rains disrupt flowering, and pests such as the berry borer spread to higher altitudes. India saw this in 2025, when an unusually long monsoon caused black rot and berry drop. Adaptation is already under way:

- **Shade and soil cover**: India's two-tier shade buffers heat; estates are adding mulching and cover crops to hold moisture.
- **Water management**: sprinkler irrigation to substitute for failed blossom showers, farm ponds and water harvesting.
- **Resilient planting material**: CCRI's rust- and drought-tolerant selections, and renewed interest in robusta and other hardy species (liberica, excelsa).
- **Diversification**: pepper, cardamom, oranges and agro-tourism (estate stays) spread income risk.
- **For buyers**: expect more volatile prices and quality; long-term relationships with estates are the best hedge.

2.14 An atlas of India's coffee regions

Chikmagalur & Baba Budan Giri — Karnataka	
Altitude	1,000-1,600 m
Rain	~1,500-2,500 mm
Coffee	Arabica (S.795, Sln 9, Chandragiri) and robusta
Process	Washed, natural, micro-lots
Estates	Ratnagiri, Baarbara, Kelagur, Thogarihunkal
Cup	Chocolate, spice, citrus, full body

Coorg (Kodagu) — Karnataka	
Altitude	750-1,100 m
Rain	~1,500-4,000 mm (wetter west)
Coffee	Mostly robusta (S.274, CxR); arabica higher up
Process	Natural 'cherry' and washed 'parchment'
GI	Coorg Arabica
Cup	Robusta: cocoa, nutty; arabica: mild, sweet

Biligiris (BR Hills) — Karnataka	
Altitude	1,000-1,500 m
Coffee	Arabica
Process	Washed
Estates	Attikan
Grade	Source of Mysore Nuggets EB
Cup	Balanced, sweet, clean

Wayanad — Kerala	
Altitude	700-1,200 m
Coffee	Robusta, smallholder-grown
Process	Mostly natural
GI	Wayanad Robusta
Cup	Heavy body, chocolate, earthy

Travancore & Nelliampathy — Kerala	
Altitude	500-1,600 m
Coffee	Both species
Farming	Intercropped with pepper, cardamom
Cup	Varied; spice notes

Pulneys — Tamil Nadu	
Altitude	1,000-2,000 m (India's highest)
Coffee	Arabica
Process	Washed
Cup	Bright acidity, fruit, complexity

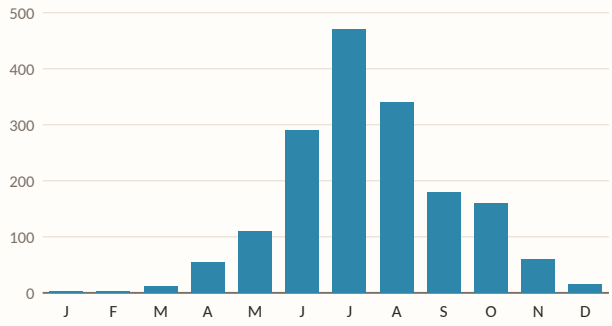
Nilgiris & Shevaroys — Tamil Nadu	
Altitude	900-1,800 m
Coffee	Mostly arabica
Place	Shevaroys around Yercaud
Cup	Mild, sweet, gentle acidity

Araku Valley — Andhra Pradesh	
Altitude	900-1,100 m (Eastern Ghats)
Coffee	Arabica
Growers	Tribal smallholders, organic-leaning
GI	Araku Valley Arabica
Cup	Fruity, sweet; award-winning lots

Koraput & North-East — Odisha, NE states	
Altitude	600-1,200 m
Coffee	Arabica
Status	Non-traditional, expanding; Koraput won KYK awards (2024)
Cup	Light, clean, developing

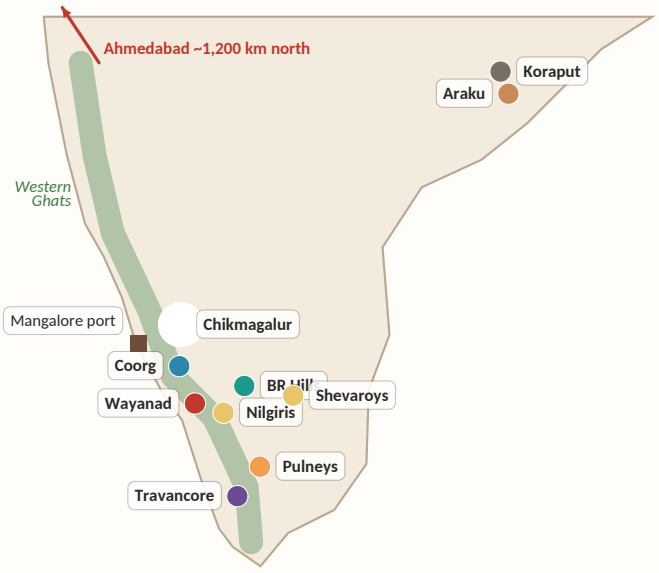
Source: Coffee Board region profiles and GI registry; CCRI; estate and roaster listings. Altitude and rainfall ranges are indicative.

Chikmagalur: approximate monthly rainfall (mm)



The rainfall pattern of a typical coffee district explains the calendar in Figure 2.5: a dry winter (harvest and drying weather), a few critical showers in March-April (blossom and backing showers), and a heavy south-west monsoon from June to September that fills the berries but can also cause rot. Western slopes facing the monsoon can receive two or three times the rain of eastern estates a few kilometres away, one reason micro-climates and estates differ so much.

Source: Approximate long-term averages for Chikmagalur town (India Meteorological Department climate normals, rounded).



Schematic, not to survey accuracy

Figure 2.6. Where India's coffee grows: the Western Ghats of Karnataka, Kerala and Tamil Nadu, plus the Eastern Ghats of Andhra Pradesh and Odisha.

2.15 How an estate is managed through the year

Practice	What estates do	Why
Nutrition	Fertiliser in three splits (pre-blossom, pre-monsoon, post-monsoon); lime every few years to correct acidic soils; compost and coffee pulp returned to the field	Yield and bean size
Bush training	Arabica on a single stem 'centred' (topped) at about 1.2-1.5 m; robusta on multiple stems; 'handling' removes unwanted suckers after harvest	Pickable height, productive wood
Pruning	Cut out old, dead and crossing branches after harvest; periodic rejuvenation of exhausted bushes	Air, light, fewer pests
Shade regulation	Lop shade trees before the monsoon so light and air reach the bushes; keep two tiers	Balance of yield and quality
Water	Sprinkler 'blossom' irrigation in dry years; mulching; farm ponds	Flowering, crop security
Weeds and soil	Slashing or manual weeding 2-3 times a year; cover crops; contour trenches	Soil conservation on slopes
Pest and disease watch	Trace and remove white-stem-borer plants; traps for berry borer; fungicide for rust and black rot when needed	Protects the crop and the tree

Source: CCRI and Coffee Board package of practices; TNAU Agritech portal.

3 Harvesting

Ripeness, picking methods, the yield arithmetic and what happens on picking day

Quality is mostly made or lost on picking day. A cup can only be as good as the ripest, cleanest cherries that went into it.

3.1 Why ripeness matters

A coffee cherry is a fruit, and like any fruit it is sweetest when fully ripe. As it ripens from green to deep red, starches turn into sugars (ripe cherries typically measure 18-24° Brix, similar to grapes), acids mellow, and the seed finishes developing. Unripe cherries give grassy, astringent, "peanutty" coffee and pale beans that refuse to roast evenly (roasters call them *quakers*). Overripe or dried-on-the-bush cherries start to ferment and add winey, boozy or mouldy notes.

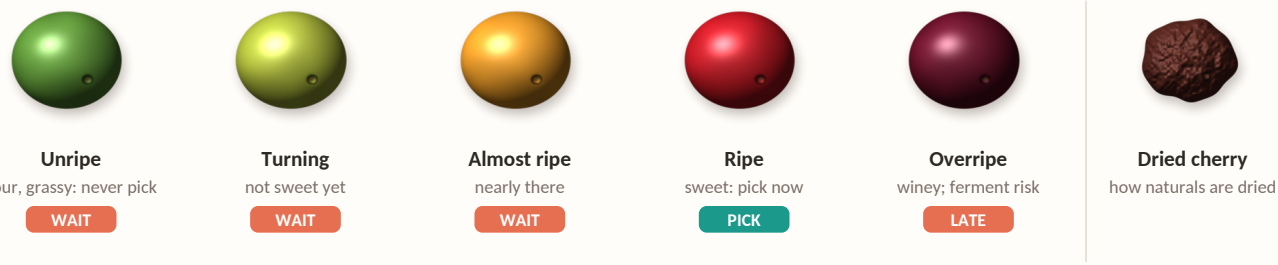


Figure 3.1. Ripeness stages. Only the ripe red (or ripe yellow in yellow varieties) cherries should go to the pulper.

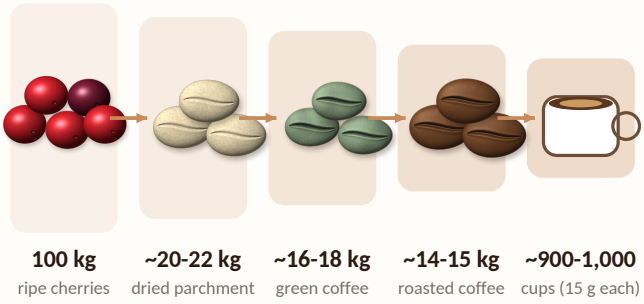
3.2 How coffee is picked

Method	How it works	Productivity	Quality and where used
Selective hand-picking	Pickers take only ripe cherries; 3-5 passes over the season	45-55 kg of cherry a day on average (50-120 kg for skilled pickers)	Highest; standard on Indian arabica estates
Strip picking	All cherries stripped off a branch onto sheets in one pass	2-3 times faster than selective	Mixed ripeness; needs sorting; common for robusta
Mechanical	Tractor-drawn machines shake the branches	Very fast; low cost per kg	Needs flat, open land (Brazil); not feasible on India's steep, shaded slopes
Gleaning	Fallen cherries collected from the ground	Cleans the field	Low; should be kept out of quality lots

Source: Springer (2026) picker study; Barista Hustle; USDA FAS.

3.3 The yield arithmetic

It takes a surprising amount of fruit to make a cup. Roughly **5-6 kg of ripe cherry** give 1 kg of green coffee; roasting then loses another 14-20% of the weight. One picker's day of about 50 kg of cherry therefore becomes roughly 9 kg of green coffee, about 7.5 kg roasted, which is enough for around 420 double espressos.



The arithmetic explains two things. First, why careful selective picking is expensive: labour is paid per kilo of cherry, and every pass through the field costs money. Second, why the grower's share of a café's cup price is small: the coffee in an 18 g double shot costs the café ₹21-30, and the green coffee inside it, which is what the grower and curer are paid for, is only ₹13-20 of a ₹220 drink.

3.4 On picking day

Good estates process cherries the same day, ideally within 6-12 hours, before they start to ferment in the sacks. On arrival cherries are tipped into a water tank: ripe, dense cherries sink, while **floaters** (unripe, insect-damaged or dried cherries) rise and are skimmed off and sold as a lower grade. Many estates also hand-sort on tables or sheets. Pickers are paid by weight, so the weighing shed doubles as the estate's quality checkpoint.

What this means for a café buyer

Ask roasters and estates about **picking and sorting**: selective picking and floating are the cheapest insurance against defects in the cup. Robusta is often strip-picked. Washed, selectively picked robusta (sold as "fine robusta" or Kaapi Royale) is a different, much cleaner product, and is worth paying for in an espresso blend.

3.5 Measuring ripeness and harvest quality

Specialty estates measure what they pick. A **colour count** of a sample (the share of ripe red cherries; specialty lots aim for 90-95% or more) and a **Brix refractometer** reading of the fruit's sugar (about 18-24° when ripe) tell the supervisor whether pickers are being selective enough. Because altitude delays ripening, the harvest moves uphill through the season: lower blocks are picked first, high blocks last, and estates make three to five passes through each block.

Good practice	What to check
Ripe-only picking	Share of ripe cherries ≥ 90-95% in a sample
Same-day processing	Cherries pulped or laid out to dry within 6-12 hours
Floating and sorting	Floaters removed and sold as a separate lower grade
Lot separation	Each day's picking, block and variety kept apart and labelled
Clean equipment	Bags, tanks and pulpers washed daily to avoid ferment taints
Records	Weight, ripeness, Brix and processing notes per lot for traceability

3.6 Labour, cost and mechanisation

Harvesting is the single largest labour job on an Indian estate, and labour is about 70% of the cost of growing coffee. Pickers are paid by the kilo of cherry, so selective picking, which yields fewer kilos per day, must be rewarded with higher rates or quality bonuses. Labour has become scarcer and more expensive, and many estates rely on seasonal workers from other states. The Coffee Board has published planting designs that would allow small machines and mechanical aids between rows, but on steep, shaded, terraced slopes the full mechanical harvesting used in Brazil remains impractical. Expect Indian specialty coffee to stay hand-picked, and labour costs to keep rising faster than general inflation.

3.7 A harvest day at an Indian estate

Time	Step	What happens
7:00-12:30	Picking	Gangs work assigned blocks; selective picking of ripe cherries into baskets and sacks
12:30-14:00	Weighing	Each picker's cherries weighed at the field shed; ripeness spot-checks; piece-rate recorded
14:00-17:00	Second round & transport	Afternoon picking; cherries trucked to the pulper house
17:00-21:00	Floating & pulping	Siphon tank removes floaters; pulper runs into the night; pulped beans go to fermentation tanks
Overnight	Fermentation	12-36 hours depending on temperature; checked for 'grit'
Next morning	Washing & grading	Channels wash off mucilage; heavy beans separated
Following 1-2 weeks	Drying	Parchment spread on concrete 'barbecues' or raised beds, raked, covered at night
Weeks later	Rest & dispatch	Parchment bagged, rested, then sent to curing works with lot records

Naturals follow a simpler path: after floating, cherries go straight to drying yards for 12-15 days. For smallholders without a pulper, selling fresh cherry or dried cherry to a trader is the norm, which is one reason most Indian robusta is natural.

4 Processing

Turning a juicy cherry into a dry, stable green bean: washed, natural, honey, monsooned and the new experimental methods

4.1 What processing has to do

A fresh cherry is 55-60% water and spoils within a day. Processing removes the fruit layers around the seed and dries the seed to about **10-12% moisture**, at which point it can be stored and shipped for a year. How the fruit is removed, and how long the seed sits in contact with sugary fruit and microbes while drying, shapes the flavour as much as variety or altitude.

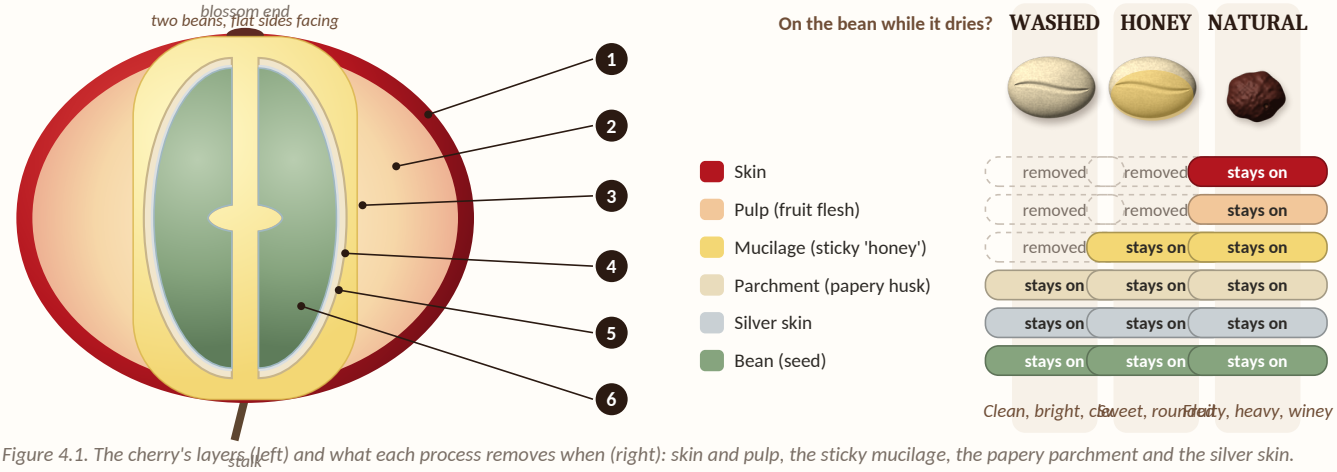


Figure 4.1. The cherry's layers (left) and what each process removes when (right): skin and pulp, the sticky mucilage, the papery parchment and the silver skin.

4.2 The washed (wet) process, step by step

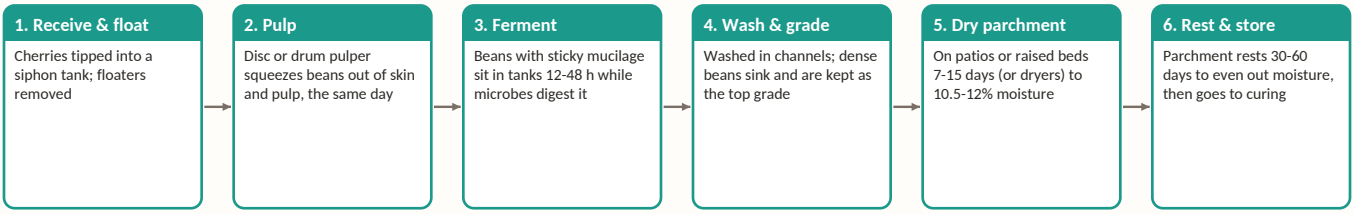


Figure 4.2. Washed processing. In India, washed arabica is sold as 'Plantation' and washed robusta as 'Parchment'.

Fermentation is the delicate step. Natural yeasts and bacteria break down the mucilage (a pectin-rich layer) so it can be washed off; workers rub a handful of beans and stop when they feel gritty rather than slippery. Too short and mucilage remains; too long and the coffee turns sour, vinegary or onion-like. Some mills skip fermentation altogether and scrub the mucilage off mechanically ("demucilaged" or semi-washed). Washed coffees taste **clean, bright and transparent**: you taste the variety and the place rather than the process. Washing needs water and creates effluent, which modern eco-pulpers and recirculation reduce.

4.3 Natural (dry) process

The oldest method: whole cherries are dried with the fruit on. They are spread 3-4 cm deep on patios or raised beds, raked several times a day so they dry evenly, and covered at night and in rain. In bright Indian winter sun this takes 12-15 days; in humid places three to four weeks. The dried husk is removed later at the curing works. Because the seed dries inside the sugary fruit, naturals taste **fruity, sweet and heavy-bodied**, sometimes winey, with less clarity. The risk is mould or over-fermentation if drying stalls. Most Indian robusta, and arabica sold as 'Cherry', is natural.

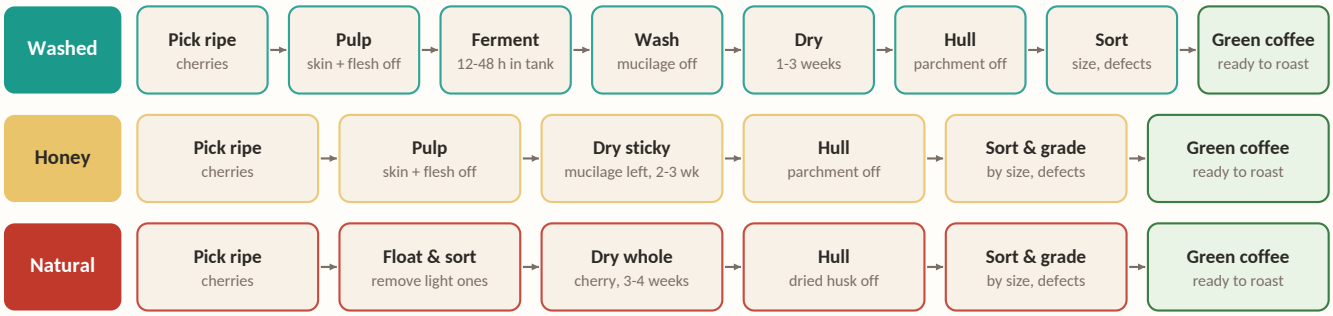


Figure 4.3. Washed, honey and natural processing side by side.

4.4 Honey, semi-washed and wet-hulled

Honey (or pulped natural) coffee is pulped like a washed coffee but dried with some or all of the sticky mucilage left on. Yellow, red and black honey indicate increasing amounts of mucilage and slower drying; the cup sits between washed and natural, sweet with a rounded body. **Wet-hulled** coffee (Indonesia's *giling basah*) is hulled while still at 30-35% moisture, which gives bluish beans and an earthy, heavy, low-acid cup.

4.5 Monsooned Malabar: India's own process

In the age of sail, Indian coffee spent up to six months in damp ships' holds on the way to Europe and arrived swollen, pale and mellow. When the Suez Canal (1869) and steamships shortened the voyage, European buyers missed that taste, so Indian exporters recreated it on land.

Today naturally processed coffee is spread in open-sided warehouses on the Malabar coast (around Mangalore and Tellicherry) during the south-west monsoon. For **12-16 weeks** (June-September) the beans absorb moist sea air, are regularly raked and re-bagged, and swell to nearly twice their size, turning pale gold. The result has almost no acidity, a syrupy body and earthy, spicy, woody notes. It carries a GI tag and is prized in espresso blends.



Before: dry-processed
green-yellow, firm

After 12-16 weeks
swollen, pale gold

4.6 Experimental processes

Process	What happens	Typical flavour
Anaerobic fermentation	Cherries or pulped beans sealed in tanks without oxygen; temperature and time controlled	Intense fruit, cinnamon, tropical, sometimes boozy
Carbonic maceration	Whole cherries in CO2-flushed tanks, a method borrowed from Beaujolais wine	Red fruit, bubble-gum, wine-like
Yeast / culture inoculation	Selected wine or brewing yeasts or lactic bacteria added	More control and repeatability
Thermal shock, co-ferments	Hot and cold water treatments; fermenting with fruit, spices or hops	Very loud flavours; co-ferments are controversial because flavour is added, not grown

These methods command large premiums in competitions and specialty cafés. Several Indian estates now sell anaerobic and carbonic lots, but they suit pour-over menus and retail shelves more than a café's everyday espresso.

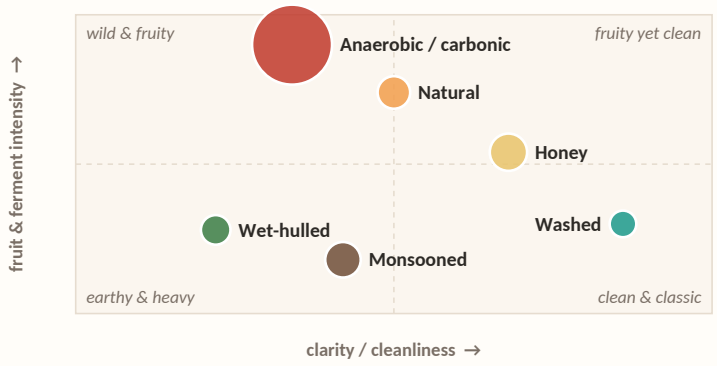
4.7 Drying, storage and moisture

Method	Strength	Watch-out
Patio (concrete or brick)	Cheap, large volumes	Slow and uneven if not raked; dust
Raised beds (African beds)	Air flows underneath; even, clean drying	Labour; more space per kg
Polytunnels / solar dryers	Protect against rain; useful in monsoon regions	Can overheat if not ventilated
Mechanical dryers	Rotating drums at 40-45°C; weather-proof	Fuel cost; can 'bake' the coffee if too hot

Target moisture under Coffee Board standards is **10.5%** for Plantation and Robusta Parchment and **11.5%** for Cherry grades. Too wet (above about 12.5%) and the coffee grows mould; too dry and it tastes papery and roasts unevenly. Green coffee is stored in jute or in hermetic liners (such as GrainPro) and keeps well for 6-12 months; older "past crop" coffee tastes woody and flat.

4.8 Process and flavour

Processing flavour map (bubble size = typical price premium)



The map summarises the trade-off: washed coffees win on clarity, naturals and honeys on fruit and sweetness, and experimental processes push fruit and fermentation to the extreme. Monsooned and wet-hulled coffees trade acidity for body and earthiness.

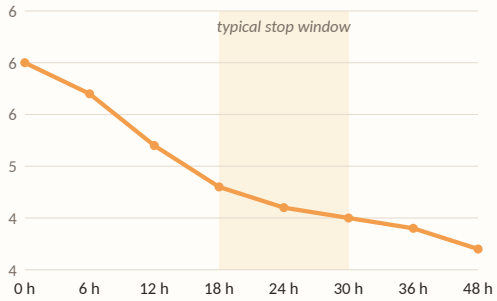
For a café menu, a washed Indian arabica with a share of washed robusta gives a clean, chocolatey espresso that works with milk; a natural or anaerobic micro-lot is best kept for pour-over and retail.

4.9 The microbiology of fermentation

Fermentation is not just 'rotting off' the mucilage. Yeasts (such as *Saccharomyces*, *Pichia* and *Candida*), lactic acid bacteria and, later, acetic acid bacteria feed on the sugars and pectin. They produce organic acids, alcohols and fruity esters, and the pH of the fermenting mass falls from about 6 to 4-4.5. Some of these compounds migrate into the bean and change its flavour.

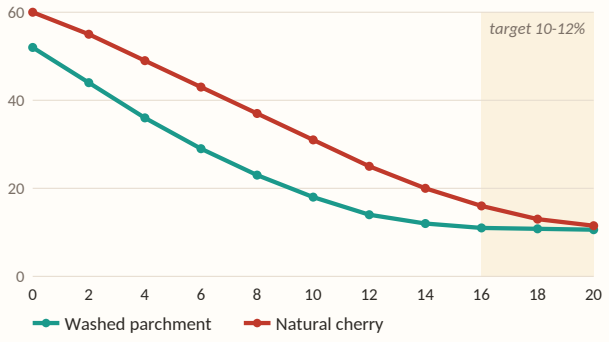
Time and temperature decide the outcome: a cool, short ferment gives clean coffee; a warm, long one gives fruit, then wine, then vinegar. Over-fermentation creates acetic, butyric and propionic acids (sour, cheesy, rancid). That is why producers now monitor pH and temperature, and some add selected yeast or bacterial cultures for repeatable results.

pH falls as microbes work (illustrative)



4.10 Drying curves: slow and steady wins

Moisture content while sun-drying (%, illustrative)



Coffee should dry evenly over one to three weeks. Drying too fast on hot concrete cracks the parchment and 'bakes' flavour; drying too slowly invites mould and ferment. Producers turn the coffee several times a day, cover it at night, and stop at 10-12%, measured with a moisture meter.

After drying, parchment rests for 30-60 days so moisture evens out between beans before hulling. Water activity (a_w, ideally 0.50-0.60) is increasingly measured alongside moisture because it predicts shelf life.

4.11 Water, waste and by-products

By-product	Where it comes from	Uses
Pulp and skin	~40% of the cherry's weight	Compost, biogas, or dried as **cascara** (a fruity 'coffee cherry tea')
Mucilage and wash water	Acidic, high in sugars; pollutes streams if untreated	Settling ponds, biodigesters, recirculating eco-pulpers
Parchment and husk	Removed at curing	Fuel for dryers and boilers, mulch
Silver skin (chaff)	Comes off in the roaster	Compost; must be collected (fire risk)
Spent coffee grounds	~1 kg wet grounds per 55 doubles in a café	Compost, mushroom growing, biofuel pilots

Traditional wet mills used large volumes of water per kilo of green coffee; modern eco-pulpers and recirculation cut this sharply, and Indian pollution rules require effluent treatment. For a café, spent grounds are a visible sustainability story: partner with a composting or urban-farming group.

4.12 Indian processing in practice

Trade name	Species	Process	Where	Cup
Plantation (washed arabica)	Arabica	Washed	Estates in Chikmagalur, Biligiris, Pulneys, Shevaroys	Clean, balanced, mild acidity
Arabica Cherry	Arabica	Natural	Smallholders and estates	Heavier, fruit, less clean
Robusta Parchment	Robusta	Washed	Coorg, Wayanad estates	Clean, chocolate, low bitterness; Kaapi Royale
Robusta Cherry	Robusta	Natural	Most smallholders	Heavy, woody; bulk for instant and blends
Monsooned Malabar	Both	Natural, then monsooned	Malabar coast warehouses	No acidity, syrupy, spicy
Micro-lots	Mostly arabica	Honey, anaerobic, carbonic	Specialty estates	Fruit-forward, premium-priced

4.13 The equipment of processing

Equipment	Job	Where
Siphon / flotation tank	Separates floaters from dense ripe cherries	Estate pulper house
Disc or drum pulper	Squeezes beans out of skin and pulp	Estate
Eco-pulper / demucilager	Pulps and scrubs off mucilage mechanically with little water	Modern estates, co-operatives
Fermentation tanks	Concrete or steel tanks where mucilage breaks down	Estate
Washing channels	Water channels that wash and density-grade parchment	Estate
Drying yards ('barbecues')	Concrete or brick patios for parchment and cherry	Estate, smallholders
Raised beds and polytunnels	Mesh beds with airflow; rain protection	Specialty estates
Mechanical dryers	Rotating drums or vertical dryers at 40-45°C	Large estates, curers
Moisture meter	Checks 10-12% target	Everywhere
Huller and polisher	Remove parchment / husk and silver skin	Curing works
Screens, gravity table, optical sorter	Grade by size and density; reject defects by colour	Curing works

4.14 Where quality is lost in processing

Failure point	What goes wrong	Taste	Prevention
Delay before pulping	Cherries ferment in sacks	Sour, vinegar	Pulp within 6-12 h
Unsorted cherries	Unripe and damaged mixed in	Grassy, astringent, dirty	Float and hand-sort
Over-fermentation	Too long or too warm	Sour, onion, cheesy	Monitor pH and time
Dirty tanks and water	Microbial taints	Phenolic, medicinal	Clean equipment daily
Drying too fast	Hot concrete, thick layers not turned	Baked, cracked parchment	Thin layers, raking, shade at noon
Drying too slow / rain	Mould	Musty, earthy	Covers, raised beds, dryers
Re-wetting in storage	Damp godowns	Mouldy, faded	Dry stores, hermetic bags, pallets
Long storage	Ageing green coffee	Papery, woody ('past crop')	Ship and roast within the year

5 Milling, grading and quality

Curing works, India's grade names, defects and how professionals score coffee

5.1 The curing works

In India, dry mills are called **curing works**; there are clusters around Chikmagalur, Hassan, Kushalnagar, Mangalore and Coimbatore. They turn parchment or dried cherry into export-ready green coffee and grade it. The **outturn** (the share of clean coffee that comes out) is roughly 80% for parchment and about 50% for dried cherry.

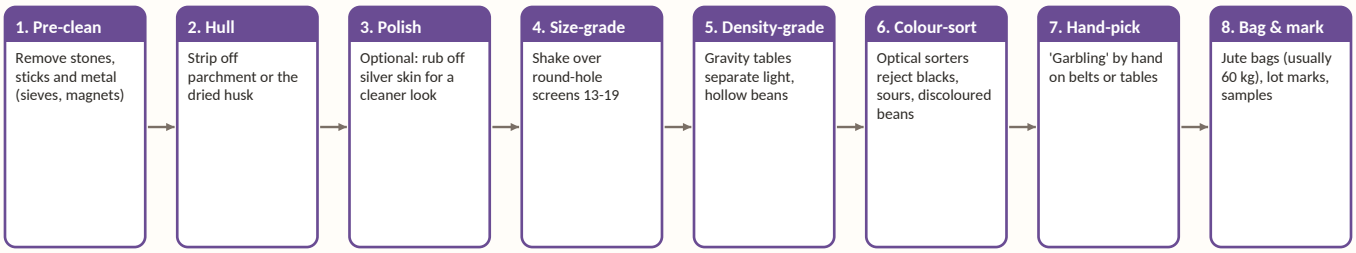
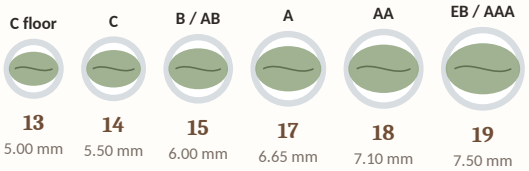


Figure 5.1. Inside a curing works.

5.2 India's grade names, decoded

Indian grades combine a **processing name** with a **size letter**. Washed arabica is **Plantation**, washed robusta is **Parchment**, and naturals of either species are **Cherry**. The letters describe bean size, measured by the round-hole screen the beans sit on, plus limits on defects ("clean garbled" means hand-picked clean).

Screen holes to scale (mm)



Screen number = hole size in 64ths of an inch; Indian specs use millimetres.

Grade	Sieve requirement
Plantation AA	≥90% on screen 18 (7.10 mm)
Plantation A	≥90% on screen 17 (6.65 mm); ≤1.5% through 15
Plantation B	≥75% on screen 15 (6.00 mm)
Plantation C	≥75% on screen 14 (5.50 mm); 100% on 13
Plantation PB	Peaberry; no sieve rule; ≤2% flats
Arabica Cherry AB	≥90% on screen 15
Robusta Parchment AB	≥90% on screen 15; the base of Kaapi Royale
Robusta Cherry AB	≥90% on screen 15; India's commonest robusta
Mysore Nuggets EB	Plantation A from Mysore, Biligiris, Shevaroys; ≥90% on screen 19 (7.5 mm)
Robusta Kaapi Royale	≥90% on screen 17; whole, defect-free washed robusta
Monsooned Malabar AA	Monsooned arabica, screen 18+

Source: Coffee Board of India quality specifications ('Coffee Karma'); Josuma Coffee; trade grade guides.

Size is not flavour. A Plantation AA from a low, hot farm can taste plainer than a Plantation B from a cool, high one. But even size matters to a roaster, because small and large beans roast at different speeds. A lot with one screen size roasts evenly.



Figure 5.2. Green coffee as it leaves the curing works: washed and natural arabica, robusta, peaberry and monsooned beans look clearly different.

5.3 Defects: what they are and what they taste like

The Specialty Coffee Association (SCA) grades green coffee by counting defects in a 350 g sample. A **specialty grade** lot may have **no primary defects and at most five full defects** in total; premium grade allows up to eight.

Defect	Beans = 1 defect	Cause	Taste in the cup
Full black	1	Dead or over-fermented cherries	Sour, phenolic, dirty
Full sour	1	Over-fermentation, slow drying	Vinegar, onion, rotten fruit
Dried cherry / pod	1	Poor hulling	Fermented, mouldy
Fungus damage	1	Mould in drying or storage	Musty, earthy, risk of toxins
Severe insect damage	5	Berry borer holes	Dirty, rioy, sour
Partial black / sour	3	As above, milder	Muddy or sour notes
Immature / unripe	5	Unripe cherries picked	Grassy, astringent; pale 'quakers'
Broken, chipped, cut	5	Pulper or huller damage	Uneven roast, scorching
Floater, shell, parchment	5	Light, deformed or unhulled beans	Stale, uneven

Source: SCA Green Coffee Grading Handbook (defect equivalents).

5.4 Cupping: how coffee is scored

Professionals judge coffee by **cupping**, a standardised tasting. The SCA protocol uses 8.25 g of lightly roasted coffee per 150 ml of water (1:18), ground just before tasting, steeped at 93°C for about four minutes. The taster breaks the crust of grounds, smells the released aroma, skims, and then slurps from a spoon as the coffee cools, because flavours change with temperature.

Each coffee is scored on fragrance and aroma, flavour, aftertaste, acidity, body, balance, uniformity, clean cup, sweetness and overall impression, for a total out of 100. **80+ points is 'specialty'**, 85+ excellent and 90+ outstanding. In 2023-24 the SCA introduced the Coffee Value Assessment, which separates descriptive notes from preference scores. In India the Coffee Board runs the **Flavour of India: Fine Cup Award** (since 2002) and Know Your Kaapi competitions.



Quality control checklist for a café

On arrival: check roast date, bag integrity, aroma; keep a sample.

Cup every new lot and every roast batch, ideally blind against last week's coffee.

Track three numbers per blend: roast date to use (target 7-30 days), shot recipe (dose, yield, time) and customer complaints. Drift in any of them is an early warning.

5.5 Inside a cupping session

1. Roast samples lightly (8-12 minutes) the day before; rest 8-24 hours.
2. Weigh 8.25 g per cup (five cups per coffee to check uniformity); grind just before; smell the dry fragrance.
3. Pour 150 ml of water at 93°C; start the timer; smell the wet aroma.
4. At about 4 minutes, break the crust with a spoon, smell, then skim the foam.
5. From about 70°C, slurp from the spoon and score; re-taste as it cools to room temperature.
6. Compare scores and notes; flag any defective cup.

Attribute	What is judged
Fragrance / aroma	Dry and wet smell
Flavour	Overall taste + aroma
Aftertaste	Length and pleasantness
Acidity	Brightness, quality
Body	Texture, weight
Balance	Harmony
Uniformity, clean cup, sweetness	Per cup, 2 points each
Overall	Taster's judgement
Defects	Taints and faults subtracted

Typical Indian coffee	Cup profile	Indicative score	Best use
Commercial robusta cherry	Heavy, woody, bitter, earthy	~70-78 (robusta protocol)	Instant, low-cost blends
Washed Plantation A arabica	Chocolate, nuts, mild citrus, clean	~80-83	House espresso base

Typical Indian coffee	Cup profile	Indicative score	Best use
Washed fine robusta (Kaapi Royale)	Dark chocolate, malt, low bitterness	~80-83 (Fine Robusta scale)	Espresso blends, cold brew
Estate natural S.795 micro-lot	Berries, jaggery, wine, heavy body	~84-86	Pour-over, retail
Anaerobic or carbonic micro-lot	Tropical fruit, spice, boozy	~86-88	Showcase, competition

Source: Score ranges are indicative, based on published cupping scores of Indian lots in roaster and competition listings; robusta is scored on the CQI Fine Robusta protocol.

5.6 Specialty versus commercial coffee

	Specialty	Commercial
Defects (per 350 g)	0 primary, ≤5 full defects	Many tolerated
Cup score	80+ on the SCA scale	Usually 70-79
Traceability	Estate, lot, variety, process known	Region or country at best
Picking and processing	Ripe-only, controlled processing	Strip-picked, variable
Roast	Light to medium, roast date shown	Often dark, 'best before' only
Price to grower	Premium over futures, often fixed by contract	Futures ± differential
Where sold	Specialty cafés, roasters' websites	Supermarkets, instant, vending

5.7 Grading and quality institutions in India

- **Coffee Board of India**: sets grade specifications, runs quality labs and liquoring (cupping), certifies exports, and organises the Flavour of India Fine Cup Award (since 2002) and Know Your Kaapi.
- **Q Graders**: professional cuppers certified by the Coffee Quality Institute; India now has a growing pool, many at exporters and specialty roasters.
- **Competitions**: the India Barista, Brewers and AeroPress championships (run with the Specialty Coffee Association of India) push café standards upward.

5.8 What a green-coffee offer sheet tells you

Field	Example (illustrative)	Why it matters
Lot	Ratnagiri-type estate lot #24-117, 20 bags x 60 kg	Traceability; buy a whole lot for consistency
Origin & altitude	Chikmagalur, Karnataka; 1,300-1,450 m	Terroir and density
Variety	S.795 and Chandragiri	Flavour potential
Process	Washed, 36 h dry fermentation, raised-bed drying 14 days	Clean, bright profile
Grade & screen	Plantation A; 90% on screen 17	Even roasting
Moisture / water activity	10.8% / 0.55 aw	Stability in storage
Density	~720 g/L (free-flow)	Higher = usually higher grown, roasts slower
Defects (350 g)	0 primary, 3 secondary	Specialty grade
Cup score & notes	84.25: milk chocolate, orange, cane sugar	Menu description
Price & terms	₹ per kg ex-curing works; delivery; payment	Budget and cash flow

6 Trade, prices and the money

How Indian coffee is sold, how world prices are set, and who earns what from farm to cup

6.1 How Indian coffee changes hands

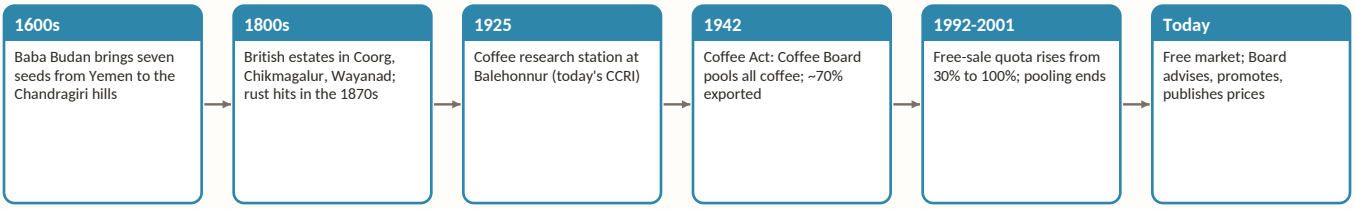


Figure 6.1. From a state monopoly to a free market.

For half a century every grower had to deliver coffee to the Coffee Board's pool, which sold about 70% abroad and auctioned the rest at home; growers received a pooled price months later. Liberalisation in the 1990s let growers sell freely, and pooling ended in 1996. Today the conventional chain runs **grower to village consolidator to trader to huller or curing works to exporter or roaster to retailer**. Larger estates sell directly to exporters, and a growing number sell micro-lots straight to Indian specialty roasters, who publicise the estate name.

6.2 How the world price is set

Physical coffee is priced off two futures markets. **Arabica** trades on ICE's Coffee "C" contract in New York: one contract is 37,500 lb (about 17 tonnes or 283 bags), quoted in US cents per pound, and each 0.05¢ tick is worth $18.75. **Robusta** trades in London in 10-tonne lots priced in US dollars per tonne. A physical lot is priced as **futures plus or minus a 'differential'** that reflects origin, quality and timing. India's coffee is deliverable on the C contract at a discount, and Indian exporters hedge with futures. At home, prices track world prices converted at the rupee rate, which is why a weak rupee (about ₹96 per dollar in October 2026) props up Indian farm-gate prices.

Worked example: from New York to Chikmagalur

ICE C price, 9 Oct 2026: about 285 US¢/lb = US$6.28 per kg = **₹603 per kg** at ₹96/US$.

Indian arabica parchment, 6 Oct 2026: ₹24,100-24,800 per 50 kg = ₹482-496 per kg of parchment. At an 80% outturn this is **₹600-620 per kg of clean coffee**, almost exactly the New York price.

That is why Part 2 assumes ₹600/kg for commodity (Plantation A) arabica and ₹950/kg for estate specialty lots that earn a quality premium.

Coffee prices tripled in six years (ICO composite, monthly average, US cents per lb)

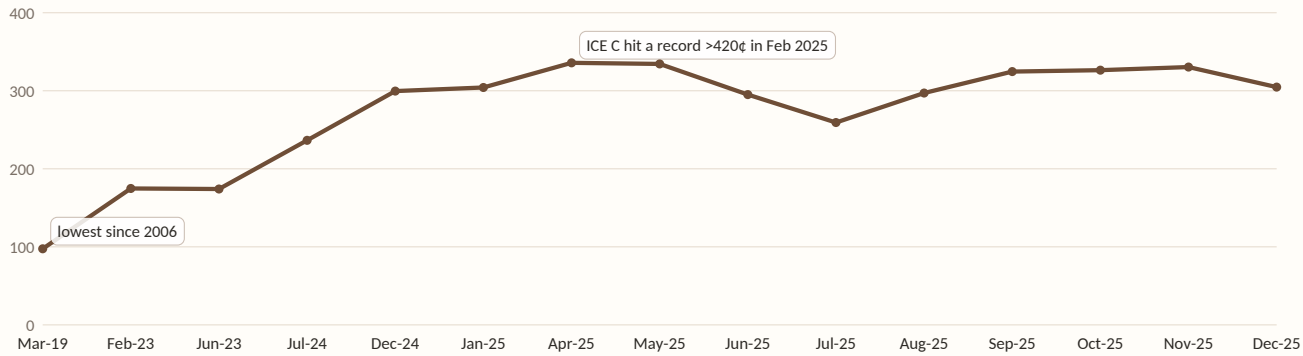


Figure 6.2. Selected monthly averages. Drought in Brazil and Vietnam, four years of falling stocks and a strong dollar pushed prices to records in 2025; a larger 2026 crop has since pulled ICE arabica back to about 285¢ (Oct 2026).

Source: ICO Indicator Prices monthly sheets (2019-2025); ICE; Trading Economics.

6.3 From farm to cup: where the money goes

₹ per kg of roasted specialty coffee

Green cost is per kg roasted (after 16% roast loss)

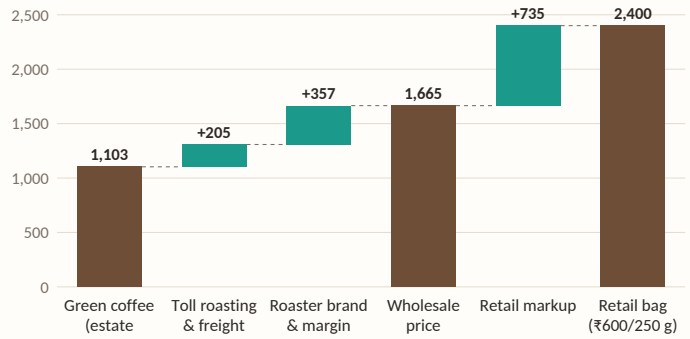
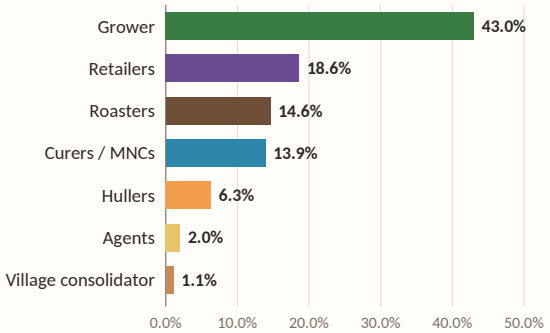


Figure 6.3. Price build-up for a Specialty-mix coffee (Part 2 sourcing assumptions).

The waterfall shows a familiar pattern: the grower's green coffee is less than half of the retail shelf price. In a café the multiplier is far larger. One kilogram of roasted coffee makes about 55 double shots, and at an average ₹220 per drink that is **about ₹12,000 of beverage sales per kg**. The coffee in an 18 g double shot costs ₹21-27; the grower's share of a ₹220 cappuccino is roughly **₹13-20, or 6-9%**. Most of the café's price pays for labour, rent, milk and service, as the Part 2 model shows.

Who keeps the margin? Indian robusta chain (% of total margin)



A study of India's conventional robusta chain found the grower keeping about 43% of the total margin, with roasters and retailers together taking a third. Global studies put farmers' share of the final retail value of roasted coffee at between 3% and 15%; for India one estimate is about 15.7%.

Certification premiums are small (₹50-150 per 50 kg bag, or 2-3%). The bigger gains come from quality and from selling direct: estate-branded Indian coffees now fetch prices close to export levels at home.

Source: AgEcon Search study of India's robusta value chain; Selina Wamucii; GCP, IDH and Solidaridad, 'The Grounds for Sharing' (2024); USDA FAS (May 2026).

6.4 Logistics and trade rules

Topic	What to know
Bags & containers	Green coffee ships in 60 kg jute bags; a 20-ft container holds 275-320 bags (16.5-19.2 t)
Ports	About 61% of India's exports leave through Mangalore; 3-6 weeks to Europe
What India exports	6.22 million bags in 2026/27: 59% green beans, 41% instant coffee; Europe takes 44-45%
Imports	~1.39 million bags, mostly robusta for instant-coffee re-export; 100% duty on green/roasted coffee imported for the home market
Trade deals	India-UK CETA: duty-free roasted and instant coffee; India-EFTA TEPA (Oct 2025): zero duty in Switzerland, Norway, Iceland
EU deforestation rule	Shipments to the EU must prove no deforestation after 31 Dec 2020; the rule has been simplified and postponed
Taxes at home	GST on instant coffee cut from 18% to 5% in 2025, cutting retail prices by 11-12%

Source: USDA FAS (May 2026); PIB (Nov 2025); EU EUDR updates (Dec 2025).

6.5 Price risk for a café, in rupees

A busy specialty café uses about 1,500 kg of roasted coffee a year including retail bags, i.e. roughly **1,786 kg of green coffee** (about 30 bags of 60 kg). At the House blend's ₹1,073/kg green cost:

- a **+30% price shock** costs about ₹5.7 lakh a year, around 2-3 points of EBITDA margin;
- a **-15%** move saves about ₹2.9 lakh.

Small buyers cannot use futures easily (one contract is 17 tonnes), so the practical hedges are **fixed-price contracts** with estates or curing works for 6-12 months of supply, holding 2-3 months of green stock, and menu prices reviewed every year.

Price driver	Why it matters
Brazil weather	Frost (2021) and drought (2023-24) hit a third of world supply
Vietnam robusta	Drought cut output and lifted robusta prices
Stocks	Five years of falling stocks (to 20.1 m bags)
Currencies	A weak real boosts Brazil's sales; a weak rupee lifts Indian prices
Funds	Speculative positions amplify moves
Freight, rules	Shipping costs, EUDR compliance, tariffs

6.6 How an export shipment works

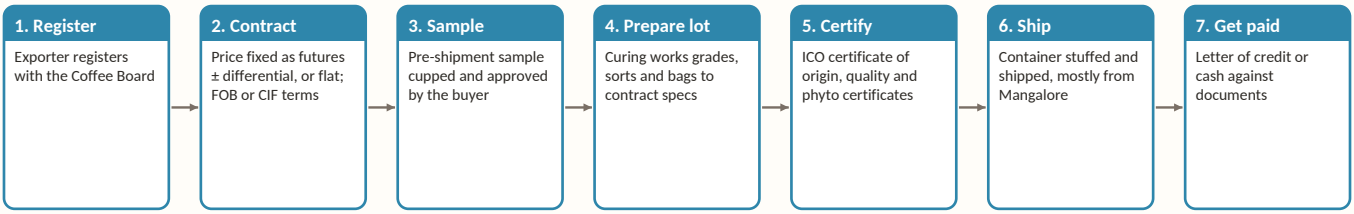


Figure 6.4. The steps of an Indian green-coffee export.

6.7 The cast of characters

Actor	Who	Role
Growers	~2.5 lakh, 98% smallholders; a few large estate companies	Sell cherry/parchment or branded estate lots
Consolidators & traders	Village agents, dealers in coffee towns	Aggregate, finance and transport small lots
Curing works	Dry mills in Chikmagalur, Hassan, Kushalnagar, Mangalore	Hull, grade, sort; often also trade
Exporters	Global trading houses and Indian exporters	Contract, hedge, ship to 128 countries
Instant makers	Tata Consumer, CCL Products, Nestlé India, HUL	41% of exports; most domestic consumption
Roasters	National brands (Coffee Day, Bru, Narasu's…) and 300+ specialty roasters	Blend, roast, brand
Cafés and retailers	Chains, independents, supermarkets, online, quick commerce	Brew, serve, sell to the end customer
Institutions	Coffee Board, CCRI, planters' associations, SCAI	Research, standards, promotion

6.8 A century of boom and bust

When	What happened	Price outcome
1975-77	Brazil's 'black frost' destroys trees	New York prices top $3/lb in 1977
1962-89	International Coffee Agreements set export quotas	Prices held in a band
1989	Quota system collapses	Prices roughly halve within months
2001-02	Vietnam's robusta expansion + Brazil oversupply	'Coffee crisis': ~45-50¢/lb, a century low in real terms
2011	Tight supply, strong demand	C contract peaks near $3/lb
2019	Big Brazilian crops, weak real	ICO composite falls below $1/lb (lowest since 2006)
2021-25	Brazil frost and drought, Vietnam drought, falling stocks	Record >$4.20/lb (Feb 2025)
2026	Record Brazilian crop expected	C around $2.85/lb (Oct 2026)

The pattern repeats: a weather shock in Brazil or Vietnam lifts prices; high prices encourage planting; three to four years later new trees bear fruit and prices fall. Growers live with this volatility; a café should simply expect it.

6.9 Direct trade and specialty contracts

Specialty buyers increasingly agree prices directly with estates instead of buying anonymous lots through traders. A typical Indian direct-trade arrangement for a café or roaster looks like this:

Term	Typical content
Pricing	Fixed rupee price per kg of green (e.g. ₹900-1,100 for an 84+ point lot), agreed before harvest; sometimes a floor plus a quality premium
Quality clause	Pre-shipment sample must match the approved cup profile and score; moisture 10-11.5%; defect limits
Volume and timing	e.g. 20 bags (1.2 t) from the 2026-27 crop, delivered in two lots to the toll roaster
Payment	Advance at contract (helps the grower finance harvest), balance on delivery
Story and rights	Estate name, photos and story may be used on menus and bags
Relationship	Multi-year intent; visits at harvest; feedback on cupping

For a single café the volumes are small (about 30 bags a year), so many cafés buy through their toll roaster's estate relationships at first and contract directly once volumes grow.

7 Roasting

The chemistry in the drum, roast levels, machines, profiles, freshness and what roasting costs

7.1 What happens inside the roaster

Roasting heats green beans to 180-240°C for roughly 8-15 minutes in a drum roaster. In that time the bean loses 12-20% of its weight (mostly water), nearly doubles in volume, turns from green to brown and develops the aromas we call coffee. Roasters describe four stages:

- **Drying (0-5 min).** Water evaporates; the bean turns yellow and smells of hay, then bread, at around 150-160°C.
- **Browning (Maillard) phase.** From about 150-160°C, sugars react with amino acids (the Maillard reaction) to form brown melanoidins and hundreds of aroma compounds; above about 170°C sucrose caramelises. Chlorogenic acids break down into bitter compounds; trigonelline yields niacin and roasty aromas.
- **First crack (~196-205°C).** Steam and CO2 pressure inside the cells fracture the bean with an audible pop, like popcorn. Light roasts are dropped shortly after it.
- **Development and second crack.** The time after first crack (typically 15-25% of the roast, the 'development time ratio') sets sweetness and body. Near 224-230°C a second, quieter crack begins, oils reach the surface and roasty, smoky, bitter flavours take over: dark roast territory.

A typical 10.5-minute drum roast

Bean temperature (brown), rate of rise (teal, °C per minute) and the stages

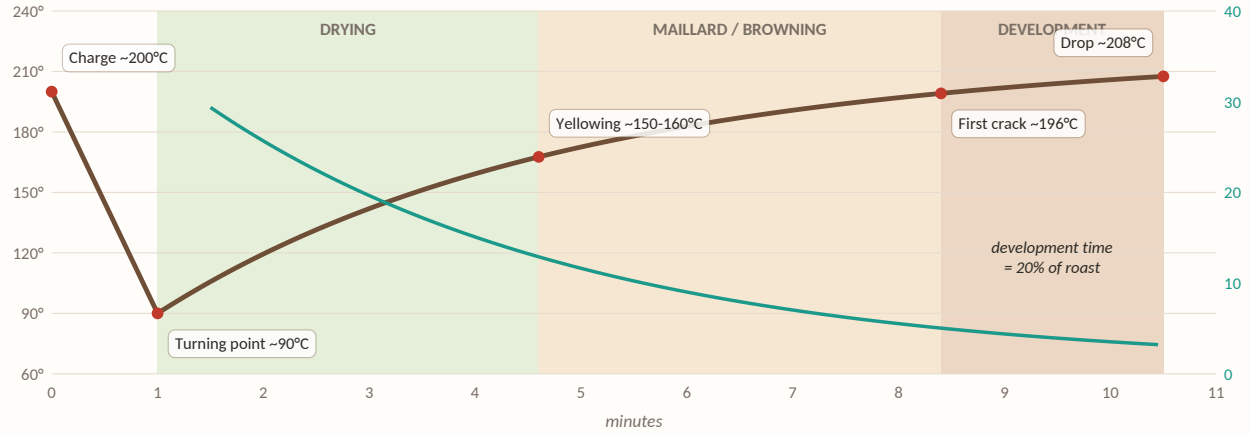
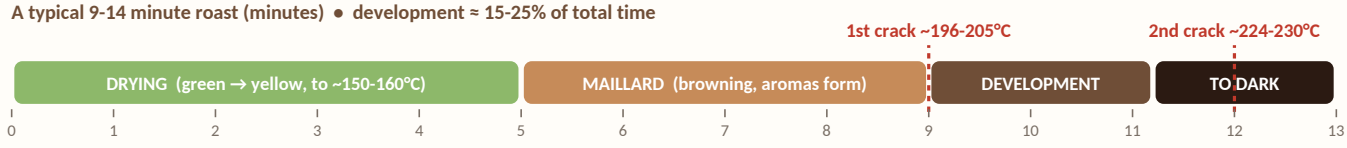
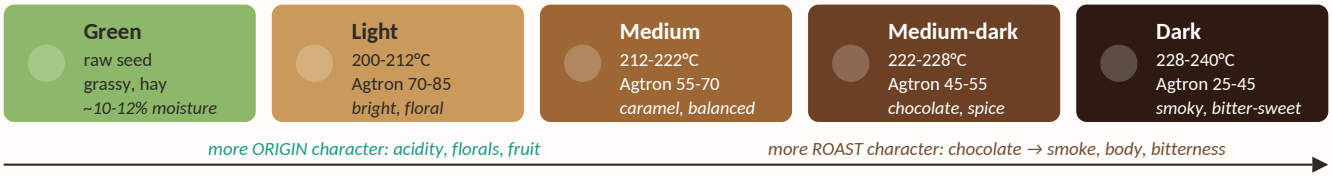


Figure 7.1. Roasters steer by a steadily falling rate of rise. A rate that spikes scorches beans; one that crashes 'bakes' them flat.

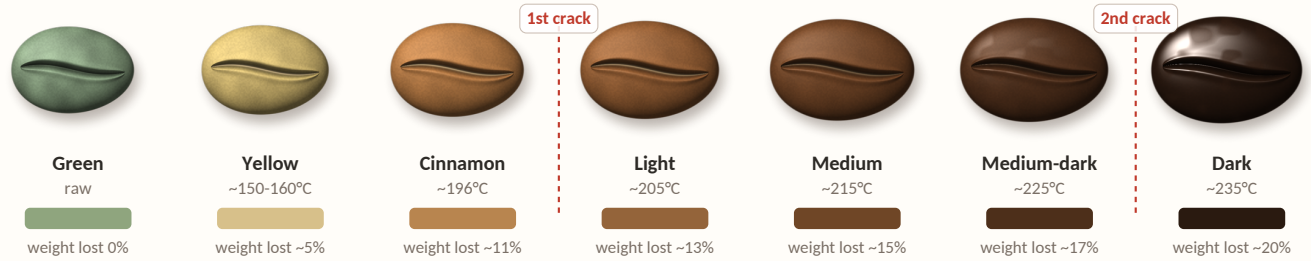
7.2 Roast levels and flavour

A typical 9-14 minute roast (minutes) • development ≈ 15-25% of total time





Level	Drop temperature	Taste	Best for
Light (cinnamon to city)	Shortly after first crack, ~200-210°C	Bright acidity, florals, fruit, tea-like; origin shows clearly	Pour-over, AeroPress, cupping
Medium (city+ to full city)	~210-220°C	Balanced: caramel, nuts, chocolate, gentle acidity	Filter and espresso; crowd-pleasing
Medium-dark (Vienna)	Start of second crack, ~224-230°C	Bittersweet chocolate, spice, heavier body, oil spots	Espresso with milk, filter kaapi
Dark (French, Italian)	Into second crack, 235°C+	Smoky, roasty, bitter, low acidity; origin hidden	Traditional espresso, kaapi, instant



Swatches = colour of the ground coffee. Beans grow ~50-90% in volume but lose weight as water and gases leave.

Figure 7.2. The same bean from green to dark roast.

7.3 Roasting machines

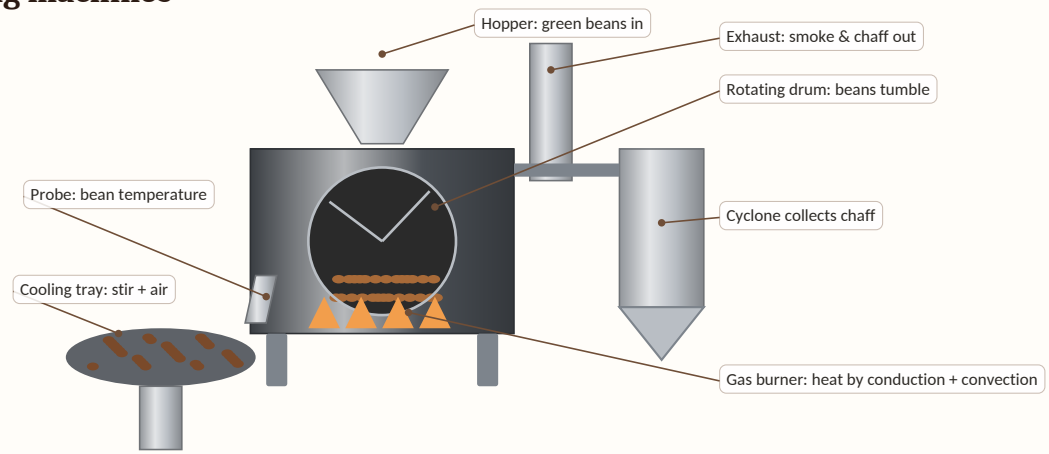


Figure 7.3. A drum roaster and its main parts (schematic).

Drum roasters (above) tumble beans in a rotating drum heated by gas or electricity; heat arrives by conduction from the hot drum and convection from hot air drawn through it. They range from 500 g shop roasters to 120 kg production machines and dominate specialty roasting. India has local makers such as Aaroasta in Rajkot (0.5-50 kg); imported brands include Probat, Giesen, Diedrich and Loring.

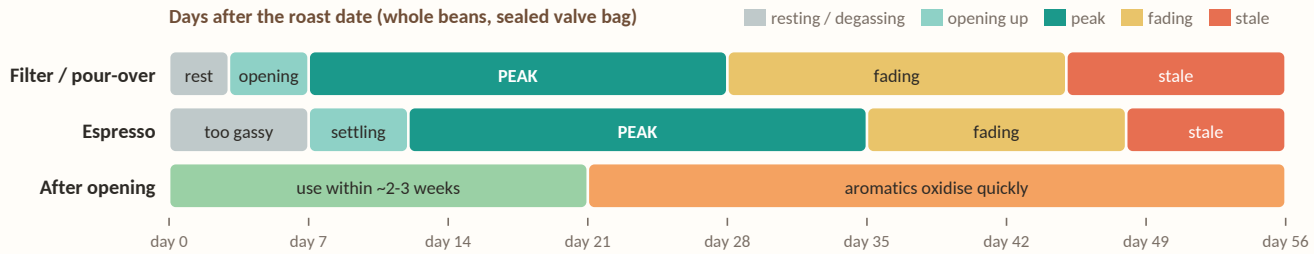
Fluid-bed (hot-air) roasters float the beans on a stream of hot air: fast, even and bright-tasting, popular for sample roasting. Industrial plants use continuous roasters for instant and supermarket coffee. In a city, an **afterburner** or catalytic converter is needed to burn off smoke and smell.

7.4 Profiles, blends and consistency

A **roast profile** is the recipe: charge temperature, when gas and airflow change, and when to drop. Software such as Artisan or Cropster logs bean temperature every second so a profile can be repeated batch after batch; colour meters (Agtron scale) and cupping check the result. Espresso roasts usually develop longer than filter roasts so they extract sweetly at high pressure. Blends may be roasted together (*pre-roast*) or separately and then mixed (*post-roast*), which lets each component reach its ideal level. Indian and Italian espresso blends often contain 10-30% robusta for crema, body and lower cost.

7.5 Resting, packing and freshness

Freshly roasted coffee is full of carbon dioxide and must rest before brewing: about 2-7 days for filter and 7-14 days for espresso, or the gas makes extraction uneven and flavours sharp. After that, aroma fades steadily as the coffee oxidises. Good roasters pack into **one-way valve bags** (the gas gets out, air does not), sometimes flushed with nitrogen. Whole beans are at their best roughly 7-30 days after roasting and noticeably flat after 4-6 weeks once the bag is opened. Ground coffee loses most of its aroma within an hour, which is why cafés grind to order.



7.6 What roasting costs

The economics hinge on the **roast loss** (14-20%; Part 2 uses 16%) and on how full the roaster is kept. A 3 kg drum roaster does three or four batches an hour, i.e. roughly 8-12 kg of roasted coffee per hour. A busy café uses about 1,100-1,600 kg a year including retail bags, which is only 100-200 hours of roasting. That is why most cafés buy roasted coffee or pay a roaster to **toll-roast** their own green beans; owning a roaster pays only when there is wholesale volume to fill it.

Route	Cost per kg (Specialty / House mix)	Capital tied up	Trade-off
Buy roasted	Roaster's price, ₹1,650-1,850/kg for a specialty blend	None	Least control; roaster's brand on the bag unless private label
Toll roasting	Green cost ÷ 0.84 + ₹180 fee + ₹25 freight ≈ ₹1,300-1,500/kg	Green stock (40 days)	Your own green coffee, profile and label; minimum batch sizes
In-house roastery	Green ÷ 0.84 + ₹60 variable ≈ ₹1,160-1,340/kg, plus roaster salary, AMC, rent	₹13 lakh roaster, afterburner, QC kit + fit-out	Full control, theatre and a wholesale business; needs roughly 450-500 kg/month of wholesale to beat toll roasting

Source: Part 2 model (Sourcing and Capex sheets); Aaroasta product range; roaster price lists.

7.7 Roast defects and how to spot them

Defect	Cause	Taste
Underdeveloped	Too little time after first crack; centre still pale	Grassy, peanut, sour, astringent
Baked	Rate of rise stalled or crashed; roast dragged on	Flat, bready, papery, dull
Scorched	Drum too hot at charge; burn marks on the flat side	Smoky, ashy, bitter
Tipped	Too much heat early; burnt tips of the bean	Burnt, sharp
Overdeveloped / dark	Too long or too hot after first crack	Bitter, carbon, origin lost
Quakers	Unripe beans that will not brown	Peanut, cereal; pick them out

7.8 Building an espresso blend

Blends exist to produce a consistent, balanced espresso all year as single lots come and go. Each component has a job. A good Indian espresso blend can be built entirely from Indian coffees:

Role	Component	Contribution	Indian example
Base (50-70%)	Washed Plantation arabica, medium roast	Chocolate, nuts, sweetness, balance	S.795 or Chandragiri from Chikmagalur
Fruit / acidity (10-20%)	Natural or honey arabica, or high-grown washed	Lift, berry, citrus	Natural micro-lot, Pulneys arabica
Body / crema (10-30%)	Washed fine robusta	Crema, body, cocoa; cuts through milk	Robusta Kaapi Royale
Spice / depth (0-10%)	Monsooned Malabar	Syrupy body, spice, low acid	Monsooned Malabar AA

The Part 2 'House blend' (15% washed robusta, 70% estate arabica, 15% micro-lot) follows this logic: chocolatey and milk-friendly, with a micro-lot share that justifies a specialty price.

7.9 Decaffeination

Method	How	Notes
Swiss Water / water process	Green beans soaked in a caffeine-free extract of green coffee; caffeine filtered out with carbon	Chemical-free; premium
Supercritical CO2	CO2 under high pressure dissolves caffeine selectively	Good flavour retention; industrial
Ethyl acetate ('sugarcane')	Solvent derived from fermentation (often of sugarcane)	Common for Colombian decaf; fruity
Methylene chloride	Direct or indirect solvent method	Cheapest; residues tightly limited

Decaf is not caffeine-free: EU rules require roasted decaf to contain no more than 0.1% caffeine (about 97% removed). Indian demand is small, so most cafés buy imported decaf beans; it is still worth offering for evening customers.

7.10 Packaging and shelf life

Format	Freshness	Used for
Whole beans, one-way valve bag	Best 7-30 days after roast; 2-3 months sealed	Specialty retail, cafés
Ground, valve bag	Best within 1-2 weeks of opening	Convenience; filter kaapi powder
Nitrogen-flushed tins / bags	6-12 months sealed	Supermarkets
Drip bags and steep bags	Weeks to months (single-serve, sealed)	Offices, travel, gifting
Capsules / pods	Months (oxygen barrier)	Home and office machines
Cold brew bottles / cans	Days (fresh) to months (retorted)	Retail fridges, quick commerce
Instant jars and sachets	12-24 months	Mass market

7.11 Starting-point roast profiles

Style	Total time	First crack at	Development	Drop temp.	Weight loss	Aim
Filter (light)	9-10 min	7:30-8:00	1:15-1:30 (15-18%)	200-205°C	12-13%	Origin character, acidity, florals
Espresso (medium)	10.5-12 min	8:15-8:45	2:00-2:30 (18-22%)	210-218°C	14-16%	Sweetness, chocolate, body for milk
Kaapi / dark	12-14 min	8:30-9:00	into second crack	225-232°C	17-19%	Roasty, bittersweet, low acidity; stands up to chicory and milk

These are typical starting points for a 3-5 kg drum roaster; every machine, batch size and bean behaves differently. A toll roaster will run sample roasts of your green coffee, cup them with you and log the agreed profile, so every later batch can be repeated.

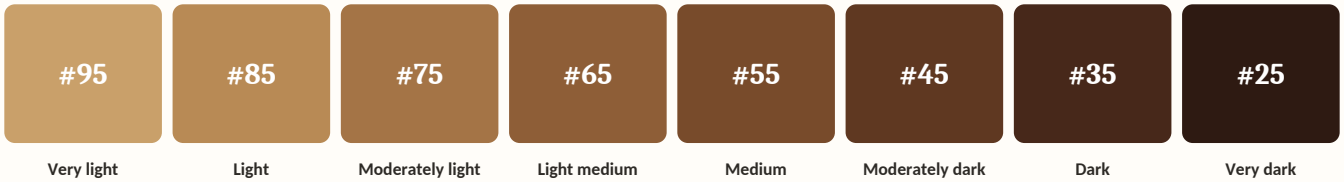
7.12 Setting up a small roastery

Item	What it takes
Space	150-500 sqft in light-industrial or commercial premises; floor loading, drainage
Roaster	3-5 kg drum roaster (Indian makers in Rajkot and elsewhere; imported options); the Part 2 model budgets ₹9 lakh for a 3 kg machine
Smoke control	Afterburner or catalytic unit, chimney and ducting (₹2.5 lakh in the model); essential in a city
Utilities	LPG/PNG line or electric supply; three-phase power; ventilation
Approvals	FSSAI manufacturing licence; pollution-control consent (GPCB) where required; fire NOC; trade licence
Quality kit	Sample roaster, moisture meter, scales, sealer, cupping set (₹1.5 lakh)
People	A trained roaster who also manages green buying, QC and wholesale accounts
Throughput	~8-12 kg roasted per hour; one shift a week covers a single café's needs

The numbers explain the Part 2 verdict: the roastery's fixed costs (salary, AMC, extra space, depreciation) are about ₹7-9 lakh a year, while roasting a single café's 1,500 kg in house rather than by toll saves only about ₹2 lakh. Wholesale volume is what fills the machine and the profit gap.

7.13 Measuring roast colour

Roast colour on the Agtron / SCA tile scale (higher = lighter)



Filter roasts typically sit around #65-75, espresso #50-60, traditional kaapi #35-45.

Figure 7.4. Roasters measure colour with a near-infrared meter (Agtron or similar) so a 'medium' roast means the same thing every batch.

8 Grinding

Why particle size matters, burrs versus blades, the right grind for each method, and how to dial in

8.1 Why grinding matters

Water can only dissolve what it can reach. A whole bean exposes very little surface, so grinding is what makes brewing possible: cut a 6 mm bean into 0.3 mm espresso particles and the surface area per gram rises about twentyfold. **Finer grinds extract faster and slow the water down**; coarser grinds extract slowly and let water run through. So grind size must match contact time: 25-30 seconds of espresso needs fine sand; four minutes in a French press needs coarse breadcrumbs.

Every grinder makes a **spread** of sizes. Tiny **fines** (under ~100 µm) over-extract and add bitterness and mud, and they can clog paper filters; big **boulders** under-extract and add sourness. A good grinder keeps the spread narrow, so the whole dose extracts at the same pace.

Particle-size distribution, same average grind (illustrative)

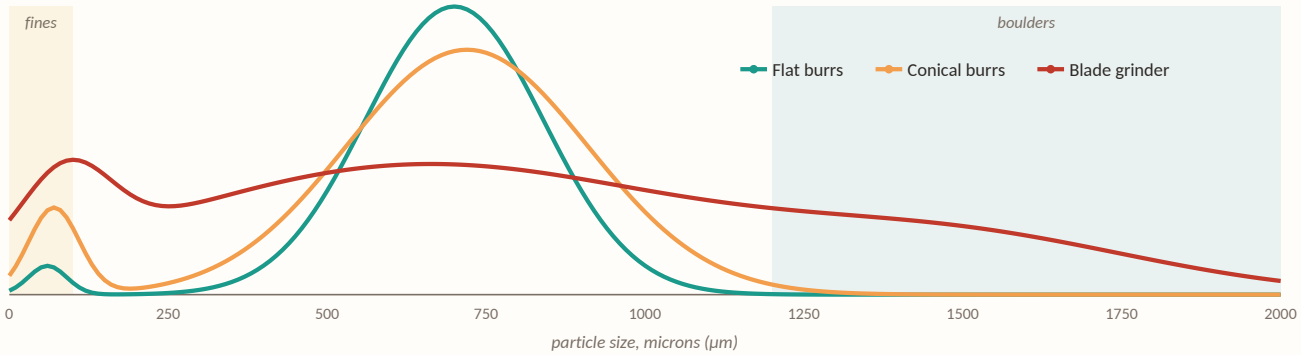


Figure 8.1. Flat burrs give a tight, single peak; conical burrs a second 'fines' peak (more body in espresso); blade grinders a smear of dust and chunks.

8.2 Types of grinder

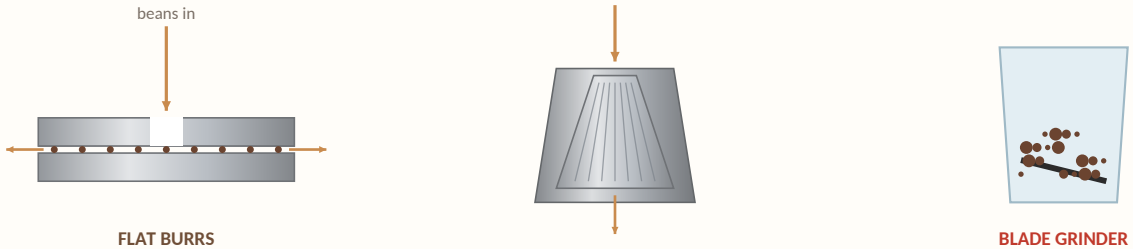
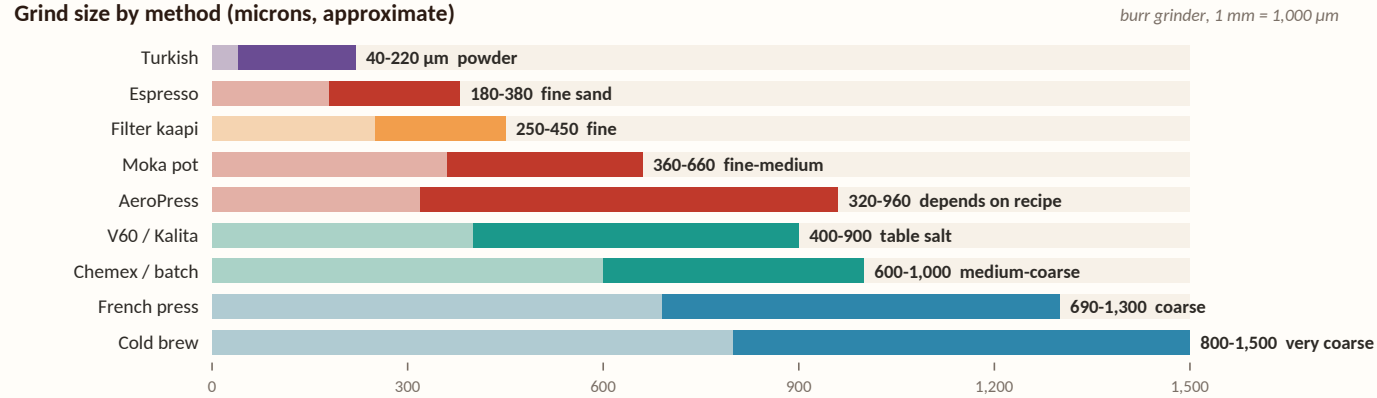


Figure 8.2. Flat burrs, conical burrs and a blade grinder.

Type	How it works	Typical cost (India)	Character
Blade	A spinning blade smashes beans; size depends on time	Cheap (₹1,000-3,000)	Very uneven; avoid for anything but spices
Conical burr	A cone turns inside a ring; grounds fall straight down	Hand grinders ₹3,000-20,000; café ₹1-2 lakh	Low heat and retention; bimodal grind adds body
Flat burr	Two parallel rings; beans travel outwards between them	Café on-demand grinders ₹1.2-2.5 lakh+	Very uniform; clarity; can retain coffee
Roller	Pairs of rollers crack beans in stages	Industrial	Most uniform; used for packaged ground coffee

Other details matter in a café: **burr size** (64-98 mm flat burrs grind fast and cool), motor speed and heat, **retention** (grounds left inside that go stale and mix into the next dose), stepless versus stepped adjustment, and **burr life**. Burrs dull after a few hundred kilograms to about a tonne, depending on their size, after which the grind gets dustier and shots run faster.

8.3 The right grind for each method



Source: Honest Coffee Guide, Bean Box and The Coffee Locator grind charts (ranges vary by grinder).

8.4 Dialling in: let taste steer the grinder

'Dialling in' means adjusting the grind (and then dose and time) until the coffee tastes right. The rules are simple:

- **Sour, salty, thin, quick** = under-extracted: grind **finer**.
- **Bitter, dry, hollow, slow** = over-extracted: grind **coarser**.
- Change one thing at a time, purge the grinder after each change, and taste.
- Re-check every morning: humidity, bean age and burr temperature all shift the grind.

Espresso: 18 g in, 36 g out	Taste	Action
Runs in 18 s	Sour, thin	Finer
Runs in 25-30 s	Sweet, balanced	Keep
Runs in 40 s	Bitter, harsh	Coarser
Channels or spurts	Sour and bitter	Fix distribution and tamping

8.5 A café grinding set-up

- Two on-demand espresso grinders: one for the house blend, one for a guest single origin or decaf. Grinding by timer keeps doses consistent.
- One bulk grinder for batch brew, pour-over and grinding retail bags for customers.
- Daily: brush out the chute, purge stale grounds. Weekly: grinder-cleaning tablets. Keep only a day's beans in the hopper; check burr alignment and wear every few months.

8.6 Buying a café grinder: what to look for

Feature	Good spec	Why
Burr size and type	64-98 mm flat or 60-80 mm conical	Speed, heat, uniformity
Dosing	By time (on-demand) or by weight (grind-by-weight)	Consistency at peak hours
Retention	Grams left inside after grinding	Stale grounds in the next shot
Adjustment	Stepless micrometric adjustment	Fine control when dialling in
Heat and speed	Cooling fans, slower RPM	Stable grind through rush hours
Service	Local spares, burr availability, AMC	Uptime in India

Practical tips: a light spritz of water on beans before grinding (the 'Ross droplet technique') reduces static and clumping; single-dose grinders suit pour-over bars; and for filter kaapi grind fine but not powdery, adding chicory after grinding if used.

8.7 Grinding at scale: packaged ground coffee and kaapi powder

Supermarket ground coffee and South Indian filter powder are ground on industrial roller or large disc grinders that produce very uniform particles at high speed, often followed by 'normalising' to set density. Because ground coffee stales fast, packers flush bags or tins with nitrogen and seal immediately; filter-coffee shops in South India traditionally roast and grind daily in small batches, which is why freshly ground kaapi powder from a neighbourhood shop tastes so different. For a café selling retail coffee, the best practice is to sell whole beans and offer to grind on request for the customer's brewer.

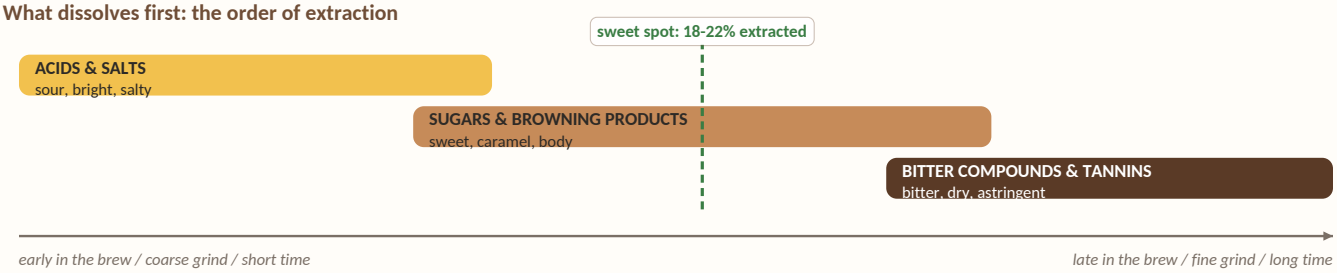
9 The science of brewing

Extraction and strength, the variables you control, water, why so many methods exist, and milk

9.1 Extraction: dissolving the right amount

About 28-30% of a roasted bean's weight can dissolve in water, but a delicious cup takes only **18-22%** of it. Compounds dissolve in a predictable order: acids and salts first, then sugars and browning products, and bitter compounds last. Stop too early and the cup is sour and thin (**under-extracted**); go too far and it turns bitter and drying (**over-extracted**).

What dissolves first: the order of extraction



Two numbers describe any cup. **Strength** is the concentration of dissolved coffee, measured as total dissolved solids (TDS) with a refractometer: about 1.15-1.35% for filter coffee and 8-12% for espresso. **Extraction yield** is the share of the dry coffee that ended up in the cup. Example: 30 g of coffee and 500 g of water give about 450 g of brew; at 1.30% TDS it holds 5.85 g of coffee solids, an extraction of 19.5%. The SCA's 'Golden Cup' standard combines the two: 18-22% extraction at 1.15-1.35% TDS, with roughly 55 g of coffee per litre of water (1:18).

The brewing control chart: strength vs extraction

Filter coffee. Diagonal lines = brew ratio (grams of water per gram of coffee)

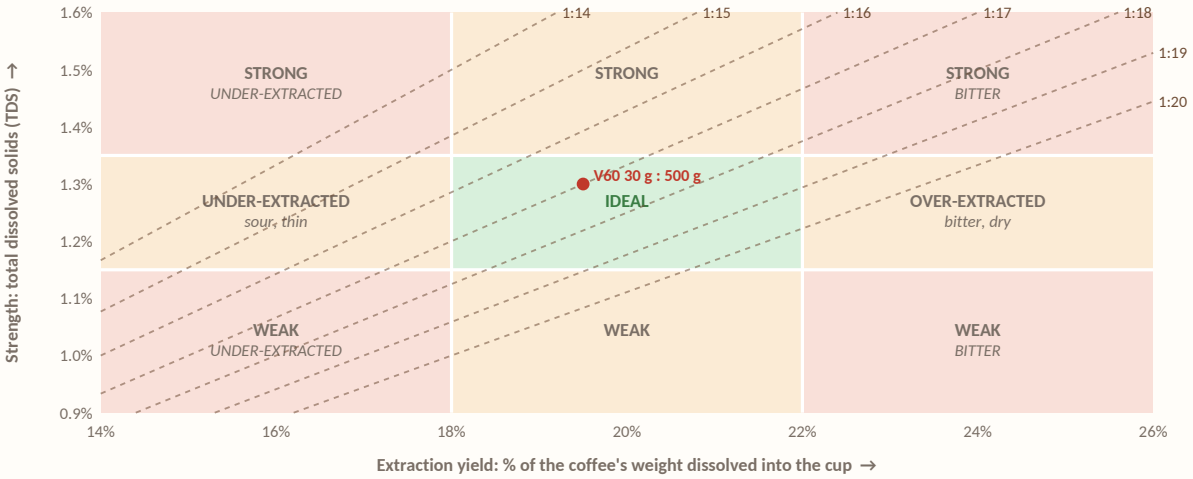


Figure 9.1. Strength is mostly set by the ratio; extraction by grind, time and temperature. The goal is the green box.

9.2 The variables you control

Variable	Mostly affects	Typical setting	Note
Ratio (coffee : water)	Strength	1:15-1:17 filter; 1:2 espresso; 1:8 cold-brew concentrate	Weigh, don't scoop
Grind size	Extraction speed, flow	See Figure 8.3	The main dial-in lever
Contact time	Extraction	25-30 s espresso; 3-4 min pour-over; 12-24 h cold brew	Linked to grind and method
Water temperature	Extraction speed	90-96°C hot brews; cooler for dark roasts	Hotter extracts more, faster
Agitation	Evenness, speed	Stirring, swirling, pouring height	More agitation = more extraction
Filter medium	Body versus clarity	Paper (clean), metal (oily, full), cloth (in between)	Paper removes oils and fines
Freshness	Aroma, evenness	7-30 days from roast; grind just before	Stale coffee tastes flat whatever you do

9.3 Water: 98% of the cup

Parameter (SCA)	Target	Acceptable
Total dissolved solids	150 mg/L	75-250
Total hardness (as CaCO3)	68 mg/L	50-175
Alkalinity (as CaCO3)	40 mg/L	40-75
pH	7.0	6.5-7.5
Sodium	10 mg/L	≤30
Chlorine	0	0

Minerals do real work: calcium and magnesium help pull flavour out of the coffee, while bicarbonate (alkalinity) buffers acidity. Water that is too soft tastes flat and sour; too hard tastes chalky and furs up boilers; high alkalinity mutes acidity. Chlorine ruins aroma.

Ahmedabad's municipal and bore-well water varies a lot in hardness, so cafés use reverse osmosis followed by remineralisation to hit the target and protect the espresso machine. The Part 2 model budgets ₹1 lakh for it.

Source: SCA water quality standard as reproduced by industry guides (Pact Coffee, Bartovation).

9.4 Why are there so many brewing methods?

Every method is a different answer to four questions: **how water meets the coffee** (soaking or flowing through), **what filters it** (paper, metal, cloth or nothing), **whether pressure is used**, and **how long and how hot**. These choices trade clarity against body, strength against volume, and control against convenience. Culture and history did the rest: Italy's espresso bars, Turkey's coffee houses, Tamil Nadu's tumblers, Vietnam's condensed-milk phin and Japan's pour-over craft.

Family	Principle	Examples	Typical result
Immersion (steeping)	Coffee soaks in water, then is separated	French press, cold brew, Clever, siphon, cupping	Even extraction, full body, forgiving
Percolation (gravity)	Water flows through a bed of coffee	V60, Kalita, Chemex, batch brew, filter kaapi, phin	Clarity and sweetness; technique matters
Pressure	Water forced through compacted coffee	Espresso (9 bar), moka pot (1-2 bar), AeroPress (low)	Concentrated, syrupy, crema
Boiling / decoction	Coffee simmered with water	Turkish cezve, traditional boiled	Thick, unfiltered, intense

Why there are so many methods: each trades clarity against body

Red = pressure, blue = immersion, teal = paper percolation, orange = metal-filter percolation, purple = boiled

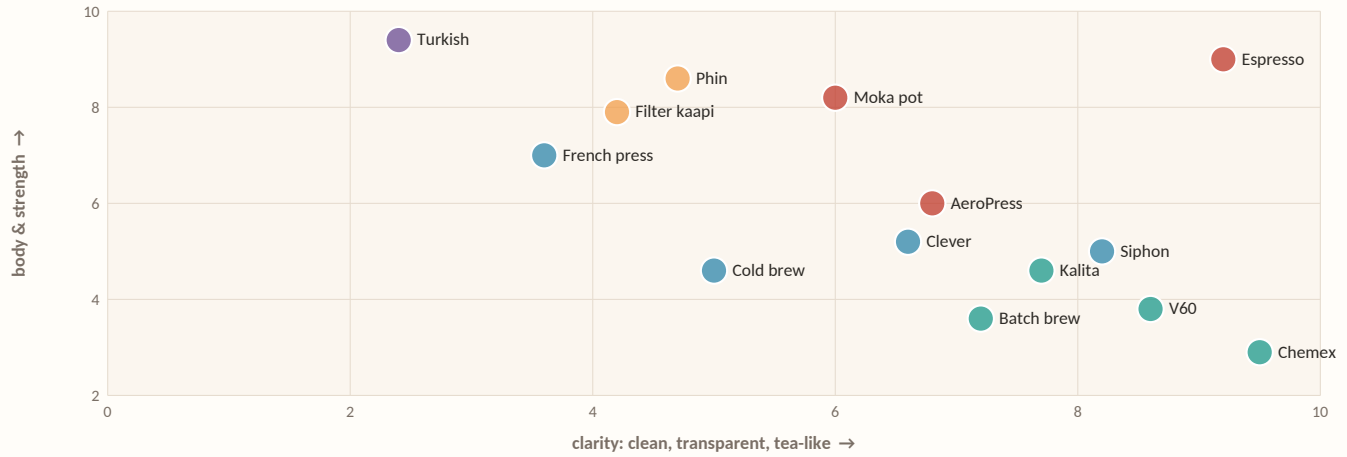


Figure 9.2. Where the main methods sit on the clarity-body map.

9.5 Milk: the other half of most Indian coffee

Most coffee sold in Indian cafés contains milk. Steaming injects air and heat: whey and casein proteins wrap tiny air bubbles to make foam, while fat adds creaminess but weakens foam. Full-cream milk (3.5% fat or more) makes the silkiest microfoam for latte art. Sweetness peaks around 55-65°C; above about 70°C milk scalds and tastes eggy. Buffalo milk (6-7% fat) is rich but foams heavily, which is why many Indian cafés use full-cream cow's milk. Among plant milks, 'barista' oat milks foam best, while soy can curdle in acidic coffee.

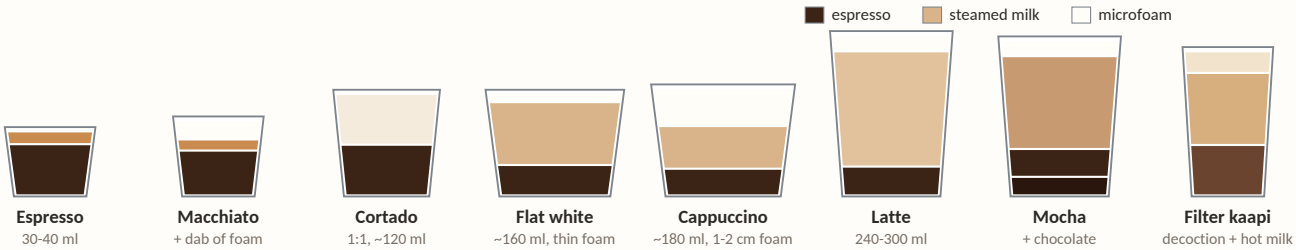


Figure 9.3. The espresso-and-milk family, shown as cross-sections.

9.6 Espresso physics

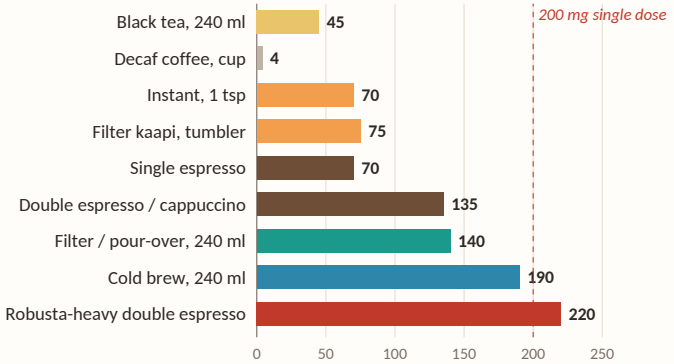
Espresso is percolation under pressure. The pump pushes water at about 9 bar into a puck of fine coffee that resists the flow; a well-made puck lets roughly 1.5-2 g of espresso through per second. Gases and oils emulsify into crema. Because contact time is short, espresso needs fine grinding, fresh coffee and very stable temperature.

Channelling is the enemy: if the puck has a weak spot (uneven distribution, a crack, a tilted tamp), water rushes through it, over-extracting that path and under-extracting the rest; the shot tastes sour and bitter at once. **Pre-infusion** (wetting the puck at low pressure for a few seconds) and **pressure profiling** (ramping pressure up and down) reduce channelling and let baristas shape flavour.

Espresso target	Typical range
Brew ratio (dose : yield)	1:1.5 to 1:2.5
Strength (TDS)	8-12%
Extraction yield	18-22%
Pressure	~9 bar at the puck
Temperature	90-96°C
Flow	~1.5-2 g per second
Time	25-30 s for 1:2

9.7 How much caffeine is in each drink?

Typical caffeine per serving (mg, midpoint of usual ranges)



Source: EFSA Scientific Opinion on caffeine safety (2015); typical ranges from USDA FoodData Central and café menus.

Caffeine varies with species (robusta has about twice as much), dose, and method. The European Food Safety Authority considers up to **400 mg a day** safe for healthy adults (about three cappuccinos or two large filter coffees), single doses up to 200 mg, and up to 200 mg a day in pregnancy.

For a café menu, label strong drinks (cold brew, robusta-heavy shots) and offer decaf and half-caf; evening trade in Ahmedabad is large.

9.8 What is in your cup: coffee chemistry

Component	Green bean	Roasted (medium)	Role in the cup
Polysaccharides (fibre)	~50%	~33-38%	Body; mostly insoluble
Lipids (oils)	15-17% (arabica)	~17%	Aroma carrier, crema, mouthfeel
Proteins & amino acids	10-13%	~7-10%	Feed the Maillard reaction
Sucrose	6-9% (arabica)	<0.5%	Caramelises into sweetness and colour
Chlorogenic acids	5.5-8%	1-3%	Bitterness, astringency, antioxidants
Caffeine	1.2-1.5% (robusta 2.2-2.7%)	~unchanged	Bitterness, stimulation
Trigonelline	~1%	~0.2-0.7%	Breaks down into niacin and aromas
Organic acids	Citric, malic ~1-2%	Citric, malic, quinic, acetic	Brightness, sourness, staleness
Melanoidins	0%	~20-25%	Brown colour, body, roasty bitterness
Minerals	~4%	~4.5%	Potassium-rich

Source: Approximate composition ranges from standard coffee-chemistry references (Clarke & Macrae; Illy & Viani) and the PMC10138461 review.

Roasting turns a dense, starchy seed into a brittle, porous, aromatic one. Sugars almost disappear, chlorogenic acids halve or more, and new brown compounds (melanoidins) and more than 800 volatile aroma compounds appear.

A handful of aroma compounds do most of the work: **2-furfurylthiol** (the 'roasted coffee' smell), **guaiacols** (spicy, smoky), **pyrazines** (nutty, roasty), **furaneol** (caramel), **β-damascenone** (fruity, honey), **2,3-butanedione** (buttery) and **3-methylbutanal** (malty). Many are volatile and fragile, which is why freshly ground coffee smells so much better.

9.9 Coffee and health

- **Moderate drinking is linked with benefits.** Large umbrella reviews of observational studies associate 3-4 cups a day with lower risks of type 2 diabetes, liver disease and Parkinson's disease and lower all-cause mortality (association, not proof).
- **Unfiltered coffee raises cholesterol.** Diterpenes (cafestol, kahweol) pass into French press, Turkish, boiled and, to a lesser degree, espresso and moka coffee; paper filters remove most of them.
- **Caffeine limits**: up to about 400 mg a day for healthy adults and 200 mg in pregnancy (EFSA). Caffeine's half-life is about 5 hours, so an evening cold brew affects sleep.
- **Acrylamide** forms early in roasting; levels in coffee are low and regulated by EU benchmark levels.
- **What goes in matters**: an Indian-style cold coffee with full-cream milk, sugar and ice cream can carry around 300 kcal; black coffee has almost none. A café can offer lighter options (less sugar, oat milk, small sizes).

9.10 Brew-ratio quick reference

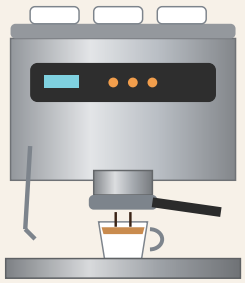
Water	1:15 (strong)	1:16	1:17	1:18 (light)
150 ml water	10.0 g	9.4 g	8.8 g	8.3 g
250 ml water	16.7 g	15.6 g	14.7 g	13.9 g
350 ml water	23.3 g	21.9 g	20.6 g	19.4 g
500 ml water	33.3 g	31.2 g	29.4 g	27.8 g

Dose = water ÷ ratio. Start at 1:16 for pour-over and French press; go stronger (1:15) for darker roasts or with milk, lighter (1:17-1:18) for delicate light roasts. A gram scale costing ₹500-800 is the single most useful brewing tool after a burr grinder.

10 Brewing methods, step by step

What each method is, where it came from, and exactly how to make it

Recipes below are starting points that professionals use. Weigh coffee and water with a scale, use filtered water just off the boil (unless stated) and adjust the grind by taste.



Dose	18-20 g (double basket)
Yield	36-40 g (ratio 1:2)
Time	25-30 s
Grind	Fine, like caster sugar
Water	92-94°C, 9 bar
Variants	Ristretto 1:1-1:1.5; lungo 1:3; americano = shot + hot water

Espresso *pressure, 25-30 seconds, the base of most café drinks*

Espresso forces hot water at about 9 bar through a compact puck of finely ground coffee, producing a small, concentrated, syrupy shot topped with crema (an emulsion of oils and CO2). Angelo Moriondo patented a steam-pressure machine in Turin in 1884; Bezzera and Pavoni added the portafilter in the early 1900s; Achille Gaggia's lever machine (1947-48) created crema as we know it; the pump-driven Faema E61 (1961) set the modern template.

A café machine has a boiler (single, heat-exchanger or one per group), group heads, portafilters with baskets, a pump and steam wands. An SCA survey of professionals found the typical recipe is an 18-20 g dose producing about 36 g in 25-30 seconds at 9 bar and about 93°C.

How to make it

1. Flush the group head for 2 seconds; wipe the basket dry.
2. Grind 18 g into the basket; level it by tapping or with a distribution tool.
3. Tamp straight and level with firm, consistent pressure; wipe the rim.
4. Lock in and start immediately; place the cup on a scale.
5. Stop at 36 g; check time (25-30 s) and taste; adjust the grind, never the tamp.

- Fresh coffee (7-30 days off roast) gives thicker crema; robusta adds crema too.
- Uneven shots (spurts, pale streaks) mean channelling: fix distribution first.

The espresso-and-milk family

Drink	Espresso	Milk and foam	Cup
Macchiato	Single or double	A spoon of foam	60-90 ml
Cortado	Double	Equal steamed milk, little foam	120 ml
Flat white	Double (often ristretto)	Thin, glossy microfoam	150-180 ml
Cappuccino	Single or double	Equal-ish milk and 1-2 cm foam	150-200 ml
Latte	Single or double	Lots of milk, thin foam	240-300 ml
Mocha	Double	Chocolate + latte milk	240-300 ml
Iced latte / cold coffee	Double	Cold milk, ice; Indian 'cold coffee' often blended with ice cream	300-400 ml





Pour-over: V60, Kalita and Chemex *percolation through paper, clean and aromatic*

Pour-over brewers hold a paper filter in a cone; you pour water by hand and it drains through the coffee bed. Melitta Bentz invented the paper filter in Dresden in 1908. Hario's **V60** (2004) has a 60° cone, spiral ribs and one large hole, so flow depends on your pouring: fast and expressive. The **Kalita Wave** has a flat bed and three small holes: slower, more even, more forgiving. The **Chemex** (Peter Schlumbohm, 1941) uses thick bonded paper for an exceptionally clean, tea-like cup.

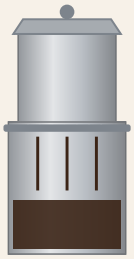
Pour-over shows off single-origin, light-to-medium roasts: florals, fruit and acidity come through clearly. James Hoffmann's widely used V60 recipe is below.

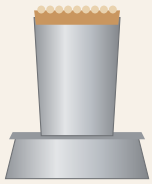
Dose	30 g coffee
Water	500 g at 93-100°C
Ratio	1:16.7
Grind	Medium-fine (table salt)
Time	~3:30 total
One cup	15 g : 250 g in five 50 g pours

How to make it

1. Rinse the paper with hot water and discard the water.
2. Add the coffee, make a small well; start the timer and pour 60 g to bloom; swirl. Wait up to 45 s while CO2 escapes.
3. Pour steadily to 300 g by 1:15 (60% of the water).
4. Pour to 500 g by 1:45, then give a gentle stir and a final swirl to flatten the bed.
5. Let it drain (done by about 3:30). Flat bed = even extraction.

- Slow drawdown and bitterness: grind coarser. Fast and sour: finer.
- Kalita: same ratio, pour in 4-5 pulses. Chemex: grind a little coarser; expect 4-4:30.





South Indian filter kaapi *slow percolation into a strong decoction, then hot milk and sugar*

Filter coffee is the everyday coffee of Tamil Nadu and Karnataka. The stainless-steel **filter** is two stacked cylinders: the upper one has a perforated base and a pressing disc (the 'umbrella'); the lower one collects the **decoction**. Fine coffee powder, often blended with roasted chicory, percolates slowly for 10-20 minutes into a thick, strong extract. It is mixed with boiling milk and sugar and served in a **tumbler and davara** (dabarah): pouring between the two from a height cools and froths the coffee ('meter coffee').

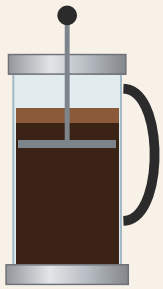
Chicory (roasted root of Cichorium intybus) adds body, colour and bitterness at lower cost; blends run 70:30, 80:20 or 90:10 coffee to chicory. Kumbakonam '**degree coffee**' is said to take its name from the lactometer 'degree' that certified pure, undiluted milk.

Powder	20-25 g (2-3 heaped tbsp)
Water	100-150 ml near boiling
Decoction	~60-80 ml, very strong
Grind	Fine, finer than pour-over
Time	10-20 min percolation
Serve	1 part decoction : 2-3 parts hot milk, sugar to taste

How to make it

1. Put the powder in the upper chamber and rest the umbrella disc on it (no hard tamping, or it will stall).
2. Pour near-boiling water over the disc, close the lid and wait 10-20 minutes.
3. Heat full-cream milk to a boil; add sugar to the tumbler.
4. Add 1 part decoction to 2-3 parts milk; 'pull' the coffee between tumbler and davara 3-4 times until frothy.

- A specialty kaapi made with estate arabica and little or no chicory is a natural menu signature for an Ahmedabad café (Part 2 prices drinks at ₹220 on average).
- Decoction keeps for a few hours; make it fresh for each service.



French press *immersion with a metal mesh: rich, heavy, forgiving*

A glass or steel jug with a mesh plunger. The coffee steeps, then the mesh separates the grounds; oils and fine particles pass through, giving a full, rounded body and a little sediment. Early versions were French (1850s); Italians Attilio Calimani (1929) and Faliero Bondanini (1958) patented the design that became the Bodum 'Chambord'.

Dose	30 g
Water	500 g just off the boil
Ratio	1:16.7
Grind	Medium to medium-coarse
Time	4 min + 5-8 min settle

How to make it

1. Add coffee, pour all the water, and leave for 4 minutes.
2. Stir the crust, then skim off the foam and floating grounds with two spoons.
3. Wait 5-8 minutes more so fines settle.
4. Press the plunger only to the surface and pour slowly.
   - Hoffmann's 'no-press' method gives a cleaner cup with less sludge.



AeroPress *immersion plus gentle pressure through paper*

Invented in 2005 by Alan Adler (of Aerobie frisbee fame), the AeroPress is a plastic syringe: coffee steeps in the chamber, then you press it through a small paper filter. It is fast, portable, almost unbreakable and endlessly adjustable; there has been a World AeroPress Championship since 2008.

Dose	11 g
Water	200 g at ~99°C
Ratio	1:18
Grind	Fine-medium
Time	2 min steep + 30 s press

How to make it

1. Fit a rinsed paper, place on a mug and add the coffee.
2. Pour all the water; insert the plunger slightly to stop dripping.
3. At 2:00 swirl gently; wait 30 seconds.
4. Press slowly for about 30 seconds.
   - Inverted brewing (upside down) avoids early dripping; many champions use it.



Moka pot *steam pressure on the stove: strong and bold*

Alfonso Bialetti's aluminium Moka Express (1933, designed with Luigi De Ponti) put strong coffee into every Italian kitchen. Water in the lower chamber boils; steam pressure of about 1-1.5 bar pushes it up through the coffee basket into the top chamber. It is not espresso (which needs about 9 bar) but it is intense, heavy and slightly bitter, and a popular home method in India too.

Water	Hot water to just below the valve
Coffee	Basket level-filled, 15-20 g
Grind	Fine-medium, not espresso-fine
Heat	Medium; lid open

How to make it

1. Fill the base with hot water (shortens the time the coffee sits on heat).
2. Fill the basket level without tamping; screw the top on firmly.
3. Heat until coffee flows; remove as soon as it starts to gurgle and sputter.
4. Cool the base under a tap to stop extraction; stir and serve.
   - Dilute with hot water for a smoother 'americano'.



Cold brew, cold drip and Japanese iced *time instead of heat*

Cold brew steeps coarse coffee in cold or room-temperature water for 12-24 hours. Little acid and bitterness is extracted, so it tastes smooth, chocolatey and sweet, though less aromatic. **Cold drip** (Kyoto towers) drips iced water through coffee over 3-12 hours. **Japanese iced coffee** brews hot pour-over straight onto ice (for example 30 g coffee, 300 g hot water, 200 g ice), keeping the aroma of hot brewing. Iced coffee is one of the fastest-growing café categories in India.

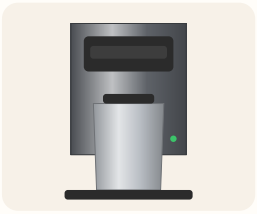
Ratio	1:8 concentrate, or 1:12-1:15 ready to drink
Grind	Coarse
Time	12-24 h, then filter
Keep	Refrigerated, ~1 week

How to make it

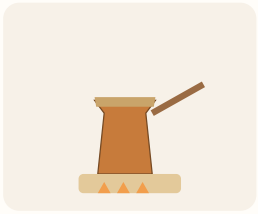
1. Mix coffee and water in a jar or a cold-brew bag; stir to wet all grounds.
2. Steep 12-18 h in the fridge (or 12 h at room temperature, then chill).
3. Filter through paper or cloth; dilute concentrate 1:1 with water or milk over ice.

- Medium-roast Indian arabica with some fine robusta makes excellent, cheap cold brew.

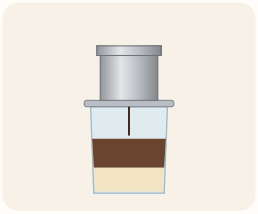
Other methods worth knowing



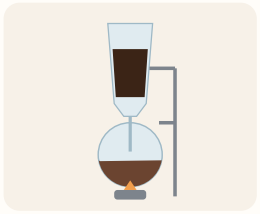
Batch brew



Cezve



Phin



Siphon



Clever

Method	What it is	Recipe	Result / use
Batch brew (drip machine)	A machine showers hot water over a basket of coffee and drips into a flask; SCA-certified brewers control temperature and contact time	60 g per litre (1:16-1:18); medium grind; 4-6 min	Volume at speed: ideal for office crowds; hold in airpots 30-60 min
Turkish (cezve / ibrik)	Powder-fine coffee, water and often sugar or cardamom heated until it foams, 2-3 times, never boiled; served unfiltered; UNESCO heritage (2013)	7-8 g per 60-70 ml	Thick, intense, aromatic; grounds settle in the cup
Vietnamese phin	A small steel drip filter on a glass; dark robusta drips slowly onto sweetened condensed milk; with ice it becomes cà phê sữa đá	20 g : 100 ml; 4-5 min	Strong, sweet, chocolatey; a natural fit for Indian robusta
Siphon (vacuum pot)	Vapour pressure pushes water into the upper chamber to steep; when heat is removed, a vacuum pulls the coffee down through a cloth filter (1830s-40s, Germany and France)	15 g : 225 g; 1-1.5 min steep	Clean, aromatic, theatrical
Clever dripper	A cone with a shut-off valve: steep like a French press, drain through paper when placed on a cup	18 g : 300 g; 3-4 min	Easy consistency with paper clarity

Instant coffee: how it is made

Instant (soluble) coffee is brewed in factories and dried. Roasted, ground coffee is extracted with hot water under pressure in counter-current columns (the extract leaves at about 15-30% solids), concentrated (up to ~60%), and then either **spray-dried** in tall towers into powder (often re-wetted and agglomerated into granules) or **freeze-dried**, which keeps more aroma. Aroma driven off during processing is captured and added back; final moisture is below 5%. Instant dominates Indian homes (roughly two-thirds to three-quarters of consumption, much of it coffee-chicory), and about 41% of India's coffee exports are instant, made by firms such as Tata Consumer, CCL Products and Nestlé.

All methods compared

Method	Family	Grind	Ratio	Water	Time	Body	Clarity	Skill	Kit cost (India)
Espresso	Pressure	Fine	1:2	92-94°C	25-30 s	●●●●●	●●	Hard	₹6-26 lakh café; ₹15k+ home
Moka pot	Pressure	Fine-med	~1:7	Stove	4-5 min	●●●●	●●	Easy	₹1.5-4k
AeroPress	Hybrid	Fine-med	1:15-18	80-99°C	2.5 min	●●●	●●●	Easy	₹3-4k
V60 / Kalita	Percolation	Med-fine	1:16-17	93-100°C	3-4 min	●●	●●●●●	Medium	₹0.8-3k
Chemex	Percolation	Medium	1:16	93-100°C	4-5 min	●	●●●●●	Medium	₹4-6k
Filter kaapi	Percolation	Fine	~1:5 dec.	~95°C	10-20 min	●●●●	●●	Easy	₹0.4-1.2k
French press	Immersion	Coarse	1:16	~95°C	4-9 min	●●●●	●●	Easy	₹0.8-2.5k
Cold brew	Immersion	Coarse	1:8-15	Cold	12-24 h	●●●	●●●	Easy	₹0.5-2k
Batch brew	Percolation	Medium	1:16-18	92-96°C	4-6 min	●●	●●●●	Easy	₹0.3-2 lakh
Turkish	Boiled	Powder	1:9	Just foaming	3-4 min	●●●●●	●	Medium	₹0.6-2k
Phin	Percolation	Medium	1:5	~95°C	4-5 min	●●●●	●●	Easy	₹0.5-1k
Siphon	Immersion	Medium	1:15	~92°C	1-2 min	●●	●●●●	Hard	₹4-10k

Source: SCA; Hoffmann recipes (Hario, 2021-23); manufacturers; indicative Indian retail prices (Oct 2026).

10.16 Espresso machines for a café

Type	How it works	Suits	Indicative price (India, Apr 2026)
Single boiler	One boiler for brewing and steam; switch between them	Home only	₹15,000-60,000
Heat exchanger (HX)	Steam boiler heats brew water through a coil	Small cafés; needs flushing	₹2-6 lakh (commercial)
Multi-boiler	Separate, temperature-controlled boiler per group + steam boiler	Specialty cafés	Brewlyn 2GR ₹6.3 lakh; La Marzocco Linea PB 2GR ₹11.3 lakh (+GST)
Pressure / flow profiling	Programmable pressure or paddle-controlled flow	High-end specialty	La Marzocco PBX ₹13.7 lakh; Strada X ₹26 lakh+
Super-automatic	Grinds, tamps, brews and steams at a button	Offices, QSR, hotels	₹1.5-8 lakh

Source: West End Coffee price comparison (Apr 2026); Kaapi Machines; distributor listings.

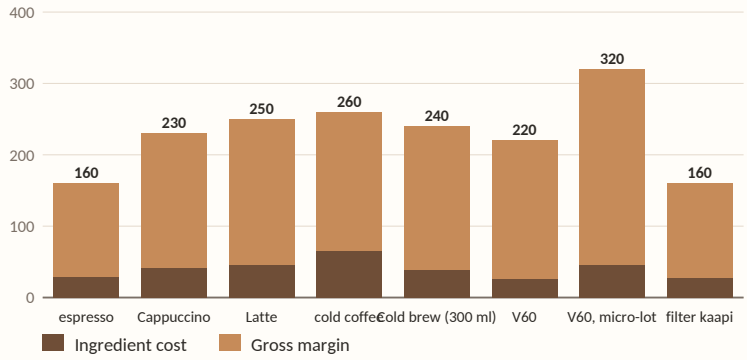
10.17 Troubleshooting

Problem	Likely cause	Fix
Espresso sour, fast	Grind too coarse, dose too low, coffee too fresh or stale	Grind finer; check dose; rest beans 7+ days
Espresso bitter, slow	Grind too fine, dose too high, water too hot	Grind coarser; lower dose or temperature
Pour-over stalls	Too fine, fines clogging paper, too much agitation	Coarser; gentler pours
French press muddy	Too fine; plunged too hard	Coarser; skim and wait; plunge to surface only
Kaapi drips too slowly	Powder too fine or pressed hard	Slightly coarser; rest the disc lightly
Milk big bubbles	Wand too high, milk too warm	Start cold; keep tip just under surface; stop at 60-65°C

10.18 What each drink costs a café to make

Drink	Coffee	Coffee ₹	Milk	Milk ₹	Other ₹	Total cost	Menu price	Gross margin
Double espresso	18 g	₹27	–	–	₹2	₹29	₹160	82%
Cappuccino	18 g	₹27	180 ml	₹12	₹3	₹42	₹230	82%
Latte	18 g	₹27	250 ml	₹17	₹3	₹47	₹250	81%
Indian cold coffee	18 g	₹27	200 ml	₹14	₹25	₹65	₹260	75%
Cold brew (300 ml)	22 g	₹33	–	–	₹6	₹39	₹240	84%
V60, House blend	15 g	₹22	–	–	₹4	₹26	₹220	88%
V60, micro-lot	15 g	₹42	–	–	₹4	₹46	₹320	86%
Specialty filter kaapi	12 g	₹18	120 ml	₹8	₹1	₹27	₹160	83%

Cost and gross margin per drink (₹)



Coffee drinks earn gross margins of **75-90%**; 'other' covers sugar, syrups, ice cream, filter papers and cup. Milk drinks cost more but sell for more. The real costs of a café sit below the gross margin: staff, rent and overheads (Part 2).

Costing uses Part 2's House-blend, toll-roasted coffee at ₹1,482/kg (₹2,824/kg for a micro-lot) and full-cream milk at ₹68/litre; menu prices are indicative for Ahmedabad specialty cafés (ex-GST).

10.19 Getting started at home

Budget	Kit	Beans
Under ₹3,000	V60 or filter-kaapi set, paper filters, kitchen scale (₹500-800)	Pre-ground fresh coffee from a local roaster, used within 2 weeks
₹10,000-15,000	Hand burr grinder (₹4,000-8,000), V60 or AeroPress, gooseneck kettle, scale	Whole-bean single estates; grind fresh
₹50,000+	Entry espresso machine with steam wand, espresso grinder, tamper, milk jug	Medium-roast espresso blend; learn to dial in

10.20 Signature drinks: recipes for a café menu

Drink	Specification	Why
Spanish latte	Double shot + 25-30 ml condensed milk + 150 ml steamed (or cold) milk	Sweet, creamy; huge in Indian cafés
Cardamom (elaichi) latte	Double shot + 10 ml cardamom syrup + 180 ml steamed milk	Local flavour; works iced
Affogato	Double shot poured over a 60 g scoop of vanilla ice cream	Dessert menu
Espresso tonic	150 ml tonic over ice; float a double shot; orange peel	Light, bitter-sweet summer drink
Cold brew with jaggery	150 ml cold brew + 20 ml jaggery syrup + ice (+ milk)	Indian twist on sweet cold brew
Vietnamese-style iced	Phin: 20 g dark robusta : 100 ml over 30 ml condensed milk, on ice	Uses Indian robusta
Iced filter kaapi	40 ml decoction + 150 ml cold milk + sugar or jaggery, shaken with ice	Signature for an Ahmedabad café
Mocha	Double shot + 20 g chocolate sauce + 200 ml steamed milk	Crowd-pleaser
Japanese iced pour-over	30 g coffee : 300 g hot water brewed onto 200 g ice	Showcases micro-lots cold
Indian cold coffee	Double shot + 200 ml cold milk + 40 g ice cream + sugar, blended	Best-seller; highest ingredient cost

10.21 Steaming milk and latte art

1. Start with cold milk (fridge-cold, about 4°C) filled to the bottom of the jug's spout; purge the steam wand.
2. **Stretch**: tip just below the surface, steam on full; let it 'tear paper' gently to add air until the jug feels hand-warm (~37°C). Short stretch for a latte, longer for a cappuccino.
3. **Texture**: raise the jug so the tip sits a little deeper and the milk spins in a whirlpool, folding bubbles into silky microfoam, until 60-65°C (too hot to hold for more than a second or two).
4. Wipe and purge the wand; tap the jug to burst big bubbles and swirl until glossy.
5. **Pour**: start high to sink milk under the crema, then bring the spout close to the surface so foam flows on top: a dot becomes a heart; wiggle for a rosetta; stack for a tulip.

10.22 Equipment care schedule

When	What
After every shot	Flush group head; knock out and wipe the basket
After every milk drink	Purge and wipe the steam wand
Daily (closing)	Back-flush groups with detergent; soak portafilters, baskets and jugs; clean grinder chute and hopper; empty and wash drip trays; clean brewers
Weekly	Brush group gaskets and shower screens; grinder-cleaning tablets; deep-clean batch brewer
Monthly	Check water filter capacity and TDS; inspect gaskets and steam-wand tips
Quarterly / annually	Replace gaskets and shower screens; service boilers and pumps (AMC); check burr wear; replace RO membranes and remineralisation cartridges as rated

10.23 Producing cold drinks for a café

Drink	Production	Shelf life	Notes
Cold brew	Batch of 1-2 kg coffee in 10-20 litres; steep 16 h in the fridge; filter; store ≤5°C	5-7 days	Label batch date; discard after shelf life
Japanese iced	Brewed to order on ice	Immediate	Showcase drink; uses micro-lots
Iced espresso drinks	Fresh shot over ice and cold milk	Immediate	Speed; pre-portion milk and syrups
Indian cold coffee	Blended to order or from a pre-mixed base	Base: same day	Highest ingredient cost; watch portion control
Syrups	Jaggery, cardamom, vanilla made in-house	2-3 weeks refrigerated	Cheaper and a point of difference

In Ahmedabad's long hot season cold drinks can be half of beverage sales, so the bar needs ice capacity (a commercial ice machine is in the model's kitchen budget), fridge space for cold brew and a fast blender station that does not block the espresso line.

10.24 Espresso, a closer look

Modern espresso has moved away from the traditional Italian single (7 g making 25 ml) towards larger, weighed doses and ratios. Precision baskets with uniform holes, distribution tools (stirring the grounds with fine needles, known as WDT) and puck screens all aim at one thing: an even puck with no channels.

Research has pushed further. A 2020 study by Cameron and colleagues in the journal *Matter*, 'Systematically improving espresso', showed that slightly **coarser grinding with lower doses** gave more consistent, higher extractions and less waste, because very fine grounds clog unevenly. Some cafés now pull 'turbo' shots: coarser grind, ratio around 1:3, about 15 seconds.

Style	Dose : yield	Time	Taste
Traditional Italian	7 g : 25 ml (single)	20-25 s	Dense, roasty
Modern normale	18 g : 36 g	25-30 s	Balanced, sweet
Ristretto	18 g : 25 g	20-25 s	Syrupy, intense
Lungo	18 g : 54 g+	35-40 s	Lighter, more bitter
Turbo	15-18 g : 45-54 g	~15 s	Clean, bright
Allongé	18 g : 90 g	30-40 s	Filter-like, light roasts

10.25 Pour-over, a closer look

Three refinements make pour-over consistent. **Pouring**: a gooseneck kettle gives a thin, controllable stream; pour in circles that avoid the paper walls. **Paper**: rinse it to remove papery taste and pre-heat the brewer; bleached papers taste cleaner than unbleached. **Bed**: a flat bed at the end means even extraction; a crater or grounds stuck high on the walls mean channelling.

Tetsu Kasuya's 4:6 method, which won the 2016 World Brewers Cup, is a good way to learn how pours shape the cup: 20 g of coarse-ground coffee and 300 g of water poured in five equal 60 g pours, 45 seconds apart. The first 40% of the water (two pours) controls sweetness and acidity: make the first pour smaller for sweeter coffee, larger for brighter. The remaining 60% (three pours) controls strength: fewer, larger pours for a lighter cup, more pours for a stronger one.

10.26 Filter kaapi: culture, chicory and serving it at scale

Filter coffee spread through South India in the 20th century, first in middle-class homes and then in Udupi restaurants, darshinis and the Indian Coffee Houses. Its taste is defined by three choices: a **dark roast** (often of Plantation and Peaberry arabica with robusta), **chicory** (which adds bitterness, body and colour and stretches the coffee), and **full-cream milk** with sugar. Indian rules require coffee-chicory blends to be labelled as such, with the proportions declared.

Restaurants serving hundreds of cups use large multi-litre filters or brewing urns, make decoction in batches through service, and keep milk simmering. For a specialty café, the opportunity is a **premium kaapi**: estate arabica, a medium-dark roast, no or little chicory, decoction made fresh every hour and served in a steel tumbler and davara, hot or iced. At about ₹27 of ingredients and a ₹150-180 menu price (section 10.18) it is one of the highest-margin drinks a café can sell, and a strong signature in Gujarat, where most customers have never tasted good filter coffee.

10.27 Timing the brew

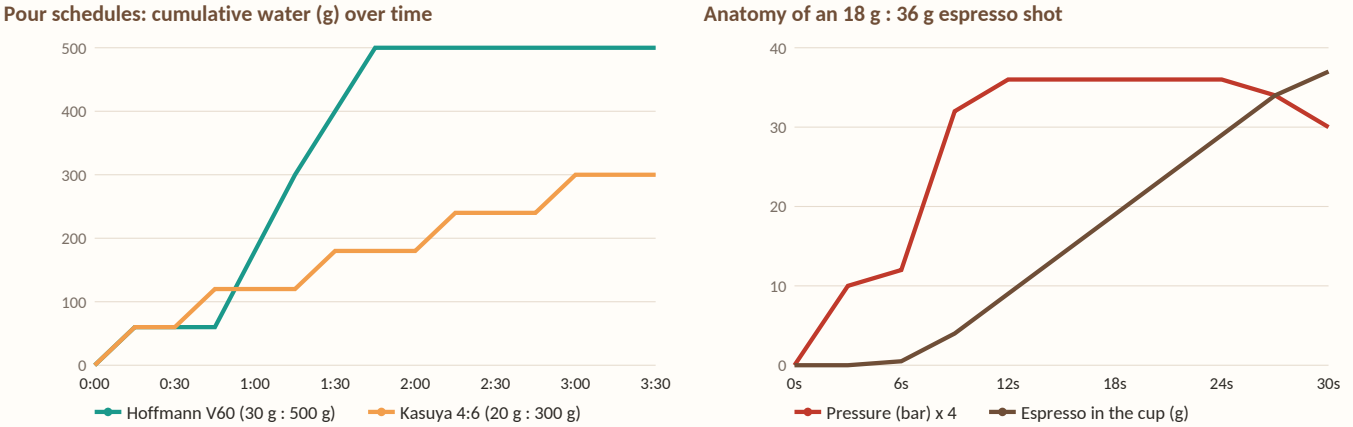
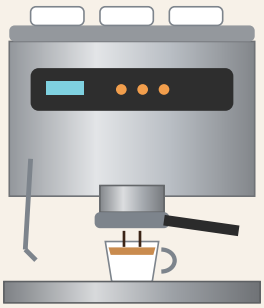


Figure 10.1. Left: two well-known pour-over schedules. Right: a typical shot with a few seconds of low-pressure pre-infusion, then ~9 bar; the first drops appear after 6-8 s and the shot reaches 36 g at about 28 s (illustrative).

10.28 Brewing cheat sheet (print me)



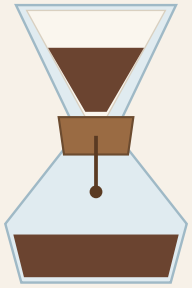
Espresso

Dose : yield	18 g : 36 g
Grind	fine
Water	93°C, 9 bar
Time	25-30 s



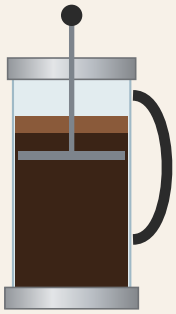
V60 pour-over

Dose : water	15 g : 250 g
Grind	medium-fine
Water	93-100°C
Time	2:30-3:00



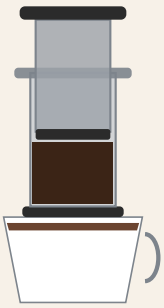
Chemex

Dose : water	30 g : 500 g
Grind	medium-coarse
Water	93-100°C
Time	4:00-4:30



French press

Dose : water	30 g : 500 g
Grind	medium-coarse
Water	just off boil
Time	4 min + 5-8 settle



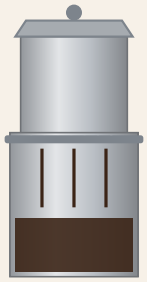
AeroPress

Dose : water	11 g : 200 g
Grind	fine-medium
Water	~99°C
Time	2:00 + 30 s press



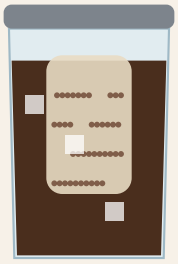
Moka pot

Coffee	basket level, 15-20 g
Grind	fine-medium
Water	hot, to the valve
Time	4-5 min, off at gurgle



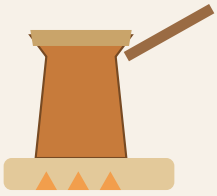
Filter kaapi

Powder : water	20-25 g : 120 ml
Grind	fine
Serve	1 decoction : 2-3 milk
Time	10-20 min drip



Cold brew

Ratio	1:8 concentrate
Grind	coarse
Water	cold / room
Time	12-18 h, fridge



Turkish

Dose : water	7-8 g : 70 ml
Grind	powder
Heat	to foam, 2-3 times
Serve	unfiltered

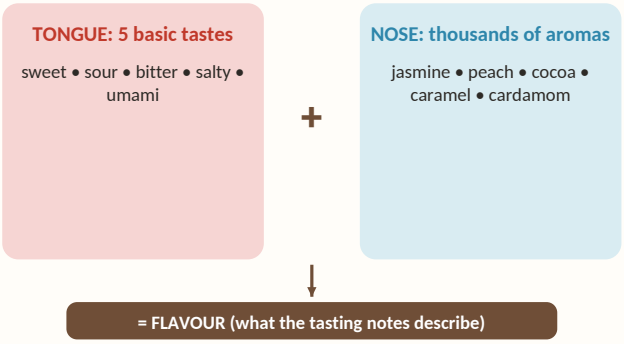
Weigh coffee and water; adjust grind by taste: sour = finer, bitter = coarser.

11 Tasting and judging quality

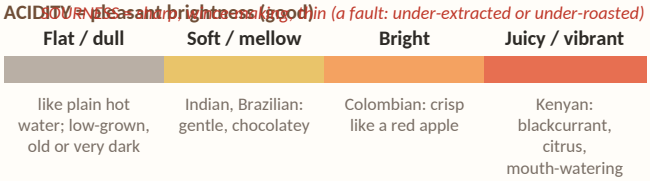
How to taste like a professional and what the words mean

Most of what we call 'taste' is smell. The tongue reads only sweet, sour, salty, bitter and umami; aromas reach the nose from the back of the mouth when you swallow or slurp (retro-nasal smell). That is why professional tasters slurp noisily: it sprays coffee across the palate and pushes aroma up to the nose.

Taste coffee at several temperatures: hot coffee hides flaws and sweetness; as it cools, acidity and defects show. Compare two coffees side by side; the differences are far easier to notice than absolute qualities.



11.1 The vocabulary





Word	Meaning	Where you find it
Acidity	Bright, lively sharpness (like apple or citrus), not sourness	Clean washed high-grown arabica
Body	Weight and texture in the mouth: tea-like to syrupy	Naturals, robusta, French press, espresso
Sweetness	Caramel, honey, ripe fruit; sign of ripe cherries and good roasting	Most prized quality marker
Balance	No one attribute shouts over the others	Good blends
Clean cup	No off-flavours from first sip to finish	Careful processing
Aftertaste / finish	How long pleasant flavour lingers; dry or bitter finish is a flaw	Well-developed roasts
Flavour notes	Descriptors like jasmine, peach, chocolate: aromas the coffee resembles, nothing is added	Use the flavour wheel

11.2 The flavour wheel and defects in the cup



Off-flavour	Likely cause
Sour, vinegar, onion	Over-fermentation, sour beans
Musty, earthy, mouldy	Damp drying or storage
Medicinal, 'rioy', iodine	Microbial taint, robusta faults
Grassy, peanut, astringent	Unripe cherries, underdeveloped roast
Papery, woody, cardboard	Old green coffee or stale roast
Burnt, ashy, smoky	Over-roasting or scorching
Hollow, flat	Stale beans, too-soft water
Bitter, drying	Over-extraction, too fine or too hot

11.3 Linking the chain to the cup

Link in the chain	What it contributes to the cup
Species & variety	Sets the potential: sweetness, acidity, complexity
Altitude & terroir	Density, acidity, aromatic complexity
Picking	Sweetness and cleanliness (ripe-only picking)
Processing	Clarity (washed) versus fruit and body (natural, honey, experimental)
Drying & storage	Freshness of the green coffee; absence of must and ferment
Roast	Which flavours are expressed: acidity and florals (light) or chocolate and roastiness (dark)
Freshness	Aroma intensity; flat after 4-6 weeks opened
Grind & brew	Whether the potential is extracted: sour, sweet or bitter
Water & milk	Clarity, sweetness, texture

11.4 A four-week tasting plan

1. **Week 1, processing:** taste a washed and a natural coffee from the same region side by side as pour-over. Note clarity versus fruit and body.
2. **Week 2, roast:** compare a light and a dark roast of similar beans as French press. Note acidity, sweetness and bitterness.
3. **Week 3, method:** brew one coffee three ways (V60, French press, AeroPress). Note how body and clarity change.
4. **Week 4, espresso and milk:** taste a shot alone, then as a cortado and a latte. Note how milk changes acidity and sweetness, and why espresso blends include robusta.
5. Throughout, write three words for each cup using the flavour wheel. Within a month most people can tell washed from natural and light from dark.

11.5 Judging a café, not just a cup

Area	Signs of a good café
Sourcing	Estate and roast date on the menu or bag; staff can explain the beans
Freshness	Beans 7-30 days off roast; ground to order
Brewing	Weighed doses and yields; clean group heads and grinders
Milk	Glossy microfoam, not scalded (no hotter than ~65°C)
Range	Espresso, at least one manual method, a cold option, decaf
Water and hygiene	Filtered water; clean cups, jugs and steam wands
Experience	Speed at rush hour, seating, service, value

11.6 Common myths

Myth	Reality
'Dark roast has more caffeine'	Roasting barely changes caffeine; by volume a dark roast has slightly less because beans are lighter
'Espresso has the most caffeine'	Per serving a 240 ml filter coffee usually has more than a single espresso
'Robusta is always bad'	Washed, carefully processed fine robusta can score 80+ and is prized in espresso
'Thick crema means quality'	Crema shows freshness and robusta content, not flavour quality
'Fresher is always better'	Coffee needs a few days to degas; very fresh coffee brews unevenly
'Store beans in the fridge'	Moisture and odours ruin them; use an airtight container in a cool cupboard (freeze only sealed, portioned bags)
'Boiling water burns coffee'	Just-off-boil water (93-96°C) is ideal for most filter brewing
'Chicory is fake coffee'	It is a traditional, labelled ingredient in South Indian filter coffee; the issue is only undeclared or excessive use

11.7 Pairing coffee with food

Coffee	Food	Why it works
Washed light-roast pour-over	Fresh fruit, citrus tarts, light cakes	Acidity echoes fruit
Indian espresso / cappuccino	Dark chocolate, croissants, almond biscotti	Chocolate notes match
Natural or anaerobic micro-lot	Berry desserts, cheesecake	Fruit-forward pairing
Filter kaapi	Idli, vada, upma, Mysore pak	The classic South Indian breakfast
Cold brew	Savoury sandwiches, khakhra, nachos	Low acidity, smooth
Monsooned Malabar / dark	Gujarati farsan, spiced cookies	Spice meets spice

11.8 The flavour wheel, category by category

Category	Notes in this family	Usually comes from
Fruity	Berry, dried fruit, citrus, stone fruit	Ethiopian and Kenyan washed; naturals; anaerobics
Floral	Jasmine, rose, black tea	Gesha, high-grown washed arabica
Sweet	Brown sugar, caramel, honey, vanilla	Ripe picking and good roast development
Nutty / cocoa	Almond, hazelnut, chocolate, dark chocolate	Brazil, India, medium roasts
Spices	Clove, cinnamon, pepper	Indian and Sumatran coffees; monsooned
Roasted	Cereal, malt, tobacco, burnt	Darker roasts; burnt = roast fault
Green / vegetative	Grassy, peapod, beany, olive oil	Underdeveloped roast or unripe beans
Sour / fermented	Winey, whisky, vinegar, over-ripe	Fermentation; positive in moderation, a defect when sharp
Other (faults)	Papery, musty, earthy, rubber, medicinal	Old coffee, storage and microbial taints

Source: SCA and World Coffee Research Coffee Taster's Flavor Wheel and Sensory Lexicon (2016).

12 From roaster to customer

Packaging, distribution channels, the café as the last step, and who India's coffee customers are

The last mile is where most of coffee's value is added. Once roasted, coffee is a perishable, aromatic product that has to reach the customer fresh, in the right format and through the right channel.

12.1 How coffee reaches the customer



Figure 12.1. The downstream chain. Freshness is lost at every hand-off.

Channel	Typical products	What wins	Notes
Cafés and restaurants (HoReCa)	Whole beans, 1 kg bags; often with equipment on loan or lease and barista training	Consistency, service, price per kg	Largest specialty channel; sticky accounts
Hotels and offices	Beans for bean-to-cup machines, drip bags, batch brew	Uptime, simplicity	GIFT City-type office clusters
Supermarkets and kiranas	Instant jars and sachets, filter-coffee powder, ground coffee	Price, brand, shelf life	Mass market; dominated by Nescafé, Bru and filter brands
E-commerce marketplaces	Beans, ground, drip bags, instant	Ratings, discounts	Wide reach; high commissions and returns
Roaster websites (D2C) and subscriptions	Fresh beans by roast date; gear	Freshness, story, convenience	Core of Indian specialty roasters' growth
Quick commerce (10-minute apps)	Cold brew bottles/cans, instant, drip bags, beans	Speed, impulse	Fastest-growing urban channel since 2023
Vending and kiosks	Instant premixes, bean-to-cup	Speed, price	Railways, offices, campuses

12.2 What a bag of specialty coffee pays for

Illustrative build-up of a ₹571 (ex-GST) 250 g bag sold direct to consumers

Total ₹571 (ex-GST)



Figure 12.2. Green cost uses the Part 2 Specialty mix (₹926/kg green, 16% roast loss). Other items are typical ranges for Indian D2C roasters and are illustrative.

Two lessons follow. First, specialty roasting is a thin-margin business once marketing and delivery are paid for; scale and repeat customers (subscriptions, café accounts) make it work. Second, a café that sells its own beans skips most of the distribution cost: the customer is already in the room. That is why the Part 2 model gives own-label routes a higher retail share (8% of sales against 6%).

12.3 The café: the final and most valuable step

Format	Size (sqft)	Seats	Typical capex	Offer	Where
Kiosk / cart	100-300	0-6	₹8-20 lakh	Espresso, cold coffee, grab-and-go	Malls, metro stations, campuses
Express / takeaway	300-700	8-20	₹20-45 lakh	Espresso bar, pastries, delivery	Office districts, high streets
Full-service café	800-1,500	30-60	₹50-90 lakh	Espresso, manual brew, full food menu	Neighbourhood and high-street sites
Roastery café	2,000+	60+	₹1.5 crore+	On-site roaster, retail, training	Flagship of a brand
Drive-thru	1,000+ plus lanes	Few	₹1 crore+	Speed, cold drinks	Highways, suburbs

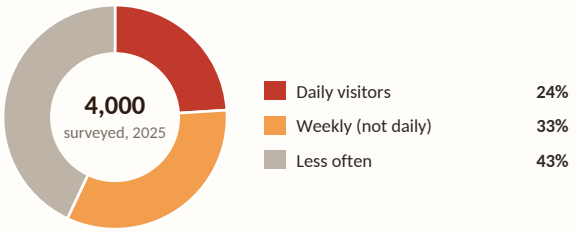
Source: Capex ranges from Part 2's equipment and fit-out costs and Indian fit-out quotes; indicative.

Customers in a café pay for far more than coffee: a place to sit, Wi-Fi, service, speed, music, food and the feeling of a treat. That is why a café can charge ₹200-350 for a drink whose ingredients cost ₹25-65 (Chapter 10), and also why its costs are dominated by staff and rent rather than beans.

12.4 Who India's coffee customers are

Segment	What they buy	When
Students and young professionals	Cold coffee, frappés, value combos; social media driven	Evenings, weekends
Office workers	Morning espresso and batch brew, delivery to desks, meetings	8-11 am, 3-5 pm
Families and friends	Desserts, food, larger groups; Gujarat's dry law makes cafés the evening social venue	After 5 pm, weekends
Enthusiasts	Pour-over, single estates, retail beans, brewing gear	Weekends
South Indian households	Filter kaapi at home; buy powder and chicory blends	Morning ritual
Remote workers	Long stays, Wi-Fi, power points; low spend per hour	Weekday daytime

How often Indians visit coffee shops



World Coffee Portal's 2025 survey of 4,000 Indian consumers found **24% visit a coffee shop daily** and 57% at least weekly, preferring to sit in with friends and family; a third of visits happen after 5 pm. Food is a big part of the bill: in Indian cafés it is typically 30% or more of sales, which the Part 2 model reflects.

Implications for the café: design for evening crowds and groups (seating, desserts), win the office morning with speed (batch brew, pre-orders), and convert enthusiasts into retail-bean customers.

Source: World Coffee Portal, Project Café India 2025 (via Comunicaffe, Sep 2025).

12.5 Freshness from roaster to cup

Stage	Good practice	Timing after roast
Roaster	Roast to order; rest; pack in valve bags with roast date	Day 0-2
Logistics	Ship within days; avoid heat (Ahmedabad summers exceed 40°C)	Day 2-5
Café storage	Cool, dark, sealed; first-in-first-out; small hopper loads	Day 5-30
Grinding	Grind to order; never pre-grind for the day	Seconds
Retail customer	Buy 250 g at a time; finish within 3-4 weeks	Day 7-40

Why this matters for the café plan

Toll roasting near Ahmedabad or fortnightly deliveries keep coffee in the 7-30 day sweet spot without owning a roaster.
Retail beans and drip bags sold at the counter add high-margin revenue and keep the brand at customers' homes.
Quick-commerce listings of cold brew and drip bags can follow once the café brand is known.

12.6 Running the bar: workflow and capacity

A double espresso takes 25-30 seconds to brew but about a minute of barista time with grinding, tamping, milk and hand-off. One barista on a two-group machine comfortably makes **60-90 drinks an hour**; two baristas 120-150. The Part 2 base case for Prahladnagar has about 140 bills and 125 drinks a day, so even if a fifth of them arrive in the busiest hour (25 drinks), one skilled barista copes. Weekend evenings on Sindhu Bhavan Road need two.

Speed comes from layout and preparation: a straight line from **POS to espresso to milk to hand-off**, batch-brewed coffee and pre-made cold brew for queues, syrups and garnishes prepared off-peak, and a separate pour-over station so manual brews do not block the espresso line.

Peak-hour check	Prahladnagar	SBR
Bills per day (Y3, Base)	140	170
Drinks per day	~126	~174
Peak hour (20%)	~25 drinks	~35 drinks
Weekend peak (x1.8)	~45 drinks	~63 drinks
Baristas needed at peak	1-2	2

12.7 Menu engineering

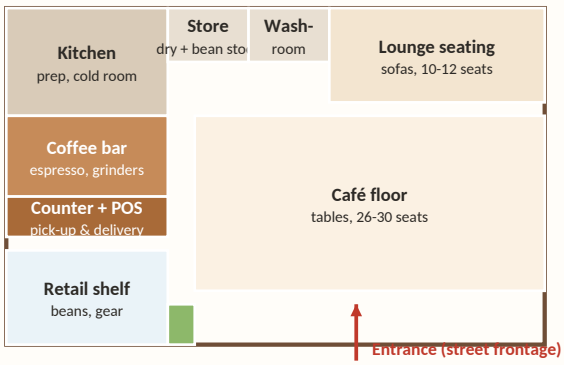
	High margin	Low margin
High popularity	**Stars**: promote, protect quality (cappuccino, cold brew, Spanish latte)	**Plough-horses**: re-price or re-cost (Indian cold coffee with ice cream)
Low popularity	**Puzzles**: reposition, train staff to suggest (pour-over micro-lots, kaapi)	**Dogs**: drop or redesign

Menu engineering sorts items by popularity and margin per item. Coffee drinks are mostly 'stars'; food items and elaborate cold drinks need closer costing because ingredients are a larger share of price.

Practical rules: keep the drinks menu to 15-20 items; put the highest-margin signatures at the top right; pair coffee with food (food is ~30% of café sales in the model); and revisit prices every year, because wages rise faster than menu prices in the model's base case.

12.8 Designing the café

Schematic 1,000 sqft café (not to exact scale)



A 1,000 sqft specialty café (the Part 2 size for Prahladnagar) typically gives about 60% of its area to customers and 40% to the bar, kitchen, storage and washroom. Design choices that matter most:

- **Bar visible from the entrance**: the espresso machine is theatre and signals quality.
- **Separate order, pick-up and delivery points** so aggregator riders do not crowd seated guests.
- **Seating mix**: two-tops for work and dates, a long communal table, a lounge for evening groups; power points and good Wi-Fi.
- **Acoustics and light**: soft surfaces, warm light in the evening, daylight in the morning.
- **Back of house**: RO plant, ice machine, cold room, dishwash and bean storage away from heat.

12.9 Licences, GST and the rules of the game

Area	What is needed	When
FSSAI	Registration (turnover up to ₹12 lakh) or State licence (₹12 lakh-₹20 crore)	Mandatory before opening; display the number
Shop & Establishment	Gujarat Shops and Establishments registration	Within 30 days of opening
GST	Restaurant services at 5% without input-tax credit (standalone cafés); packaged beans and instant coffee at 5%	GST registration above the turnover threshold
Fire NOC	From the fire department for eating houses above size thresholds	Before opening
Trade / eating-house licence	From the municipal corporation (AMC) and police where applicable	Annual renewal
Signage & music	Signage permission; IPRS/PPL licences for background music	Before use
Labour	PF, ESI registration and minimum wages (Gujarat schedule)	On hiring

Source: FSSAI licensing thresholds; GST rates for restaurant services; Gujarat labour rules. Check current rules with a consultant before opening.

12.10 The footprint of a cup

Life-cycle studies of coffee consistently find two hot spots: **farming** (fertiliser, land-use change if forests are cleared, processing energy and water) and, for milk drinks, **the milk itself**, which usually accounts for most of a latte's carbon footprint. Brewing energy, cups and waste come next. India's shade-grown, largely deforestation-free coffee starts with an advantage.

- Offer plant milks and smaller milk drinks; price them fairly.
- Avoid single-use cups for dine-in; offer a discount for reusable cups.
- Compost or donate spent grounds; recycle valve bags where possible.
- Buy from estates that can document shade cover and deforestation-free land (EUDR data helps).

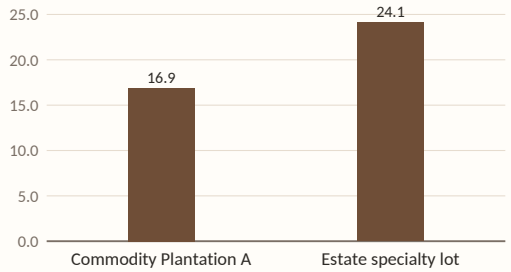
12.11 Case study: one bag from Chikmagalur to Ahmedabad

Follow one 60 kg bag of washed arabica from a Chikmagalur estate to cups in an Ahmedabad café, using the prices and yields in this report.

When	What happens	Who	Money
Mar 2025	Blossom shower; flowers open, fruit sets	Estate	-
Dec 2025	~330 kg of ripe cherry picked over several passes (5.5 kg cherry per kg green)	6-7 picker-days	Labour paid per kg of cherry
Dec 2025	Pulped the same evening, fermented ~24 h, washed, dried 10-12 days to 10.5%	Estate pulper house	~75 kg of parchment
Jan-Feb 2026	Parchment rests, then sold to a curing works	Grower → curer	₹24,500 per 50 kg = ~₹36,750
Feb 2026	Hulled, graded Plantation A, colour-sorted, bagged: 60 kg green	Curing works	~₹612/kg green (estate specialty lot: ~₹950/kg)
Mar 2026	Trucked ~1,200 km to Gujarat; 3-4 days	Transporter	₹25/kg = ₹1,500
Mar-Apr 2026	Toll-roasted in batches to the café's espresso profile; 16% loss → 50.4 kg	Roaster	₹180/kg roasted = ₹9,072
Apr-May 2026	Rested 7 days, delivered fortnightly, ground to order	Café	-

When	What happens	Who	Money
Apr-Jun 2026	50.4 kg ÷ 18 g = ~2,800 double-shot drinks at ~₹220	Customers	~₹6.2 lakh of beverage sales (ex-GST)

Coffee cost per double-shot drink (₹)



From flower to cup took about 14 months, and from tree to cup five to six. The coffee inside each drink cost the café about **₹17** for commodity Plantation A or **₹24** for an estate specialty lot, against a ₹220 menu price. The grower's ₹36,750-57,000 for the bag compares with roughly ₹6.2 lakh of drinks sold: about **6-9%**, consistent with the value-chain estimates in Chapter 6.

The extra ₹7 a drink for an estate lot is why paying for quality is almost free for a café, and why specialty cafés can afford to pay growers well above commodity prices.

12.12 A first-time café owner's coffee FAQ

Which beans should I start with? A house blend of washed Indian estate arabica (about 70%), washed fine robusta (about 15%) for body and crema, and a micro-lot share for character; toll-roasted to your profile. Add one rotating single origin for pour-over.

Do I need my own roaster? Not at the start. Part 2 shows toll roasting beats an in-house roastery until you sell roughly 450-500 kg a month wholesale.

How fresh should the beans be? Use them 7-30 days after roasting; order every one to two weeks; never pre-grind.

Which espresso machine and grinders? A two-group multi-boiler machine (₹6-14 lakh + GST) and two on-demand espresso grinders plus one bulk grinder. Budget for AMC and spares.

How much coffee will I use? About 700-950 kg a year for drinks at 140-170 bills a day, plus 450-650 kg for retail bags if you sell your own label (Part 2, year 3).

How many baristas? A head barista plus four baristas covers two shifts and peaks; train them with your roaster and SCA courses, and dial in together every morning.

What water set-up? RO with remineralisation to about 150 mg/L TDS; about ₹1 lakh; change cartridges as rated.

How should I price drinks? Ingredients cost ₹25-65 a drink; price at ₹160-320 so beverage gross margins stay at 75-85%, and review prices every year.

Is robusta or chicory a problem? Not if declared and used for a purpose: fine robusta improves espresso and cold brew; a premium filter kaapi can use little or no chicory.

How do I know my coffee is good? Cup new lots blind, keep recipes written and weighed, track freshness, and read reviews weekly. A refractometer helps but is optional.

12.13 Buying coffee for home in India

If you brew with	Buy	Typical price
V60, Chemex, AeroPress	Light to medium single estates; washed or natural	₹500-900 per 250 g
French press, cold brew	Medium to medium-dark; Indian arabica or blends	₹400-700
Espresso at home, moka pot	Medium-dark espresso blend, some robusta	₹450-750
Filter kaapi	Filter-coffee powder: pure coffee or 80:20 / 90:10 with chicory	₹250-450
Instant	Freeze-dried for best flavour; specialty instant sachets	₹200-800 per 100 g

- Buy whole beans with a roast date; grind just before brewing if you have a burr grinder, or ask the roaster to grind for your method.
- Buy 250 g at a time and finish within 3-4 weeks; store airtight, away from heat and light.
- Try a sampler pack: three estates or processes side by side is the fastest way to learn your taste.

13 The coffee market

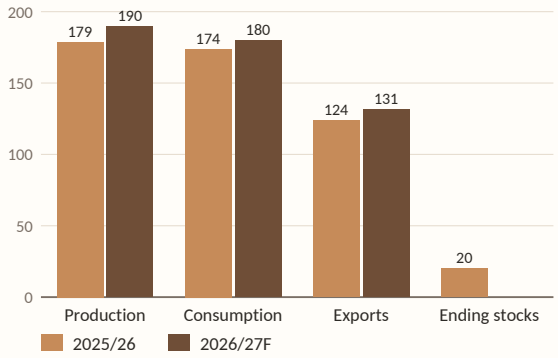
Global supply and demand, India's production, exports and drinking habits, the café boom, and Ahmedabad

13.1 Global supply, demand and price

World production reached a record 178.8 million bags in 2025/26 and USDA expects another record in 2026/27, led by Brazil (71.9 million). Consumption keeps growing about 1-2% a year to a forecast 179.7 million bags, driven by Asia: China, Indonesia, the Philippines, Vietnam and India are the fastest-growing markets. Stocks fell for five years to 20.1 million bags, which together with Brazilian and Vietnamese droughts pushed prices to records in early 2025.

For a café the lesson is that **green-coffee prices swing by 30-50%** within a couple of years. Part 2's model carries a green-price index with Best (0.90), Base (1.00) and Worst (1.25) starting points and a sensitivity to ±30%.

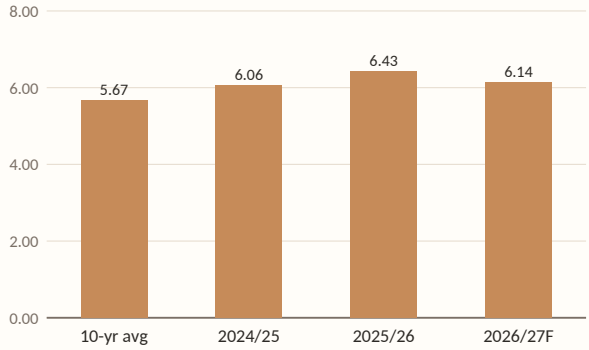
World balance, million 60-kg bags



Source: USDA FAS Coffee: World Markets and Trade (Dec 2025; Jun 2026). 2026/27 production = 2025/26 + 10.8 million bags.

13.2 India: production and exports

India's crop, million 60-kg bags (USDA)



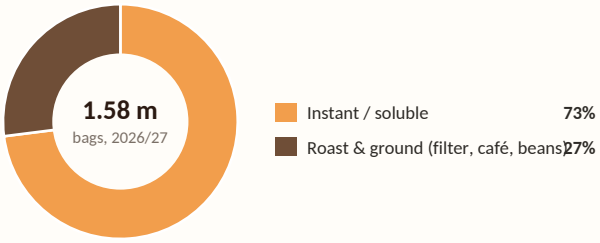
Source: USDA FAS India Coffee Annual (May 2026); PIB (Nov 2025).

India grows about 6-6.4 million bags (360-385 thousand tonnes) a year, about three quarters robusta. Area planted is about 4.75 lakh ha with some 645 million trees; production has grown only about 2% a year over the past decade because yields are flat and replanting is slow. The Coffee Board's goal is 9 lakh tonnes by 2047, mostly by expanding into non-traditional areas such as Odisha and the North-East.

Exports reached **6.22 million bags** forecast for 2026/27: 59% green beans and 41% instant coffee. About 61% leaves through Mangalore, and Europe takes 44-45%.

13.3 India: who drinks coffee and how

Domestic consumption by type



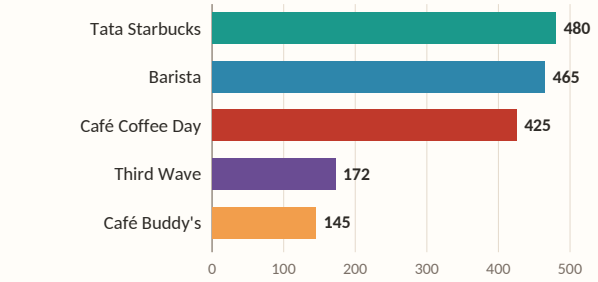
India drinks only about 1.6 million bags (about 95,000 tonnes) a year: **0.04 kg per person**, against a world average of 1.3 kg. Consumption has grown from about 80,000 tonnes in 2020 to 90-95,000 tonnes in 2024. Instant coffee, much of it blended with chicory, accounts for roughly two-thirds to three-quarters of the volume; Nestlé (Nescafé) and Hindustan Unilever (Bru) hold more than 70% of the instant market.

Geography matters: the south drinks filter coffee at home, while the north, east and west mostly drink instant, and increasingly cafés. The growth is young and urban: a 2025 World Coffee Portal survey of 4,000 Indians found 24% visit a coffee shop daily and 57% weekly, and a third of visits happen after 5 pm.

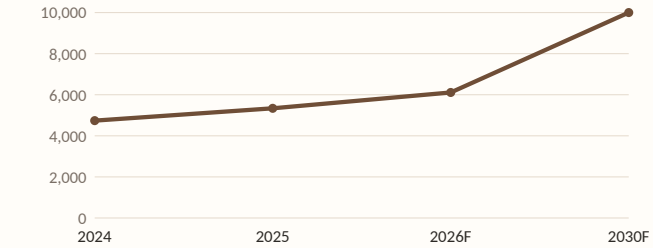
Source: USDA FAS (May 2026); Ken Research; Coffee Board consumption surveys; World Coffee Portal Project Café India 2025.

13.4 India's café boom

Largest branded chains (outlets, 2025)



Branded coffee-shop outlets in India



India had **5,339 branded coffee shops** in 2025, 600 more than a year earlier (+12.7%), and World Coffee Portal expects more than 10,000 by 2030. Specialty chains grow fastest: Blue Tokai and Third Wave each added 56 stores in a year.

Chain	Latest revenue	Note
Tata Starbucks	₹1,277 cr (FY25)	Loss ₹136 cr; ~₹2.7 cr per store; 500th store Apr 2026
Café Coffee Day	₹1,035 cr (FY25)	Restructured chain
Blue Tokai	₹341 cr (FY25)	+58%; FY24 loss ₹63 cr
Third Wave	₹241 cr (FY24)	+67%

The leaders are growing quickly but most still lose money at company level once head-office and expansion costs are included. Individual stores, however, can earn healthy margins, which is what Part 2 models.

Source: World Coffee Portal via Comunicaffe (Sep 2025); company filings and press summaries (Inc42, The Arc Web, LinkedIn).

13.5 Ahmedabad's coffee scene

Ahmedabad is a city of about 8-9 million with high incomes, a young workforce, a strong vegetarian food culture and, under Gujarat's prohibition law, no bars. Cafés and dessert places therefore do the work that pubs do elsewhere: they are where friends meet in the evening. The specialty scene has grown from Zen Café (2008) to national chains (Tata Starbucks, Third Wave, Blue Tokai, Barista, CCD) and local brands such as Slay Coffee, The Dark Roast, OCD Culture and Kaffa, with new specialty openings on Sindhu Bhavan Road (for example Beanora in 2025-26).

Hub	Character	Ground-floor rent (indicative)
Sindhu Bhavan Road (Bodakdev)	Premier F&B; high street; evening and weekend crowds; highest spend	₹85-220/sqft/month
Prahladnagar / Corporate Road	Office belt and upscale homes; dense chain competition	₹55-120
SG Highway, Thaltej, Vastrapur	Malls, offices, IIM-A belt; established cafés	₹80-180
CG Road, Navrangpura, Law Garden	Older commercial heart, students	₹70-160
Bopal, South Bopal, Shela	Fast-growing western suburbs, young families	₹40-80
GIFT City (Gandhinagar)	International finance hub; captive office workforce; few cafés	~₹110 + high maintenance
Sanand	Semiconductor and auto cluster 25 km west; Micron plant opened Feb 2026	~₹40-60 (estimate)

Source: 99acres, MagicBricks, housing.com listings (Oct 2026); Micron investor relations; GIFT City releases; Perfect Daily Grind.

13.6 India's retail coffee market structure

Segment	Leading names	Channels	Notes
Instant coffee	Nescafé (Nestlé), Bru (HUL), Tata, Continental (CCL), Davidoff	Supermarkets, kiranas, quick commerce	~65-75% of volume; GST cut to 5% in 2025
Filter coffee powder	Bru filter, Narasu's, Cothas, Leo, Coffee Day, Hatti Kaapi	South Indian retail, kiranas	Coffee-chicory blends; regional brands
Specialty roasters	Blue Tokai, Subko, KC Roasters, Black Baza, Araku, Corridor Seven, Devans and hundreds more	D2C, marketplaces, own cafés	Single estates, roast dates
Café chains	Tata Starbucks, Barista, CCD, Third Wave, Blue Tokai, Café Buddy's	Own outlets, delivery apps	5,339 branded outlets (2025)
Ready-to-drink	Cold brew cans and bottles from specialty brands and chains	Quick commerce, modern trade	Small but fast-growing

13.7 Ahmedabad's specialty cafés, compared

Café	Area	Beans	Roasting	Methods	For two
OCD Culture	Bodakdev	Rotating guest Indian roasters	Curates guest roasts	V60, AeroPress, espresso	~₹500
Blue Tokai	CG Road	Direct-sourced single estates	Own national roasteries	Espresso, pour-over, flights	₹700-900
Kaffa Coffee Roasters	Vastrapur / Thaltej	In-house roasted	Live roastery	Manual brew, cold brew	₹1,400
Roastery Cultúr	Bodakdev	Indian beans	Own roasting	Espresso, pour-over	₹800-1,500
Mainline Coffee	Navrangpura	Not specified	Buys in	Manual brew, barrel-aged cold brew	₹500-800
Third Wave Coffee	CG Road, Prahladnagar	~15 Indian estates	Central roasting	Espresso, cold brew	₹700-900
Tata Starbucks	Several	Global + India Estates blend	Central, dark style	Espresso, Frappuccino	₹800-1,000
Slay, The Dark Roast, Fick's, Grace…	Prahladnagar, Bodakdev	Mostly bought-in	Buy in	Espresso and iced drinks	₹500-1,400

Source: Café menus, aggregator listings (District, Swiggy, Zomato) and press up to Oct 2026; see the companion field guide (Final_Coff.pdf) for ratings.

The pattern is clear: Ahmedabad already has chains and a handful of serious specialty players, mostly in Bodakdev, Vastrapur and CG Road, but few cafés that combine **Indian estate traceability, an own-label roast and a full food menu** on the city's best evening street, and almost none in GIFT City. That gap shapes the Part 2 recommendation.

13.8 Trends to watch

- **Cold coffee everywhere**: iced lattes, cold brew and Indian 'cold coffee' lead café menus for eight months of Ahmedabad's year.
- **Indian estates as heroes**: roasters and cafés increasingly name the estate, variety and process; the Coffee Board's GI tags and awards support this.
- **Filter kaapi goes premium**: specialty kaapi made with estate arabica and little chicory is appearing on metro café menus.
- **Quick commerce and RTD**: 10-minute delivery has made cold brew cans and drip bags an impulse buy.
- **Plant milks and lower sugar**: oat milk for an extra ₹40-60 is now standard in specialty cafés.
- **Sustainability and traceability**: EUDR, shade-grown and bird-friendly claims, compostable packaging and spent-ground recycling.

13.9 Coffee culture in India

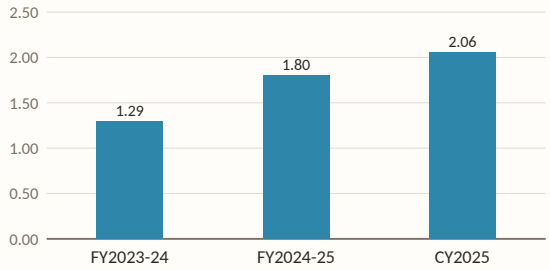
India remains a tea country: it drinks more than ten times as much tea as coffee by weight. Coffee culture developed in two separate streams. In the **south**, filter kaapi has been a daily ritual for a century, served at home and in 'darshinis', Udupi restaurants and the Indian Coffee House co-operatives. In the **north and west**, coffee meant instant (Nescafé arrived in the 1960s and became the household name) and, from 1996, cafés: Café Coffee Day's slogan 'A lot can happen over coffee' captured how cafés became meeting places for young Indians at a time when there were few others.

Since the 2010s a third stream has grown: specialty coffee led by Indian roasters who name estates, varieties and processes, and international chains that made Rs 300 drinks normal in big cities. In Gujarat, prohibition and a strong evening food culture give cafés an unusually central social role, while the state's large office and student populations in Ahmedabad and Gandhinagar sustain daytime trade.

Source	Measure	Estimate
IMARC	India café market	US$425 m (2025) to US$1.15 bn (2034), ~11% CAGR
Grand View Research	Coffee retail chains	US$564 m (2023), 8.1% CAGR
Expert Market Research	India coffee market	US$754 m (2025) to US$1.8 bn (2035), 9.1% CAGR
PIB / Coffee Board	Out-of-home coffee	15-20% CAGR to US$2.6-3.2 bn by 2028
World Coffee Portal	Branded coffee-shop outlets	5,339 (2025) to >10,000 (2030), 13.2% CAGR
NRAI	Indian food services (all)	~₹5.69 lakh crore, to ₹7.77 lakh crore by 2028

Source: Market sizing estimates differ in scope; use them for direction, not precision.

India's coffee export earnings (US$ bn)

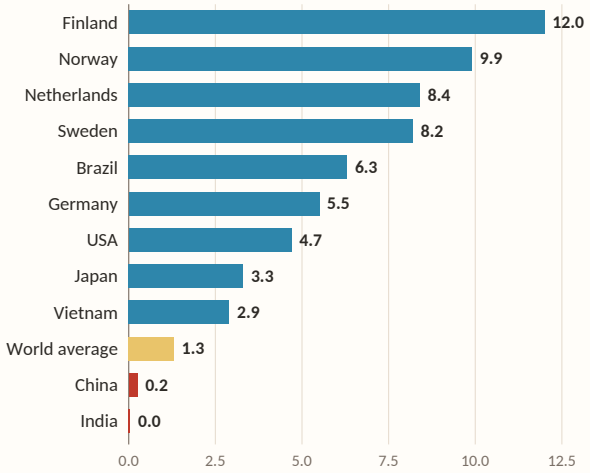


Source: PIB (Nov 2025); export reports (Jan 2026); USDA FAS (May 2026).

Export earnings jumped as world prices rose: from US$1.29 billion in FY2023-24 to a record US$1.8 billion in FY2024-25 and about US$2.06 billion in calendar 2025, even though export volume fell 4.5% in 2025. Value-added (mostly instant) coffee is about 38-41% of exports, a sign of India's strength in soluble manufacturing.

13.10 How much coffee the world drinks per person

Coffee per person per year (kg of green coffee, approximate)



Nordic countries drink the most coffee in the world, about 10-12 kg per person a year, or three to four cups a day. India drinks about 0.04 kg per person, roughly one three-hundredth of Finland's level and a thirtieth of the world average. China, starting from a similar base, has seen explosive café growth over the past decade.

The arithmetic of India's opportunity is simple: if Indian consumption rose to just 0.1 kg per head, the domestic market would more than double, absorbing part of today's exports. Café culture, specialty coffee and cold coffee are the channels through which that is happening in cities like Ahmedabad.

Source: Nordic countries and Germany: ICO and industry estimates (rounded); others: USDA FAS 2025/26 consumption divided by population; India: USDA Post (May 2026).

13.11 Who trades and roasts the world's coffee

Layer	Examples	Role
Green-coffee traders	Neumann Kaffee Gruppe, ECOM, ofi (Olam), Louis Dreyfus, Sucafina, Volcafe	Buy from exporters and co-operatives, finance, hedge, ship and blend to order
Global roasters	Nestlé, JDE Peet's, Starbucks, Lavazza, Tchibo, Strauss, Keurig Dr Pepper, UCC	A handful of firms roast a large share of the world's coffee
Instant manufacturers	Nestlé, JDE, Tata Consumer, CCL Products and others	India is a major contract manufacturer of instant coffee
Specialty importers	Small importers and roasters buying micro-lots	Direct trade, traceability, premiums
Exchanges	ICE US (arabica 'C'), ICE Europe (robusta)	Price discovery and hedging

Market power sits downstream: traders and roasters are far more concentrated than the world's 12.5 million farms. That is why value is captured mostly in consuming countries and in cafés, and why producer-country brands, including Indian estates and roasters selling at home, are a way for more value to stay at origin.

13.12 India's specialty coffee scene

Indian specialty coffee has grown from a handful of pioneers to hundreds of roasters in a decade. **Blue Tokai** (founded 2013) and **Third Wave Coffee** (2016) built national café chains on Indian estate coffee; **Araku Coffee** took tribal-grown Andhra arabica to a café in Paris (2017) before opening in India; roasters such as **Subko, KC Roasters, Black Baza, Corridor Seven and Devans** sell single estates, micro-lots and experimental processes online and through cafés. The Specialty Coffee Association of India runs national barista and brewing championships, and the Coffee Board's Flavour of India and Know Your Kaapi awards showcase estates.

Type	Examples	Model
Roaster-led cafés	Blue Tokai, Third Wave, Subko	Own roasting, estate stories, national scale
Estate and origin brands	Araku, estate-owned labels	Grow, process and sell their own coffee
Independent roasters	KC Roasters, Black Baza, Corridor Seven, Devans and many more	D2C, subscriptions, wholesale to cafés
Specialty cafés	Independents in every metro (OCD Culture, Kaffa, Roastery Cultúr in Ahmedabad)	Multi-roaster or own roast; experience-led
Ready-to-drink and instant	Cold-brew cans, specialty instant	Quick commerce and retail

Source: Company websites and press coverage; World Coffee Portal.

13.13 Chains versus independents

	National chains	A strong independent (Part 2)
Revenue per outlet	~₹2.7 crore a year (Tata Starbucks FY25: ₹1,277 cr / ~479 stores)	₹1.5-2.7 crore in the Part 2 models (Y3)
Ticket	₹350-550 per bill	₹320-360 average bill (Part 2)
Costs carried	Head office, regional teams, central roasting, brand marketing	Owner management, local marketing, toll roaster
Profitability	Store-level profit, but company-level losses while expanding (Starbucks India FY25 loss ₹136 cr; Blue Tokai FY24 loss ₹63 cr)	Store-level EBITDA 19-24% in Part 2's Base cases
Advantages	Brand pull, purchasing scale, prime sites	Agility, local menu, community, lower overheads

An independent cannot match a chain's purchasing or marketing, but it does not carry a head office either. A well-run single café on a good street can earn store-level margins similar to a chain store's, and that is the economic case for Part 2.

13.14 Outlook to 2030

Theme	What to expect	So what for a café
World prices	Record Brazilian crops in 2026/27 should ease prices from 2025's records, but low stocks and climate shocks keep volatility high	Budget a ±30% green-price range
Indian production	Slow growth (~2% a year); quality and fine robusta improve; new areas in Odisha and the North-East	More traceable Indian specialty supply
Indian demand	Café outlets to >10,000 by 2030 (13% a year); cold coffee, specialty and RTD lead	Market growth supports new cafés, but competition rises
Rules and trade	EUDR traceability, trade deals with the UK and EFTA; GST cuts on coffee products	Better export prices for estates; more documentation
Costs in cafés	Wages and rents rising faster than menu prices in many cities	Productivity and pricing discipline matter

Seed-to-cup quality checklist

Link	What to ask or check	Red flags
Variety & altitude	Which variety? How high? Shade-grown?	Unknown origin, 'blend' with no details
Picking	Selective? Ripeness checks? Floating?	Strip-picked, mixed ripeness
Processing	Washed / natural / honey; fermentation control; drying method and days	Sour, mouldy or 'ferment' notes
Grading	Grade, screen size, defect count, moisture 10-12%	Many broken or black beans
Green storage	Harvest date; hermetic bags; cool store	Past-crop, woody, papery
Roasting	Profile, roast date, roast level for the method	Scorched, baked or tipped beans
Packaging	Valve bag, roast date, 7-30 day window	Only a 'best before' date
Grinding	Burr grinder, ground to order, right size	Pre-ground, blade grinder
Water	Filtered, ~150 mg/L TDS	Hard tap water or pure RO
Recipe	Weighed dose and yield; correct temperature and time	Scoops and guesses
Serving	Clean cups and wands; milk at 60-65°C	Scalded milk, dirty equipment

The whole chain on one page



Figure 13.1. Seed to cup: who does what, how long it takes and what each step adds. Prices are Indian, October 2026.

Ten takeaways from Part 1 for a café founder

1. Quality is decided upstream: variety, altitude, picking and processing set the ceiling.
2. Indian washed arabica plus washed ('fine') robusta makes a clean, chocolatey, affordable espresso that suits milk drinks.
3. Green coffee is a volatile commodity; prices tripled from 2019 to 2025. Budget for swings.
4. Roast date matters: buy or roast little and often, and serve beans 7-30 days off roast.
5. A good grinder and treated water improve every cup more than an expensive machine.
6. Weigh everything; use recipes; dial in by taste every morning.
7. Offer a range: espresso and milk drinks for volume, pour-over for showcase lots, cold coffee for summer, and a specialty filter kaapi as a signature.
8. Coffee is only ~10% of a café's price; labour, rent and service make up most of the rest.
9. Toll roasting gives own-label control without a roaster's fixed costs; an in-house roastery needs wholesale volume.
10. India's café market is growing about 13% a year, and Ahmedabad's dry-state social life favours cafés.

Glossary of coffee terms

Term	Meaning
Arabica / robusta	The two commercial coffee species (Coffea arabica, C. canephora)
Bloom	Bubbling release of CO2 when hot water first wets fresh grounds
Body	Weight and texture of coffee in the mouth
Brix	Sugar content of cherry fruit, measured with a refractometer
Channelling	Water finding a weak path through an espresso puck
Cherry (grade)	Indian name for natural (dry-processed) coffee
Chicory	Roasted root blended with coffee in South Indian filter powder
Crema	Golden foam on espresso: emulsified oils and CO2
Cupping	Standardised professional tasting and scoring of coffee
Curing works	Indian dry mill: hulls, grades and sorts green coffee
Decoction	Strong coffee extract from a South Indian filter
Development time	Roast time after first crack (typically 15-25% of the roast)
Differential	Premium or discount of a physical lot over the futures price
Extraction yield	Share of the coffee's weight dissolved into the drink
First crack	Audible popping as beans expand at ~196-205°C
Fines / boulders	Very small / very large particles in a grind
Green coffee	Unroasted, processed and hulled coffee beans
Honey process	Pulped coffee dried with some mucilage left on
ICE 'C'	New York arabica futures contract, priced in US cents per lb

Term	Meaning
Kaapi Royale	Premium Indian washed robusta grade
Maillard reaction	Browning reaction between sugars and amino acids
Micro-lot	Small, separately processed lot from one farm block or variety
Monsooned	Indian process: coffee exposed to monsoon winds for 12-16 weeks
Mucilage	Sticky, sugary layer between pulp and parchment
Natural process	Cherries dried whole with the fruit on
Outturn	Share of clean coffee obtained from parchment or dried cherry
Parchment	Papery husk around the bean; Indian name for washed robusta
Peaberry (PB)	A single round bean in a cherry
Plantation (grade)	Indian name for washed arabica
Q grader	Cupper certified by the Coffee Quality Institute
Rate of rise	Speed at which bean temperature rises during roasting
Ratio	Coffee to water (or espresso yield) by weight, e.g. 1:16
Screen size	Sieve hole size used to grade beans (in 64ths of an inch)
Specialty coffee	Coffee scoring 80+ with minimal defects
TDS	Total dissolved solids: the strength of brewed coffee
Third wave	Coffee as craft: origin, traceability, careful roasting and brewing
Toll roasting	Paying a roaster to roast your own green coffee
Washed process	Fruit removed and mucilage fermented off before drying

PART 2

A specialty café in Ahmedabad

Inputs and outputs of the three-statement and FCFF valuation model: four sites, rent or own, three roasting routes and four bean mixes, in three scenarios.

CHAPTERS

14 The brief and the model
15 Where: choosing the hub
16 Inputs and assumptions
17 Comparing the configurations
18 The four models to pursue
19 Risk: scenarios and sensitivities
20 Recommendation and roadmap
A Appendix

14 The brief and the model

What we set out to decide, and how the Excel model is built

The question: should you open a specialty café in Prahladnagar, and if so how: rented or owned premises, which roasting route, which beans? And is there a better spot in Ahmedabad or Sanand? Part 2 answers with a full three-statement model and an FCFF valuation for every combination.

14.1 What the model decides

Decision	Options	#
Location	Prahladnagar (brief), Sindhu Bhavan Road, GIFT City, Sanand	4
Premises	Rent (lease + deposit) or own (buy the unit, 60% loan against property)	2
Roasting route	Buy roasted beans, toll-roast your own green coffee, or run an in-house 3 kg roastery that also sells wholesale	3
Bean mix	Value (commodity arabica/robusta), Specialty, Premium (40% micro-lot), House blend (recommended 15/70/15)	4
Scenario	Best, Base, Worst: demand ramp, footfall, price growth, green prices, capex overrun, wholesale volume	3

That is 96 combinations, each in three scenarios. Twenty named configurations are compared side by side, and four are examined in depth in Chapter 18.

14.2 How the workbook is built

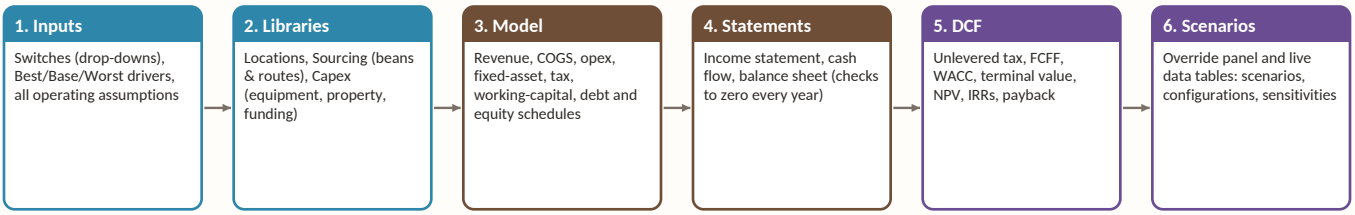


Figure 14.1. Workbook flow, following the CFI modelling guidelines and the '3-Statement Model' template structure.

- **Conventions** (CFI guidelines): blue = input, black = formula, green = link to another sheet; one font; units column on every row; every input entered once; statements contain links only; a circularity switch on the Cover (interest on opening balances by default).
- **Timeline**: Y0 = FY27 (fit-out and pre-opening, ending 31 March 2027), then ten operating years FY28-FY37. Annual periods, ₹ lakh.
- **Valuation**: free cash flow to the firm (FCFF) discounted at WACC, with a Gordon-growth terminal value at Y10 (4% nominal). Equity IRR uses free cash flow to equity. For owned premises the terminal value is split: operations after a notional market rent, plus the property's resale value net of selling costs and capital-gains tax.
- **Funding rule**: all-equity if the capital need is ₹15 lakh or less (per the brief); otherwise a term loan of 50% of capex at 11% over 5 years (rented) or 60% of capex and property at 10% over 10 years (owned). A cash-credit line covers shortfalls.
- **Checks**: balance sheet balances, cash never negative, sources equal uses, bean mixes add to 100%, WACC above growth. All pass for every configuration exported here.

14.3 Inside the schedules: the key formulas

Schedule	Logic
Bills per day	Steady-state bills x scenario footfall x bean-mix footfall effect x ramp-up x volume index, capped at seats x 6 turns / dine-in share
Revenue	Bills per day x operating days x average bill (grows with menu-price increases); split into beverages, food and retail beans; plus wholesale beans if in-house
Coffee cost	Coffee kg = beverages x 85% x 18 g; cost per kg = green cost per kg roasted x green-price index + roasting-route conversion cost x inflation
Other COGS	Milk and inputs per beverage; packaging per takeaway/delivery bill; food at 38%; retail beans at route-specific cost
Staff	Roster x location staffing factor x (1 + 15% load) x wage growth x volume scale-up
Occupancy	Area x rent per sqft x escalation (rented) or property tax (owned); CAM both
Capex & depreciation	Item list x site quantities x (1 + overrun); straight-line book depreciation by class and vintage; WDV tax depreciation; deferred tax on the difference
Tax	Book profit + book depreciation - tax depreciation; losses carried forward; 25.17%
Working capital	Receivables (delivery 7 days, wholesale 30), inventory (18 days; green stock 40 days if roasting own coffee), payables (15 days)
Debt	Term loan sized by the funding rule, equal principal repayments; cash credit draws automatically if cash would fall below the minimum
FCFF and value	EBIT - unlevered tax + depreciation - capex - change in working capital - deposit; discounted at WACC; Gordon terminal value at Y10

Using the Excel file

Open **Cafe_Financial_Model.xlsx**. On **Inputs**, pick location, premises, roasting, mix and scenario; the **Dashboard** updates instantly.

Change any blue cell: rents on **Locations**, bean prices and mix shares on **Sourcing**, equipment on **Capex**, salaries on **Inputs**.

The **Scenarios** sheet's data tables re-run the whole model for every configuration and sensitivity; **Configs** lets you define your own 20 combinations. If Excel is set to 'automatic except tables', press F9.

15 Where: choosing the hub

Screening Ahmedabad's café hubs and the four sites modelled

Eight hubs were screened on five criteria (scores 1-10, author's assessment from rent listings, competitor mapping and employment news). Weights: footfall 30%, spending power 20%, competition 15% (10 = little), rent affordability 15%, growth 20%.

Hub	Footfall	Spend	Competition	Rent	Growth	Score	Comment
GIFT City	6	8	9	6	9	7.4	Captive, fast-growing finance workforce; few quality cafés
Sindhu Bhavan Road	9	9	5	4	7	7.2	Premier F&B; street; highest footfall and spend
Sanand	4	5	9	9	8	6.5	Semiconductor & auto boom; cheap space; first mover
SG Highway / Thaltej	8	8	4	5	6	6.5	Malls and offices; crowded with chains
Bopal / Shela	5	6	7	8	7	6.4	Young western suburbs; good back-up option
Prahladnagar	7	7	4	6	6	6.2	Office belt + upscale homes; chains already present
Vastrapur / IIM Road	7	7	5	5	5	6	Students and residents; mature
CG Road / Navrangpura	7	6	5	6	5	6	Older commercial core; value-seeking crowd

Shortlist. GIFT City and Sindhu Bhavan Road score clearly above Prahladnagar. Sanand scores well on growth and cost but poorly on today's footfall; it is modelled because the semiconductor build-out (Micron's plant opened in February 2026; six approved projects in Gujarat with about 50,000 jobs) could make it a first-mover opportunity and a cheap base for a roastery. Prahladnagar is modelled as the brief's starting point. Bopal/Shela is the best unmodelled alternative.

15.1 The four sites, as modelled

Input	Prahladnagar	Sindhu Bhavan Road	GIFT City	Sanand
Café carpet area (sqft)	1,000	1,200	900	1,000
Seats (#)	40	52	36	40
Rent (₹/sqft/mth)	115	175	110	50
CAM / maintenance (₹/sqft/mth)	4	5	15	2
Security deposit (months)	4	6	6	3
Interiors, civil & MEP (₹/sqft)	2,600	3,000	2,600	2,000
Purchase price (own case) (₹/sqft)	17,500	24,000	14,000	6,500
Steady-state bills per day (#)	140	170	150	110
Average bill value (ex-GST, FY28) (₹)	320	360	330	260
Operating days per year (days)	360	360	290	350
Delivery share of café revenue (%)	15%	10%	6%	15%
Extra volume growth, Y2-Y6 (% p.a.)	0%	0%	4%	5%
Staffing factor (x)	1.00	1.20	0.95	0.90
Wholesale potential factor (x)	1.00	1.00	1.20	1.30
Property tax & society charges (own) (₹/sqft/yr)	35	40	30	15
Property appreciation (own) (% p.a.)	5%	6%	7%	6%

Source: Rents and prices: 99acres, MagicBricks, housing.com listings (Oct 2026). Demand inputs (bills/day, average bill) are judgement, triangulated with chain revenue per store (Starbucks India ~₹2.7 cr) and specialty-café spend levels in Ahmedabad listings.

15.2 What the sites imply

Base case, toll roasting, House blend	Y3 revenue	Y3 EBITDA %	Break-even bills/day	Capital: rent	Capital: own	NPV: rent	NPV: own	IRR: rent
Prahladnagar	181.4	19.5%	95	101.2	277.2	99.8	96.8	29.8%
Sindhu Bhavan Road	250.2	22.9%	108	129.9	413.3	207.2	240.8	36.3%
GIFT City	174.6	21.7%	106	97.6	220.8	159.5	198.0	35.3%
Sanand	124.1	9.5%	102	87.2	152.4	(2.2)	2.3	15.7%

Source: Model outputs, ₹ lakh. Capital = total capital requirement including pre-opening costs, deposit and a three-month ramp-up cash reserve.

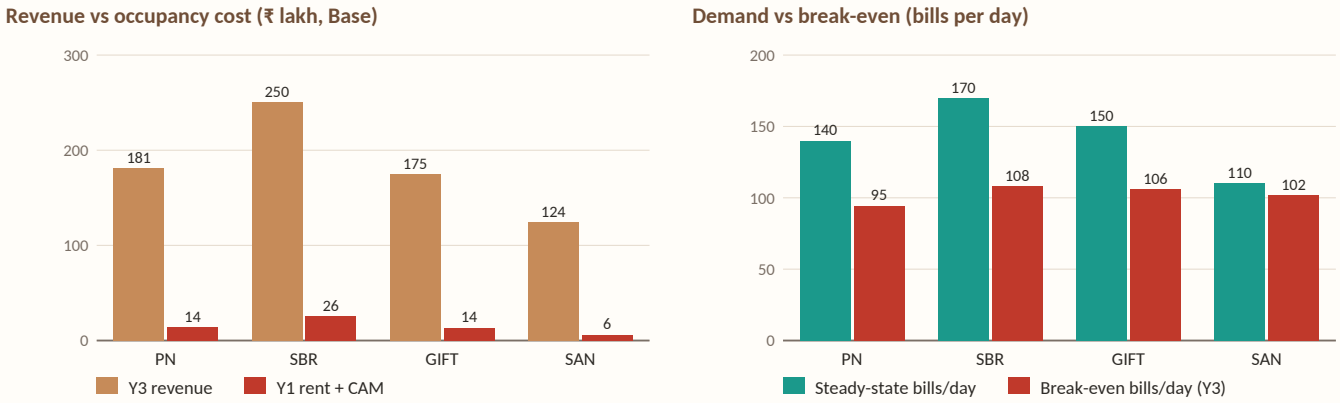


Figure 15.1. Sindhu Bhavan Road pays the most rent but earns far more; Sanand's break-even is close to its expected demand, leaving no margin for error.

Site	Competition today	Intensity	Demand pattern
Prahladnagar	Starbucks, Third Wave, Slay, The Dark Roast, Great Shakes	High	Office lunch and evening; strong delivery
Sindhu Bhavan Road	Many specialty and dessert cafés; new openings every year	Very high	Evening and weekend destination crowds
GIFT City	Few quality cafés; tenders for food zones	Low	Weekday office peaks; batch-brew and pre-order demand
Sanand	Mostly local tea stalls and QSR	Very low	Shift workers and managers; price-sensitive

Prahladnagar. Corporate Road / 100 ft road office & upscale-residential belt. Dense chain competition (Starbucks, Third Wave, Slay, The Dark Roast).

Sindhu Bhavan Road. Ahmedabad's premier F&B; high street (Bodakdev). Highest footfall and spend, evening/weekend peaks, highest rents.

GIFT City. India's IFSC: captive, fast-growing white-collar workforce (>12k, target 1 lakh jobs by 2030), few quality cafés, weekday demand.

Sanand. Industrial boom town: Micron ATMP (opened Feb 2026), Tata EV plants; ~50k semicon jobs planned in Gujarat. Cheap space, low ATV, first-mover play.

16 Inputs and assumptions

Drivers, menu, beans, people, costs, capex, funding and valuation inputs (all editable in the workbook)

16.1 Scenario drivers

Driver	Case	Y1	Y2	Y3	Y4	Y5	Y10
Ramp-up (% of steady-state bills)	Best	80%	95%	100%	100%	100%	100%
	Base	65%	85%	100%	100%	100%	100%
	Worst	50%	70%	88%	95%	95%	95%
Menu price increase	Best	0%	6%	6%	6%	6%	6%
	Base	0%	5%	5%	5%	5%	5%
	Worst	0%	4%	4%	4%	4%	4%
Volume growth after ramp (Y4+)	Best	0%	0%	0%	4%	4%	4%
	Base	0%	0%	0%	2%	2%	2%
	Worst	0%	0%	0%	0%	0%	0%
Green-coffee price index (Oct 2026 = 1.00)	Best	0.90	0.93	0.95	0.98	1.01	1.17
	Base	1.00	1.03	1.06	1.09	1.13	1.30
	Worst	1.25	1.29	1.33	1.37	1.41	1.63
Footfall factor on steady-state bills (x)	Best / Base / Worst	1.15 / 1.00 / 0.82					
Capex overrun (%)	Best / Base / Worst	0% / 5% / 15%					
Wholesale beans at maturity (in-house only) (kg/mth)	Best / Base / Worst	250 / 150 / 60					

Base assumes a café reaches 65% of steady-state bills in year 1 and 85% in year 2, raises prices 5% a year from year 2 and grows volume 2.5% a year from year 4. **Worst** is a struggling café: 82% of expected footfall, slow ramp to 95%, prices +3.5%, green coffee 25% dearer and capex 15% over budget. **Best** is a hit: 115% footfall and a fast ramp.

16.2 Revenue and menu

Item	Assumption
Revenue mix	Beverages 62%, retail beans 6% (+2% with own-label beans), food = remainder (30-32%)
Beverages	Average price ₹220 (FY28); 85% contain coffee; 18 g per coffee drink; milk & other inputs ₹24 per drink
Takeaway & delivery	Takeaway 25% of bills, delivery 6-15% of sales by site; packaging ₹15 per bill; aggregator commission 25% of delivery sales
Food & retail	Food cost 38% of food sales; bought-in retail bags at 30% margin; own-label retail at mix-specific prices
Capacity	Up to 6 seat turns a day (dine-in); demand above this is capped
Bean-mix effects	Average bill vs Specialty: Value -10%, Premium +5%, House +2%. Footfall: Value +5%, Premium -5%. Wholesale price per kg: Value ₹900, Specialty ₹1,450, Premium ₹2,300, House ₹1,700

16.3 Beans, mixes and roasting routes

Green coffee type	Green ₹/kg	Roasted, bought	Value	Specialty	Premium	House
Commodity arabica (Plantation A)	₹600	₹1,050	50%	–	–	–
Commodity robusta	₹440	₹900	50%	–	–	–
Fine robusta (washed)	₹520	₹1,150	–	20%	10%	15%
Estate specialty arabica (83-85)	₹950	₹1,700	–	75%	50%	70%
Micro-lot (86+)	₹2,200	₹3,200	–	5%	40%	15%
Blended green price, ₹/kg			₹520	₹926	₹1,407	₹1,073
Roasted cost: buy roasted			₹975	₹1,665	₹2,245	₹1,842
Roasted cost: toll roasting			₹824	₹1,308	₹1,880	₹1,482
Roasted cost: in-house (variable)			₹679	₹1,163	₹1,735	₹1,337
Coffee cost per 18 g shot (toll)			₹14.8	₹23.5	₹33.8	₹26.7

Roast loss 16%; toll fee ₹180/kg; freight ₹25/kg; in-house variable roasting ₹35/kg (LPG, power, bags) plus a roaster on ₹32,000 a month, AMC and 150 sqft of extra space. Green-coffee costs move with the scenario's price index (+3% a year in Base);

roasting and packing costs with inflation.

Source: Coffee Board indicative prices (Oct 2026); Indian green-coffee traders and roaster price lists; Part 1, Chapter 6.

16.4 People and operating costs

Role	#	₹/month	
Café manager	1	₹45,000	
Head barista	1	₹30,000	
Baristas	4	₹19,000	
Kitchen lead	1	₹28,000	
Kitchen assistants	2	₹16,000	
Service staff	3	₹15,000	
Housekeeping & dishwash	2	₹13,500	
Roaster & wholesale (in-house only)	1	₹32,000	In-house only

Cost line	Assumption
Payroll load (PF, ESI, bonus, meals)	15%
Wage growth	8% a year
Electricity	5 kWh/sqft/month at ₹9.5/kWh
Gas, water, cleaning	₹15,000/month
Marketing	5% of sales in Y1, then 3%
Repairs + AMC	1.5% of sales + ₹60k/yr
Card/UPI fees	0.7% of dine-in sales
Admin, accounts, insurance	₹3 lakh/yr
Rent escalation	5% a year; CAM by site
Pre-opening	2 months' rent, 1 month payroll, ₹3 lakh launch, ₹1.5 lakh licences

Headcount scales by a site staffing factor (SBR 1.2, GIFT 0.95, Sanand 0.9) and by half of any volume above steady state. Gujarat's minimum wage is about ₹13,300-13,900 a month (April 2026); specialty baristas earn ₹11,000-23,000.

16.5 Capex and funding

Item	Asset class	Prahladnagar ₹ lakh	SBR ₹ lakh
Interiors, civil, MEP & design	Interiors	27.3	37.8
HVAC (ductable / split ACs)	Equipment	4.7	5.7
Electrical upgrade, UPS & DG share	Equipment	2.6	2.6
Espresso machine, 2-group multi-boiler	Equipment	8.4	8.4
Grinders: 2 on-demand espresso + 1 filter	Equipment	4.4	4.4
Brew bar, cold-brew kit & water boiler	Equipment	1.6	1.6
Kitchen equipment & refrigeration	Equipment	5.8	5.8
Water treatment (RO + softener + remineralisation)	Equipment	1.1	1.1
Furniture & seating	Furniture	5.0	6.6
Crockery, glassware & smallwares	Furniture	1.6	1.6
Signage & branding	Furniture	2.1	2.1
POS, CCTV, audio & Wi-Fi	IT	1.9	1.9
In-house roastery adds (any site)	Roastery	₹13.0 lakh + 150 sqft fit-out	
Total café capex incl. 5% contingency		66.5	79.4

Book depreciation is straight-line (interiors 9 years, equipment 8, furniture 8, IT 4, roaster 10, building 60; land none). Tax depreciation uses Income-tax Act WDV block rates (furniture & fittings 10%, plant & machinery 15%, computers 40%, buildings 10%), which creates a deferred-tax balance. Maintenance capex is 1.5% of sales a year, with a 25% refurbishment of interiors and furniture in year 6. Corporate tax is 25.17% (s.115BAA) with losses carried forward.

Sources & uses, Base (₹ lakh)	Café capex	Property	Pre-opening	Deposit	Cash reserve	Total need	Loan	Equity	Debt %
C. Prahladnagar neighbourhood café	66.5	–	11.1	4.6	19.1	101.2	33.2	68.0	33%
A. Sindhu Bhavan Road flagship	79.4	–	13.6	12.6	24.3	129.9	39.7	90.2	31%
D. Sindhu Bhavan Road, owned unit	79.4	306.4	9.4	–	18.1	413.3	231.5	181.8	56%

Total capital need (₹ lakh, Base)	Rent: Buy	Rent: Toll	Rent: In-house	Own: Buy	Own: Toll	Own: In-house
Prahladnagar	101.2	101.2	122.4	277.2	277.2	324.7
Sindhu Bhavan Road	129.9	129.9	153.0	413.3	413.3	471.9
GIFT City	97.6	97.6	119.0	220.8	220.8	262.8
Sanand	87.2	87.2	106.5	152.4	152.4	181.4

Source: House blend. Capital = capex + property + pre-opening + deposit + cash reserve.

16.6 Cost of capital

Model	Debt / capital	Relevered beta	Cost of equity	Loan rate	WACC	Terminal g	TV share of value
A. Sindhu Bhavan Road flagship	31%	1.06	19.8%	11.0%	16.3%	4.0%	42%
B. GIFT City captive café	32%	1.08	20.0%	11.0%	16.2%	4.0%	44%
C. Prahladnagar neighbourhood café	33%	1.09	20.0%	11.0%	16.2%	4.0%	42%
D. Sindhu Bhavan Road, owned unit	56%	1.56	23.4%	10.0%	14.5%	4.0%	48%

Cost of equity = risk-free 7.29% (10-year G-sec, 8 Oct 2026) + relevered beta × India equity risk premium 7.08% (Damodaran, January 2026) + a 5% size and single-outlet premium, starting from an unlevered beta of 0.80. The resulting 20-24% cost of equity and 15-16% WACC are deliberately demanding for a first café.

17

Comparing the configurations

Twenty combinations of site, premises, roasting and beans, in three scenarios

Config	Site | premises | roasting | mix	Capital	NPV	Proj. IRR	Eq. IRR	Payback	Y3 EBITDA	BE bills	NPV Best	NPV Worst
PN-R-Buy	Prahladnagar | Rent | Buy roasted | Specialty	101.2	64.4	25.6%	29.3%	5.3 yrs	16.8%	99	252.5	(202.3)
PN-R-Toll	Prahladnagar | Rent | Toll roasting | Specialty	101.2	89.6	28.6%	32.9%	4.7 yrs	18.9%	96	289.3	(187.3)
PN-R-InH	Prahladnagar | Rent | In-house roastery | Specialty	122.4	40.0	20.9%	23.6%	6.8 yrs	13.7%	104	273.4	(284.1)
PN-O-Toll	Prahladnagar | Own | Toll roasting | Specialty	277.2	84.8	18.9%	24.3%	7.5 yrs	27.3%	77	322.3	(218.5)
PN-R-Toll-Val	Prahladnagar | Rent | Toll roasting | Value	101.2	53.6	24.2%	27.6%	5.7 yrs	16.7%	107	237.5	(205.0)
PN-R-Toll-Prm	Prahladnagar | Rent | Toll roasting | Premium	101.2	80.6	27.3%	31.3%	5.0 yrs	17.5%	93	282.6	(206.3)
PN-R-Toll-Hse	Prahladnagar | Rent | Toll roasting | House	101.2	99.8	29.8%	34.4%	4.6 yrs	19.5%	95	304.2	(180.9)
SBR-R-Buy	Sindhu Bhavan Road | Rent | Buy roasted | Specialty	129.9	158.1	32.5%	37.3%	4.3 yrs	20.3%	113	428.7	(196.0)
SBR-R-Toll	Sindhu Bhavan Road | Rent | Toll roasting | Specialty	129.9	192.7	35.2%	40.7%	4.0 yrs	22.4%	110	479.2	(175.4)
SBR-R-InH	Sindhu Bhavan Road | Rent | In-house roastery | Specialty	153.0	140.0	28.3%	32.1%	5.0 yrs	17.9%	118	461.7	(278.1)
SBR-O-Toll	Sindhu Bhavan Road | Own | Toll roasting | Specialty	413.3	223.5	21.7%	28.4%	6.7 yrs	33.5%	79	573.3	(195.8)
SBR-R-Toll-Hse	Sindhu Bhavan Road | Rent | Toll roasting | House	129.9	207.2	36.3%	42.2%	3.9 yrs	22.9%	108	500.3	(166.2)
GIFT-R-Buy	GIFT City | Rent | Buy roasted | Specialty	97.6	122.1	31.7%	36.2%	4.6 yrs	19.0%	111	323.9	(119.5)
GIFT-R-Toll	GIFT City | Rent | Toll roasting | Specialty	97.6	148.3	34.2%	39.3%	4.3 yrs	21.2%	108	362.2	(103.8)
GIFT-R-InH	GIFT City | Rent | In-house roastery | Specialty	119.0	98.7	26.4%	30.0%	5.8 yrs	15.0%	118	351.8	(202.2)
GIFT-O-Toll	GIFT City | Own | Toll roasting | Specialty	220.8	184.9	24.9%	32.6%	6.0 yrs	28.6%	88	439.5	(103.1)
GIFT-R-Toll-Hse	GIFT City | Rent | Toll roasting | House	97.6	159.5	35.3%	40.6%	4.2 yrs	21.7%	106	378.5	(96.7)
SAN-R-Toll	Sanand | Rent | Toll roasting | Specialty	87.2	(10.2)	14.2%	15.8%	8.6 yrs	8.8%	104	131.8	(201.3)
SAN-R-InH	Sanand | Rent | In-house roastery | Specialty	106.5	(56.1)	7.1%	3.9%	>10 yrs	3.9%	111	128.6	(292.0)
SAN-O-Toll	Sanand | Own | Toll roasting | Specialty	152.4	(6.8)	14.1%	17.0%	9.2 yrs	14.1%	93	155.6	(213.8)

Source: Model outputs, ₹ lakh, Base scenario unless stated. Payback counted from opening (start of Y1). BE bills = break-even bills per day in Y3.

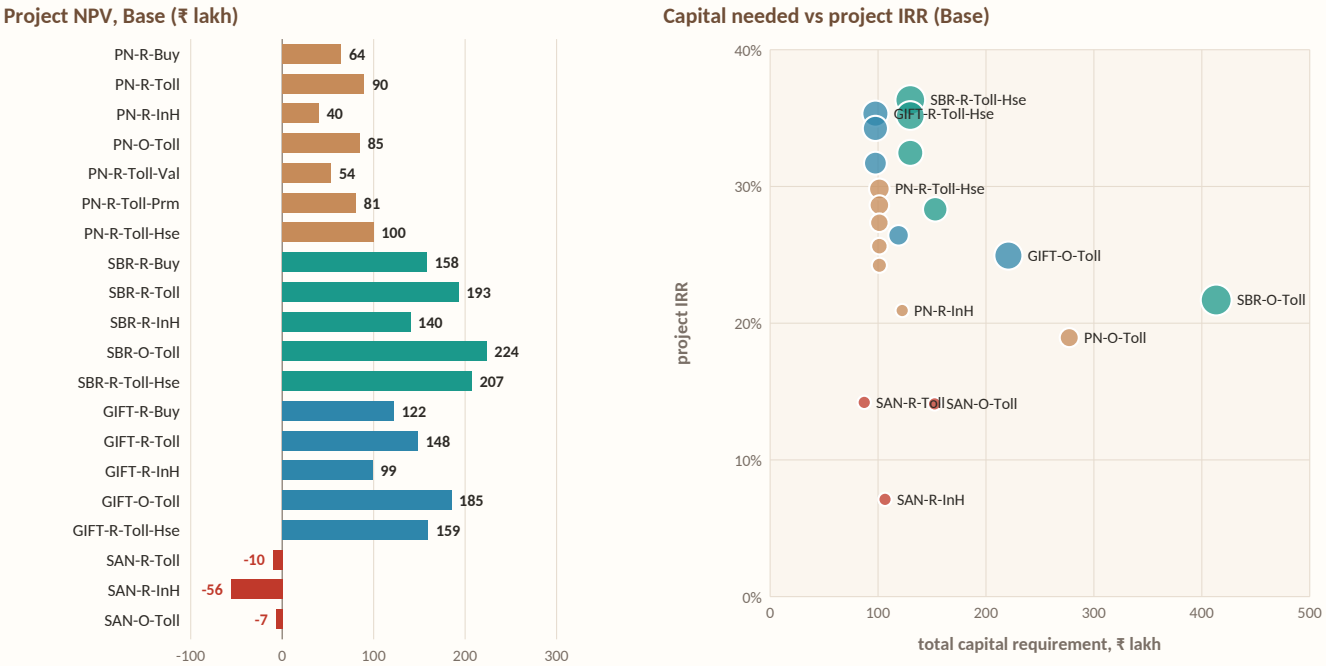


Figure 17.1. NPV by configuration (left) and capital needed versus project IRR (right; bubble size = NPV). Teal = Sindhu Bhavan Road, blue = GIFT City, caramel = Prahladnagar, red = Sanand.

17.1 What drives the ranking

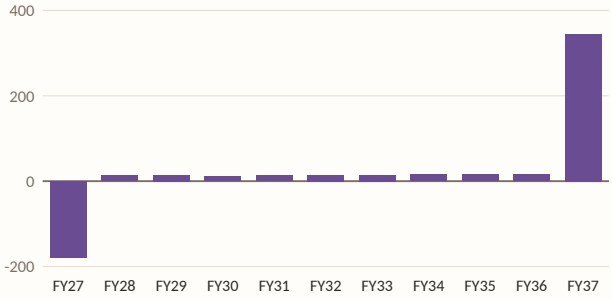
- **Location dominates.** With the same roasting route and beans, NPV ranges from about ₹-10 lakh (Sanand) to ₹193 lakh (Sindhu Bhavan Road). Footfall and average bill matter far more than rent: SBR pays the highest rent and still wins.
- **Toll roasting beats both alternatives at every site.** It is cheaper per kilo than buying roasted beans (₹1,308 against ₹1,665/kg for the Specialty mix) and lifts retail-bean margins, without the fixed costs of a roastery.
- **The House blend wins on beans**: a 15% micro-lot share supports a slightly higher average bill at a coffee cost of about ₹27 a shot. Value beans save cost but lose more in price and image; Premium (40% micro-lot) costs more than it earns.
- **Owning raises NPV at the best sites but cuts returns.** It roughly triples the capital needed (2.3-3.2 times), cuts the project IRR by about a third and depends on property appreciation.
- **Sanand is not ready.** It is NPV-negative in the Base case and positive only if the Best case's footfall arrives.

17.2 Rent versus own

Toll roasting, House blend, Base	Capital: rent	Capital: own	NPV: rent	NPV: own	IRR: rent	IRR: own	Eq. IRR: rent	Eq. IRR: own
Prahladnagar	101.2	277.2	99.8	96.8	29.8%	19.5%	34.4%	25.1%
Sindhu Bhavan Road	129.9	413.3	207.2	240.8	36.3%	22.2%	42.2%	29.2%
GIFT City	97.6	220.8	159.5	198.0	35.3%	25.6%	40.6%	33.4%
Sanand	87.2	152.4	(2.2)	2.3	15.7%	15.0%	17.6%	18.3%

Prahladnagar: extra cash flow from buying (₹ lakh)

Y10 includes the difference in terminal values



Buying means paying about ₹1.7-2.8 crore more up front to save rent, and getting the property back at the end. For Prahladnagar the **incremental IRR of buying is about 12.5%**: above the 10% cost of the property loan, but well below the ~20% cost of equity. At Sindhu Bhavan Road and GIFT City, faster appreciation makes owning add NPV.

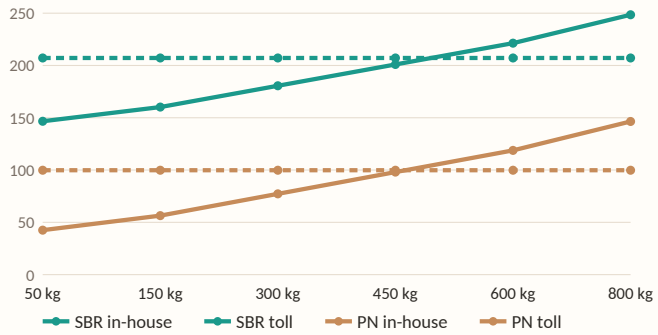
But owning concentrates two risks (café and property) in one asset, ties up ₹1.0-1.8 crore of equity, and makes the café harder to relocate if the street falls out of fashion. **Verdict: rent first**; consider buying only for a proven site with spare capital.

17.3 Roasting route and bean mix

Prahladnagar, rented, Base	NPV Value	NPV Specialty	NPV Premium	NPV House	Y3 margin Value	Y3 margin Specialty	Y3 margin Premium	Y3 margin House
Buy roasted	37.0	64.4	53.4	72.8	15.2%	16.8%	15.2%	17.2%
Toll roasting	53.6	89.6	80.6	99.8	16.7%	18.9%	17.5%	19.5%
In-house roastery	2.2	40.0	54.7	56.4	12.2%	13.7%	13.0%	14.3%

When does an in-house roastery pay? NPV (₹ lakh)

Wholesale beans sold per month at maturity



An in-house roastery adds about ₹13 lakh of equipment, a roaster's salary, AMC and extra space. Roasting the café's own needs (about 1,500 kg a year including retail bags) saves only ₹145/kg against toll roasting, which does not cover those fixed costs.

It pays once the roastery also sells **about 463-496 kg a month wholesale** to offices, hotels and other cafés: the crossover point in the chart. That makes it a phase-two decision for a brand with several outlets and wholesale accounts, not a launch decision.

17.4 Roasting route and bean mix at Sindhu Bhavan Road

SBR NPV ₹ lakh	Value	Specialty	Premium	H
Buy roasted	121	158	142	
Toll roasting	144	193	179	
In-house roastery	90	140	149	

SBR Y3 EBITDA %	Value	Specialty	Premium	House
Buy roasted	18.9%	20.3%	18.7%	20.6%
Toll roasting	20.3%	22.4%	21.0%	**22.9%**
In-house roastery	16.6%	17.9%	16.8%	18.4%

At Sindhu Bhavan Road the ranking is the same as at Prahladnagar: toll roasting with the House blend is best (boxed). Premium beans earn more per bill but lose footfall; Value beans lose price. In-house roasting narrows the gap only with the Premium mix, where wholesale margins per kilo are highest.

17.5 Every configuration in every scenario

NPV ₹ lakh	Best	Base	Worst
PN-R-Buy	253	64	-202
PN-R-Toll	289	90	-187
PN-R-InH	273	40	-284
PN-O-Toll	322	85	-219
PN-R-Toll-Val	237	54	-205
PN-R-Toll-Prm	283	81	-206
PN-R-Toll-Hse	304	100	-181
SBR-R-Buy	429	158	-196
SBR-R-Toll	479	193	-175
SBR-R-InH	462	140	-278
SBR-O-Toll	573	224	-196
SBR-R-Toll-Hse	500	207	-166
GIFT-R-Buy	324	122	-120
GIFT-R-Toll	362	148	-104
GIFT-R-InH	352	99	-202
GIFT-O-Toll	439	185	-103
GIFT-R-Toll-Hse	378	159	-97
SAN-R-Toll	132	-10	-201
SAN-R-InH	129	-56	-292
SAN-O-Toll	156	-7	-214

Equity IRR	Best	Base	Worst
PN-R-Buy	56%	29%	n.m.
PN-R-Toll	61%	33%	n.m.
PN-R-InH	50%	24%	n.m.
PN-O-Toll	39%	24%	n.m.
PN-R-Toll-Val	54%	28%	n.m.
PN-R-Toll-Prm	59%	31%	n.m.
PN-R-Toll-Hse	63%	34%	n.m.
SBR-R-Buy	65%	37%	n.m.
SBR-R-Toll	69%	41%	n.m.
SBR-R-InH	59%	32%	n.m.
SBR-O-Toll	42%	28%	-2%
SBR-R-Toll-Hse	71%	42%	n.m.
GIFT-R-Buy	61%	36%	n.m.
GIFT-R-Toll	65%	39%	n.m.
GIFT-R-InH	54%	30%	n.m.
GIFT-O-Toll	48%	33%	1%
GIFT-R-Toll-Hse	67%	41%	n.m.
SAN-R-Toll	39%	16%	n.m.
SAN-R-InH	34%	4%	n.m.
SAN-O-Toll	35%	17%	n.m.

Figure 17.2. Every configuration in every scenario: NPV (left) and equity IRR (right; n.m. = not meaningful because cash flows never turn positive).

The spread between scenarios is wide for every configuration: a single café is a high-variance bet. The **Worst case** (82% of expected footfall with a slow ramp) makes every configuration NPV-negative, because a café that never reaches break-even keeps losing money as wages outgrow prices. GIFT City has the mildest Worst case, thanks to its growing captive workforce; Sindhu Bhavan Road has the best Best case.

In practice nobody runs a loss-making café for ten years. The model therefore also reports the **cash lost if the café is shut at the end of year 3** (after equipment resale, deposit refund and working-capital release): about ₹107-118 lakh in the Worst case for the rented toll-roasted models. That is close to the whole investment, and it is the real downside to plan for.

18 The four models to pursue

Location, premises, roasting and beans: the shortlist, in depth

Four configurations stand out. All use **toll roasting** and the **House blend**; three rent their premises. They are ranked by value created per rupee of risk capital, with Prahladnagar kept as the brief's reference point.

Model (all toll roasting, House blend)	Capital	Equity	NPV	Proj. IRR	Eq. IRR	Payback	Y3 revenue	Y3 EBITDA %	NPV Best	NPV Worst
A. Sindhu Bhavan Road flagship	129.9	90.2	207.2	36.3%	42.2%	3.9 yrs	250.2	22.9%	500.3	(166.2)
B. GIFT City captive café	97.6	66.2	159.5	35.3%	40.6%	4.2 yrs	174.6	21.7%	378.5	(96.7)
C. Prahladnagar neighbourhood café	101.2	68.0	99.8	29.8%	34.4%	4.6 yrs	181.4	19.5%	304.2	(180.9)
D. Sindhu Bhavan Road, owned unit	413.3	181.8	240.8	22.2%	29.2%	6.6 yrs	250.2	33.8%	598.8	(187.5)

Source: Model outputs, Base scenario unless stated (₹ lakh).

Criterion (10 = best)	A	B	C	D
Capital needed	7	8.5	8	3
Value created (NPV)	8.5	7.5	6	9
Return on capital (IRR)	9	9	8	6
Downside resilience	6.5	8	5.5	6
Competition	5	9	5	5
Execution effort	6.5	7	7.5	5
Strategic fit (brand, phase 2)	9	8	6.5	6
Average score	7.4	8.1	6.6	5.7

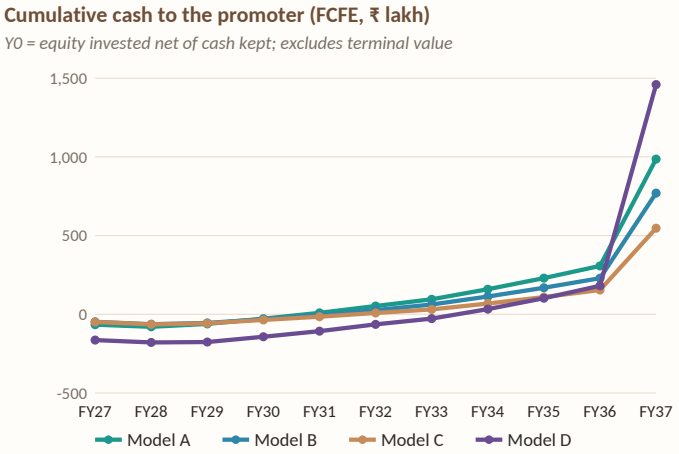


Figure 18.1. Qualitative ranking (left; author's assessment combining the model with site research) and cumulative equity cash flows (right). Model D needs the most money and takes longest to pay back.

Model A: Sindhu Bhavan Road flagship

Sindhu Bhavan Road | Rent | Toll roasting | House blend

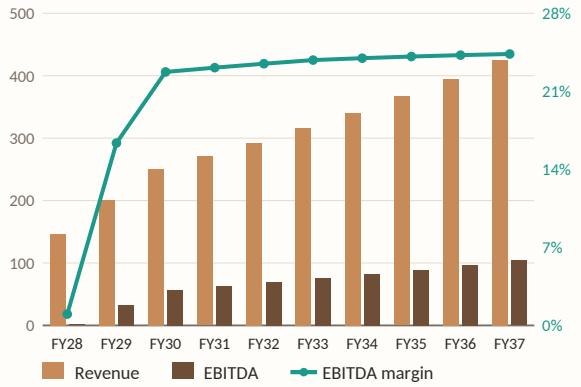
₹130 L
total capital need

₹207 L
project NPV at WACC

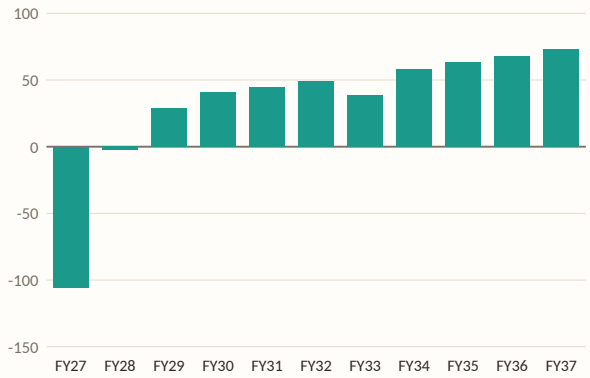
36.3% / 42.2%
project / equity IRR

3.9 yrs
payback from opening

Revenue and EBITDA, Base (₹ lakh)



Free cash flow to firm, Base (₹ lakh)



Scenario	NPV	Proj. IRR	Eq. IRR	Payback	Y3 revenue	Y3 EBITDA %	Bills/day Y3	Break-even	Peak overdraft	Loss if shut Y3
Best	500.3	57.6%	71.2%	2.4 yrs	293.3	28.3%	197	108	–	nil
Base	207.2	36.3%	42.2%	3.9 yrs	250.2	22.9%	172	108	–	nil
Worst	(166.2)	n.m.	n.m.	>10 yrs	175.5	3.9%	124	116	181.8	117.7

Why model A

Highest NPV of the rented options: Ahmedabad's strongest evening and weekend café street, the highest average bill (₹367 incl. House-blend premium) and 170 bills a day at steady state.

Rent is the city's highest (₹175/sqft) but is still only about 11% of sales.

Risks: intense competition (new specialty openings every few months), fit-out quality expectations, parking. Mitigate with a strong signature menu (specialty kaapi, cold brew) and retail beans.

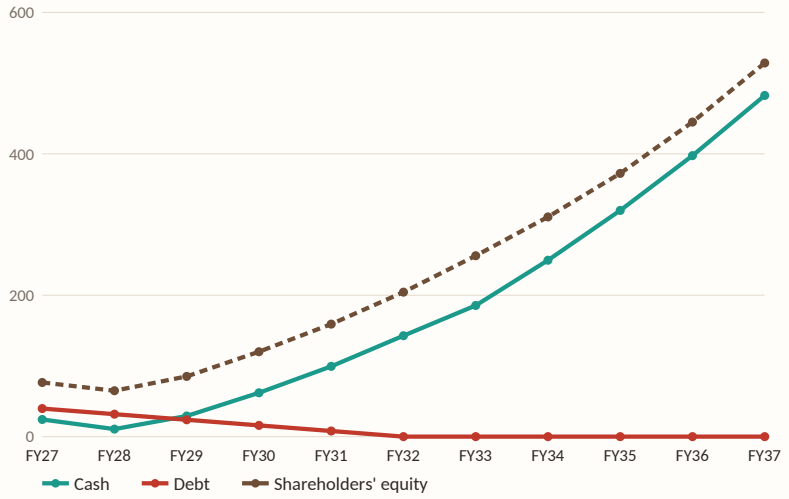
Model A in detail: statements, funding and unit economics

Condensed statements, Base (₹ lakh)	FY27	FY28	FY29	FY30	FY32	FY37
Bills per day	–	110.5	145.2	171.7	182.2	207.2
Revenue	**–**	**146.1**	**201.6**	**250.2**	**292.8**	**424.9**
Cost of goods sold	–	(44.0)	(60.4)	(74.6)	(86.3)	(122.0)
Staff	–	(46.9)	(50.6)	(54.9)	(66.0)	(103.9)
Rent, CAM & property tax	–	(25.9)	(27.2)	(28.6)	(31.5)	(40.2)
Other opex & pre-opening	(13.6)	(27.8)	(30.2)	(34.9)	(39.8)	(54.8)
EBITDA	**(13.6)**	**1.5**	**33.2**	**57.2**	**69.1**	**104.0**
Depreciation	–	(9.6)	(10.1)	(10.7)	(11.8)	(8.2)
Net interest	–	(3.4)	(3.1)	(1.5)	3.1	15.9
Tax	–	(0.1)	0.3	(10.2)	(15.2)	(28.1)
Profit after tax	**(13.6)**	**(11.7)**	**20.3**	**34.9**	**45.2**	**83.6**
Cash from operations	(13.6)	(3.5)	29.5	44.5	55.7	91.5
Capex & deposit	(92.0)	(2.2)	(3.0)	(3.8)	(4.4)	(6.4)
Free cash flow to firm	**(105.6)**	**(2.3)**	**29.1**	**40.7**	**49.0**	**73.2**
Closing cash	24.3	10.6	29.2	62.0	142.7	482.6
Fixed assets (net)	79.4	72.0	64.9	58.0	43.2	20.5
Net working capital	–	1.6	2.2	2.7	3.0	4.2
Debt	39.7	31.8	23.8	15.9	–	–
Shareholders' equity	**76.6**	**64.9**	**85.2**	**120.1**	**204.4**	**528.5**
Balance-sheet check	–	–	–	–	–	–

Unit economics, Y3 Base	
Revenue per bill	₹405
Ingredients & packaging	₹121
Staff	₹89
Rent, CAM, property tax	₹46
Utilities, marketing, fees, other	₹57
EBITDA per bill	**₹93**
Coffee cost per beverage	₹24.4
Bills per day / break-even	172 / 108

Sources & uses (₹ lakh)	
Café & roastery capex	79.4
Property incl. stamp duty	–
Pre-opening expenses	13.6
Rent deposit	12.6
Cash reserve	24.3
Total need	**129.9**
Term loan	39.7
Promoter equity	90.2
Loan rate / tenor	11% / 5 yrs
WACC / cost of equity	16.3% / 19.8%

Balance sheet trajectory, Base (₹ lakh)



Cash accumulates because the model pays no dividends (payout 0%); in practice surplus cash would fund a second outlet or be distributed. Equity IRR assumes free cash flow to equity is paid out each year.

Model B: GIFT City captive café

GIFT City | Rent | Toll roasting | House blend

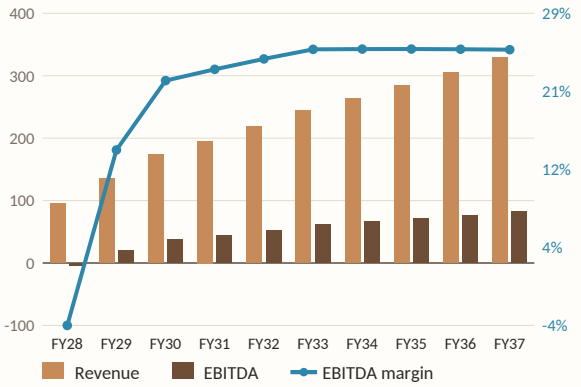
₹98 L
total capital need

₹159 L
project NPV at WACC

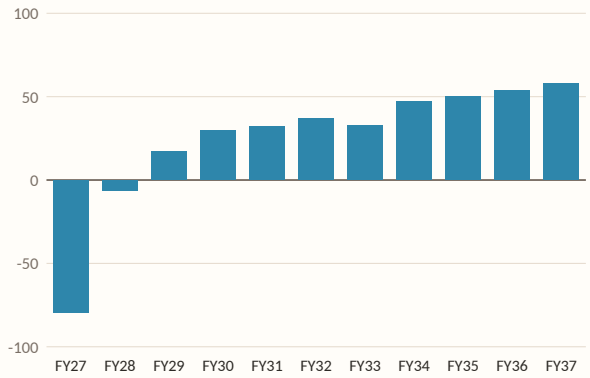
35.3% / 40.6%
project / equity IRR

4.2 yrs
payback from opening

Revenue and EBITDA, Base (₹ lakh)



Free cash flow to firm, Base (₹ lakh)



Scenario	NPV	Proj. IRR	Eq. IRR	Payback	Y3 revenue	Y3 EBITDA %	Bills/day Y3	Break-even	Peak overdraft	Loss if shut Y3
Best	378.5	54.7%	66.8%	2.7 yrs	204.6	27.1%	187	107	–	nil
Base	159.5	35.3%	40.6%	4.2 yrs	174.6	21.7%	162	106	–	12.2
Worst	(96.7)	-10.5%	n.m.	>10 yrs	122.4	3.1%	117	111	93.5	96.3

Why model B

Lowest capital and the most resilient model: captive office demand, few quality cafés and a workforce targeted to reach 1 lakh by 2030 (+4% a year volume premium in Y2-Y6).

Trade-offs: weekday-heavy (290 operating-day equivalents), high CAM in Grade-A towers, and leasing via GIFT's RFP processes.

Batch brew and pre-ordering suit office peaks; wholesale coffee to office pantries is a natural add-on.

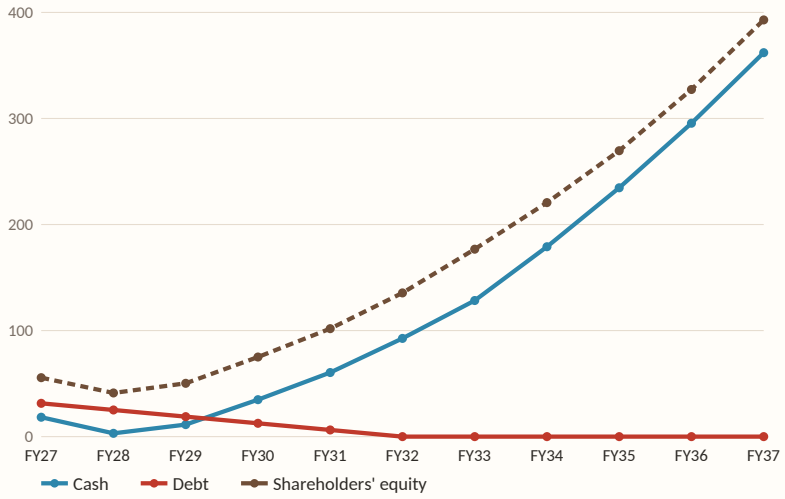
Model B in detail: statements, funding and unit economics

Condensed statements, Base (₹ lakh)	FY27	FY28	FY29	FY30	FY32	FY37
Bills per day	–	97.5	132.6	162.2	184.4	216.9
Revenue	**–**	**95.2**	**135.9**	**174.6**	**218.7**	**328.5**
Cost of goods sold	–	(28.6)	(40.7)	(51.9)	(64.4)	(94.1)
Staff	–	(37.1)	(40.1)	(45.0)	(56.3)	(90.7)
Rent, CAM & property tax	–	(13.5)	(14.2)	(14.9)	(16.4)	(20.9)
Other opex & pre-opening	(10.6)	(19.7)	(21.4)	(24.9)	(29.3)	(40.8)
EBITDA	**(10.6)**	**(3.8)**	**19.6**	**37.9**	**52.4**	**81.9**
Depreciation	–	(7.7)	(8.0)	(8.4)	(9.1)	(6.0)
Net interest	–	(2.7)	(2.6)	(1.6)	1.7	11.8
Tax	–	(0.2)	0.2	(3.0)	(11.3)	(22.1)
Profit after tax	**(10.6)**	**(14.4)**	**9.1**	**24.8**	**33.7**	**65.6**
Cash from operations	(10.6)	(7.5)	16.6	32.4	41.7	71.5
Capex & deposit	(68.7)	(1.4)	(2.0)	(2.6)	(3.3)	(4.9)
Free cash flow to firm	**(79.3)**	**(6.2)**	**17.2**	**29.6**	**37.2**	**57.7**
Closing cash	18.3	3.1	11.3	34.8	92.6	362.1
Fixed assets (net)	62.8	56.4	50.5	44.6	32.8	15.6
Net working capital	–	1.0	1.4	1.7	2.1	3.0
Debt	31.4	25.1	18.8	12.6	–	–
Shareholders' equity	**55.6**	**41.2**	**50.3**	**75.1**	**135.5**	**392.9**
Balance-sheet check	–	–	–	–	–	–

Unit economics, Y3 Base	
Revenue per bill	₹371
Ingredients & packaging	₹110
Staff	₹96
Rent, CAM, property tax	₹32
Utilities, marketing, fees, other	₹53
EBITDA per bill	**₹80**
Coffee cost per beverage	₹24.4
Bills per day / break-even	162 / 106

Sources & uses (₹ lakh)	
Café & roastery capex	62.8
Property incl. stamp duty	–
Pre-opening expenses	10.6
Rent deposit	5.9
Cash reserve	18.3
Total need	**97.6**
Term loan	31.4
Promoter equity	66.2
Loan rate / tenor	11% / 5 yrs
WACC / cost of equity	16.2% / 20.0%

Balance sheet trajectory, Base (₹ lakh)



Cash accumulates because the model pays no dividends (payout 0%); in practice surplus cash would fund a second outlet or be distributed. Equity IRR assumes free cash flow to equity is paid out each year.

Model C: Prahladnagar neighbourhood café

Prahladnagar | Rent | Toll roasting | House blend

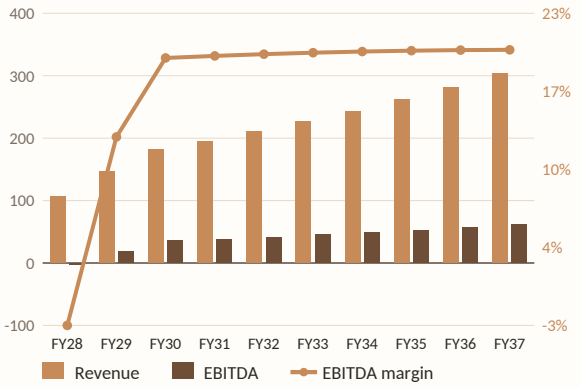
₹101 L
total capital need

₹100 L
project NPV at WACC

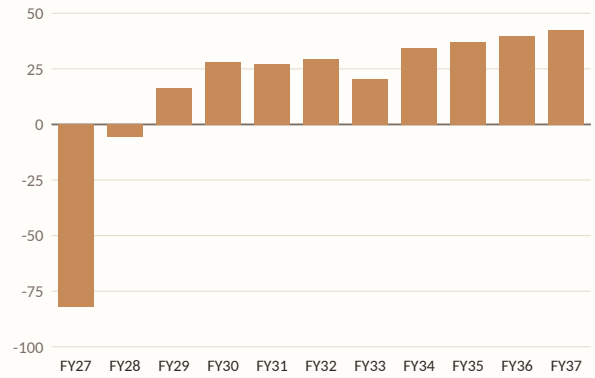
29.8% / 34.4%
project / equity IRR

4.6 yrs
payback from opening

Revenue and EBITDA, Base (₹ lakh)



Free cash flow to firm, Base (₹ lakh)



Scenario	NPV	Proj. IRR	Eq. IRR	Payback	Y3 revenue	Y3 EBITDA %	Bills/day Y3	Break-even	Peak overdraft	Loss if shut Y3
Best	304.2	50.6%	62.6%	2.7 yrs	212.6	24.9%	161	95	–	nil
Base	99.8	29.8%	34.4%	4.6 yrs	181.4	19.5%	140	95	–	16.9
Worst	(180.9)	n.m.	n.m.	>10 yrs	127.2	-0.4%	101	102	267.4	106.8

Why model C

The brief's site works: positive NPV, about 30% project IRR and payback within five years in the Base case.
But margins are thinner (19.5% EBITDA in Y3), Starbucks, Third Wave and others are already there, and the Worst case loses money.
Treat as viable but third choice, or as a second outlet once the brand is known.

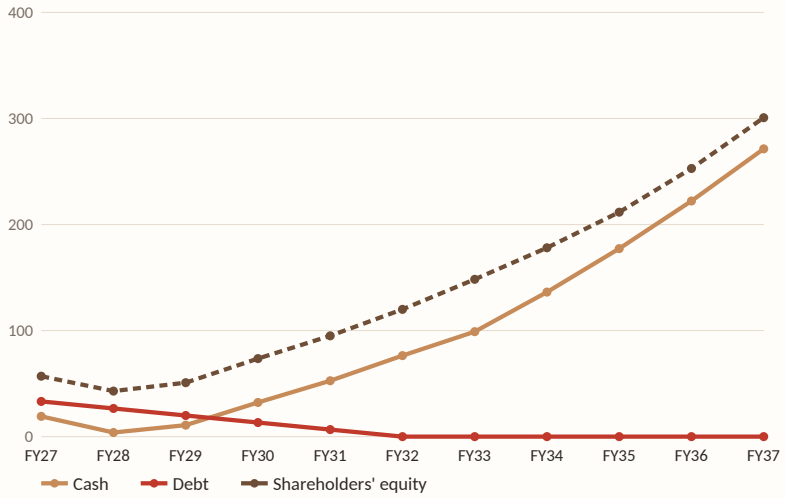
Model C in detail: statements, funding and unit economics

Condensed statements, Base (₹ lakh)	FY27	FY28	FY29	FY30	FY32	FY37
Bills per day	–	91.0	119.0	140.0	147.1	166.4
Revenue	**–**	**106.9**	**146.8**	**181.4**	**210.1**	**303.4**
Cost of goods sold	–	(32.7)	(44.6)	(54.8)	(62.8)	(88.3)
Staff	–	(39.1)	(42.2)	(45.6)	(54.5)	(85.4)
Rent, CAM & property tax	–	(14.3)	(15.0)	(15.7)	(17.4)	(22.2)
Other opex & pre-opening	(11.1)	(23.8)	(26.1)	(30.0)	(33.9)	(46.4)
EBITDA	**(11.1)**	**(2.8)**	**19.0**	**35.3**	**41.6**	**61.1**
Depreciation	–	(8.2)	(8.5)	(8.9)	(9.6)	(6.0)
Net interest	–	(2.9)	(2.8)	(1.8)	1.4	8.9
Tax	–	(0.2)	0.2	(1.9)	(8.4)	(16.1)
Profit after tax	**(11.1)**	**(14.0)**	**7.9**	**22.7**	**25.0**	**47.9**
Cash from operations	(11.1)	(7.0)	15.8	30.8	33.5	53.7
Capex & deposit	(71.1)	(1.6)	(2.2)	(2.7)	(3.2)	(4.6)
Free cash flow to firm	**(82.1)**	**(5.7)**	**16.3**	**27.9**	**29.3**	**42.5**
Closing cash	19.1	3.9	10.8	32.2	76.4	271.4
Fixed assets (net)	66.5	59.9	53.6	47.4	34.4	14.7
Net working capital	–	1.3	1.7	2.1	2.4	3.3
Debt	33.2	26.6	19.9	13.3	–	–
Shareholders' equity	**57.0**	**42.9**	**50.9**	**73.6**	**120.0**	**300.7**
Balance-sheet check	–	–	–	–	–	–

Unit economics, Y3 Base	
Revenue per bill	₹360
Ingredients & packaging	₹109
Staff	₹90
Rent, CAM, property tax	₹31
Utilities, marketing, fees, other	₹59
EBITDA per bill	**₹70**
Coffee cost per beverage	₹24.4
Bills per day / break-even	140 / 95

Sources & uses (₹ lakh)	
Café & roastery capex	66.5
Property incl. stamp duty	–
Pre-opening expenses	11.1
Rent deposit	4.6
Cash reserve	19.1
Total need	**101.2**
Term loan	33.2
Promoter equity	68.0
Loan rate / tenor	11% / 5 yrs
WACC / cost of equity	16.2% / 20.0%

Balance sheet trajectory, Base (₹ lakh)



Cash accumulates because the model pays no dividends (payout 0%); in practice surplus cash would fund a second outlet or be distributed. Equity IRR assumes free cash flow to equity is paid out each year.

Model D: Sindhu Bhavan Road, owned unit

Sindhu Bhavan Road | Own | Toll roasting | House blend

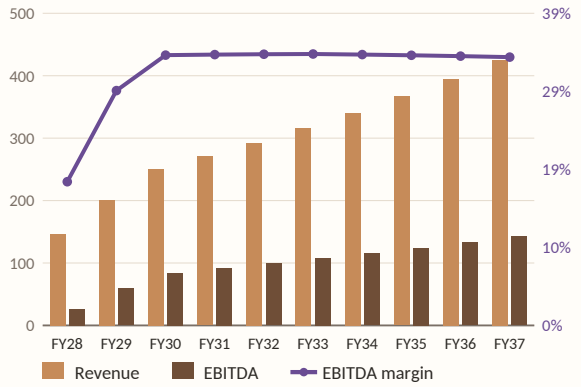
₹413 L
total capital need

₹241 L
project NPV at WACC

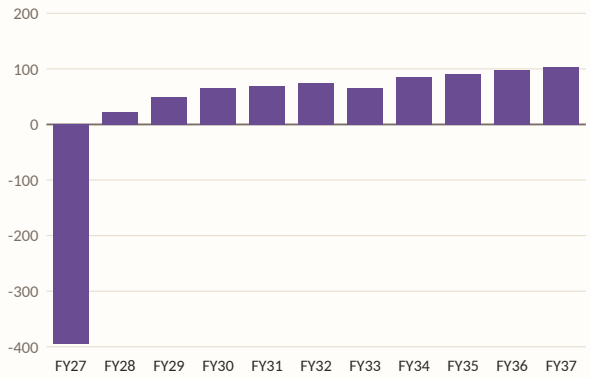
22.2% / 29.2%
project / equity IRR

6.6 yrs
payback from opening

Revenue and EBITDA, Base (₹ lakh)



Free cash flow to firm, Base (₹ lakh)



Scenario	NPV	Proj. IRR	Eq. IRR	Payback	Y3 revenue	Y3 EBITDA %	Bills/day Y3	Break-even	Peak overdraft	Loss if shut Y3
Best	598.8	30.8%	42.6%	4.9 yrs	293.3	37.5%	197	79	–	nil
Base	240.8	22.2%	29.2%	6.6 yrs	250.2	33.8%	172	78	8.1	nil
Worst	(187.5)	5.9%	0.1%	>10 yrs	175.5	19.4%	124	83	228.8	5.9

Why model D

Buying the SBR unit adds the most absolute NPV, because rent is saved and the property appreciates about 6% a year.
But it needs about three times the capital, with ₹1.8 crore of equity, a 10-year property loan, and a project IRR of about 22%.
It mixes a property investment with a café; consider it only after model A proves the site.

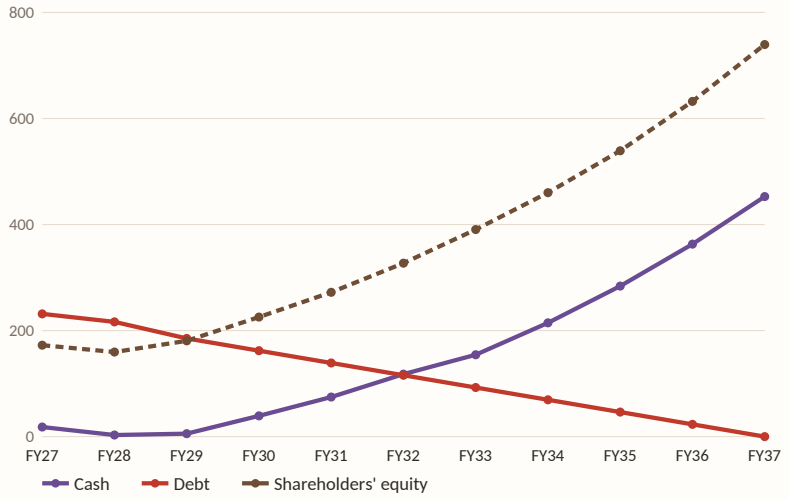
Model D in detail: statements, funding and unit economics

Condensed statements, Base (₹ lakh)	FY27	FY28	FY29	FY30	FY32	FY37
Bills per day	–	110.5	145.2	171.7	182.2	207.2
Revenue	**–**	**146.1**	**201.6**	**250.2**	**292.8**	**424.9**
Cost of goods sold	–	(44.0)	(60.4)	(74.6)	(86.3)	(122.0)
Staff	–	(46.9)	(50.6)	(54.9)	(66.0)	(103.9)
Rent, CAM & property tax	–	(1.2)	(1.3)	(1.3)	(1.5)	(1.9)
Other opex & pre-opening	(9.4)	(27.8)	(30.2)	(34.9)	(39.8)	(54.8)
EBITDA	**(9.4)**	**26.2**	**59.1**	**84.5**	**99.1**	**142.3**
Depreciation	–	(12.7)	(13.1)	(13.7)	(14.8)	(11.3)
Net interest	–	(22.4)	(21.7)	(18.3)	(10.9)	12.2
Tax	–	(4.0)	(3.1)	(7.6)	(18.5)	(36.1)
Profit after tax	**(9.4)**	**(12.9)**	**21.1**	**44.9**	**54.9**	**107.2**
Cash from operations	(9.4)	2.2	36.8	60.5	70.7	119.2
Capex & deposit	(385.9)	(2.2)	(3.0)	(3.8)	(4.4)	(6.4)
Free cash flow to firm	**(395.3)**	**22.4**	**50.0**	**64.8**	**74.5**	**103.7**
Closing cash	18.1	3.0	5.6	39.2	117.8	452.7
Fixed assets (net)	385.9	375.3	365.2	355.2	334.3	296.3
Net working capital	–	1.6	2.2	2.7	3.0	4.2
Debt	231.5	216.4	185.2	162.1	115.8	–
Shareholders' equity	**172.4**	**159.5**	**180.6**	**225.5**	**327.1**	**739.4**
Balance-sheet check	–	–	–	–	–	–

Unit economics, Y3 Base	
Revenue per bill	₹405
Ingredients & packaging	₹121
Staff	₹89
Rent, CAM, property tax	₹2
Utilities, marketing, fees, other	₹57
EBITDA per bill	**₹137**
Coffee cost per beverage	₹24.4
Bills per day / break-even	172 / 78

Sources & uses (₹ lakh)	
Café & roastery capex	79.4
Property incl. stamp duty	306.4
Pre-opening expenses	9.4
Rent deposit	–
Cash reserve	18.1
Total need	**413.3**
Term loan	231.5
Promoter equity	181.8
Loan rate / tenor	10% / 10 yrs
WACC / cost of equity	14.5% / 23.4%

Balance sheet trajectory, Base (₹ lakh)



Cash accumulates because the model pays no dividends (payout 0%); in practice surplus cash would fund a second outlet or be distributed. Equity IRR assumes free cash flow to equity is paid out each year.

19 Risk: scenarios and sensitivities

What moves value most, and how much room for error each model has

Model A (Sindhu Bhavan Road flagship): project NPV when one input moves (₹ lakh)

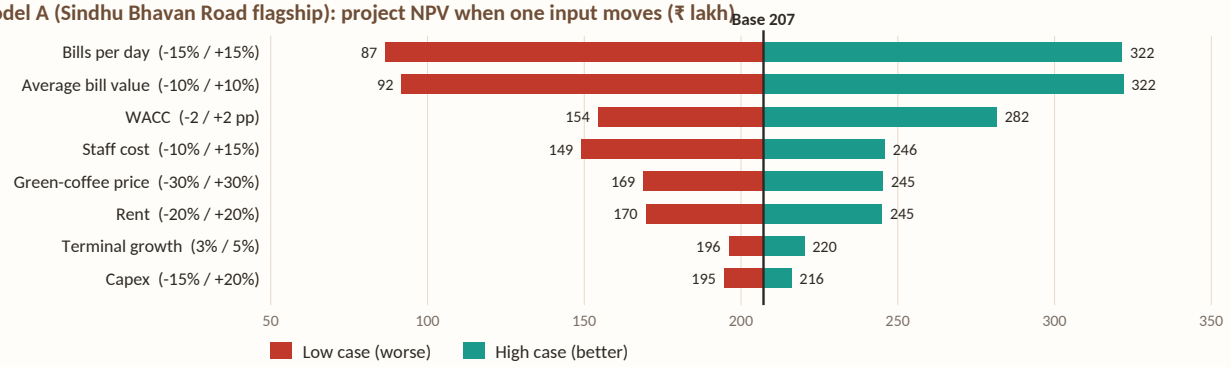


Figure 19.1. Each bar moves one input to its low or high value and keeps everything else at Base.

Model B (GIFT City captive café): project NPV when one input moves (₹ lakh)

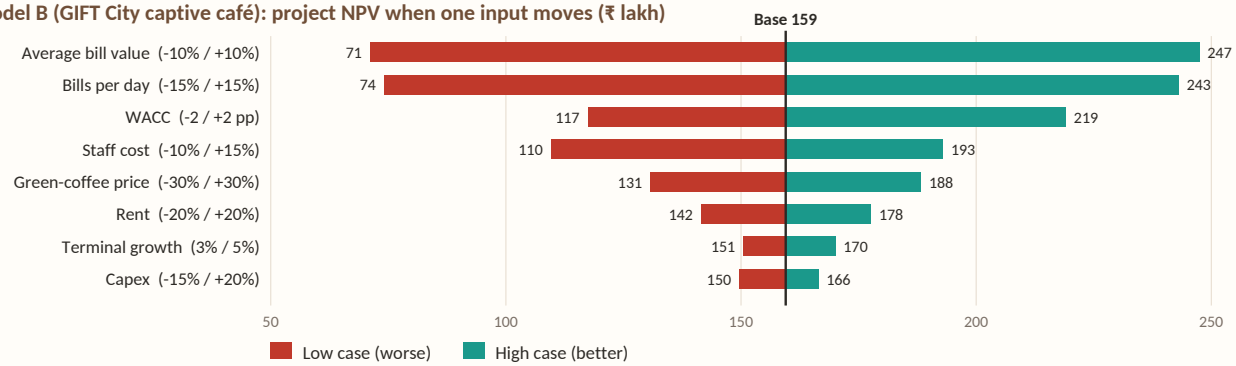


Figure 19.2. Each bar moves one input to its low or high value and keeps everything else at Base.

Model C (Prahladnagar neighbourhood café): project NPV when one input moves (₹ lakh)

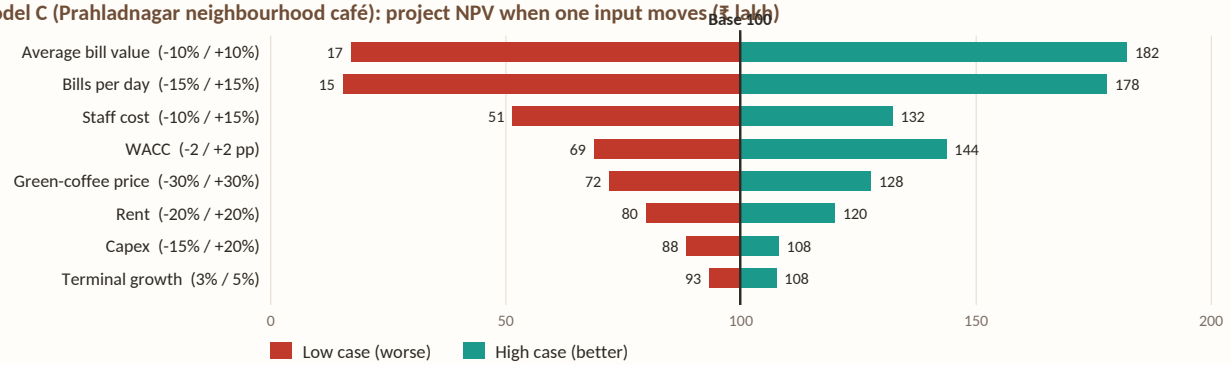


Figure 19.3. Each bar moves one input to its low or high value and keeps everything else at Base.

Model D (Sindhu Bhavan Road, owned unit): project NPV when one input moves (₹ lakh)

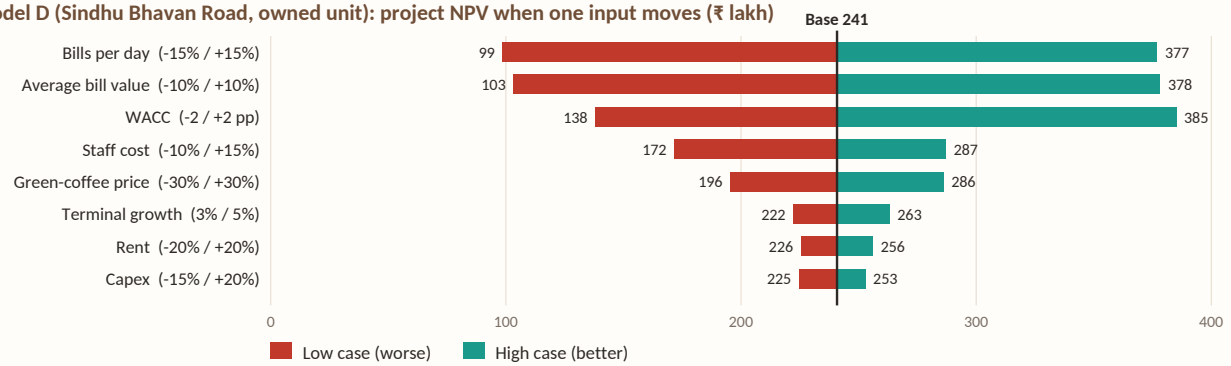


Figure 19.4. Each bar moves one input to its low or high value and keeps everything else at Base.

Demand inputs dominate: **bills per day and the average bill** each swing NPV by more than the next three inputs combined. Staff costs and WACC come next; rent and coffee prices matter, but far less. The practical message: spend money on location research (footfall counts at the shortlisted units at different times of day) and on the menu and service that drive repeat visits, rather than over-optimising the bean price.

Model A NPV	g 3.0%	g 3.5%	g 4.0%	g 4.5%	g
WACC -2pp	263	272	282	292	
WACC -1pp	227	234	241	249	
WACC as calc.	196	201	**207**	213	
WACC +1pp	170	174	179	184	
WACC +2pp	147	151	154	158	

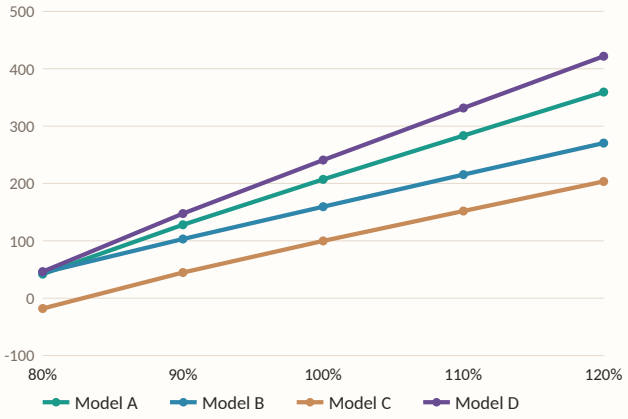
Model A project IRR	Bills -20%	-10%	Base	+10%	+20%
Rent -20%	25%	33%	40%	46%	52%
Rent -10%	23%	31%	38%	44%	50%
Rent base	21%	29%	**36%**	42%	48%
Rent +10%	19%	28%	35%	41%	47%
Rent +20%	17%	26%	33%	39%	45%

Model A NPV	Bill -10%	Bill -5%	Base	Bill +5%	Bill +10%
Bills -20%	-53	-5	42	89	
Bills -10%	23	76	128	180	
Bills base	92	149	**207**	265	
Bills +10%	157	220	283	347	
Bills +20%	222	290	359	428	

Model A Y3 EBITDA %	Buy	Toll	In-house
Green -30%	22.2%	25.5%	23.2%
Green -15%	21.4%	24.2%	20.8%
Green base	20.6%	**22.9%**	18.4%
Green +15%	19.8%	21.6%	15.9%
Green +30%	19.1%	20.2%	13.5%
Green +60%	17.5%	17.6%	8.6%

Figure 19.5. Two-way sensitivities for Model A (green = better, red = worse; the boxed cell is the Base case). Owners can re-run every table for any configuration on the Scenarios sheet.

NPV vs customers (bills/day as % of plan)

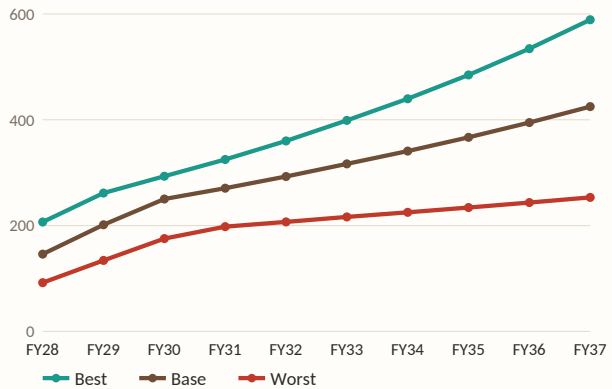


How many customers do you need? The chart re-runs each model with bills per day at 80-120% of plan, holding everything else at Base. Models A, B and D stay NPV-positive even at 80% of planned bills; Model C turns negative at about 83%. Model D, carrying property, swings the most in absolute terms.

That makes **footfall verification before signing a lease** the single most valuable piece of due diligence: count people and competitor queues at the actual unit across weekday mornings, evenings and weekends.

19.1 How the scenarios play out (Model A)

Revenue by scenario (₹ lakh)



EBITDA by scenario (₹ lakh)

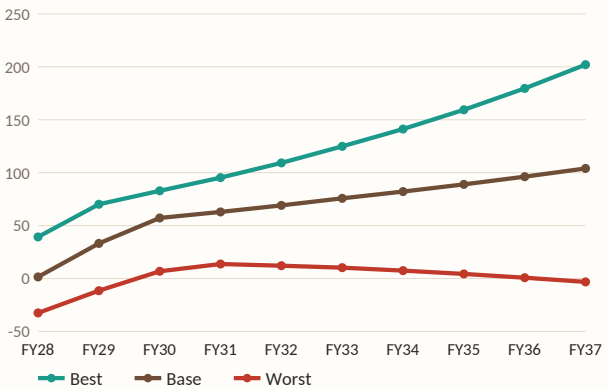


Figure 19.6. In the Worst case the café never escapes thin margins: revenue grows only with prices while wages grow faster, so EBITDA peaks in year 4 and then declines.

19.2 Where each ₹100 of sales goes

Model C (Prahladnagar), Y3: every ₹100 of sales

Total ₹100 (ex-GST)



Figure 19.7. Coffee beans themselves are only about ₹6-7 of every ₹100. Staff and ingredients are the big levers; rent is about ₹9.

19.3 Key risks and how to manage them

Risk	Why it matters	Mitigation
Demand below plan	Biggest value driver (tornado)	Count footfall at candidate units; pre-launch community and office tie-ups; menu with a signature; delivery from day one
Slow ramp-up	Year-1 losses and cash burn	Three-month cash reserve in the funding plan; rent-free fit-out period; staged hiring
Wage inflation above price growth	Margins erode over time	Annual price reviews; productivity (batch brew, POS), scheduling to footfall
Green-coffee price spike	Coffee is ~6% of sales; +30% costs ~2.5-3 points of margin	Toll roasting with 2-3 months' green stock; forward contracts with estates
Competition	New openings on SBR and Corporate Road	Differentiate on Indian estate coffees, kaapi, retail beans; loyalty app
Lease risk	Rent escalations, eviction	9-year lease with 3-year lock-in and capped escalation (5%/yr or 15% per 3 years)

20 Recommendation and roadmap

What to do, in what order, and what to check before signing

Recommendation

1. Open the first café on Sindhu Bhavan Road (Model A): rented unit, toll-roasted House blend. Capital about ₹130 lakh (₹90 lakh equity), NPV ₹207 lakh, project IRR 36.3%, payback 3.9 yrs.

2. If capital or a good SBR unit is not available, choose GIFT City (Model B): ₹98 lakh capital, NPV ₹159 lakh, the most resilient downside. It is also the natural second outlet.

3. Prahladnagar (Model C) is viable but third (NPV ₹100 lakh, IRR 29.8%): thinner margins and more chain competition. Choose it only for an exceptional unit at or below ₹115/sqft rent with strong visibility.

4. Rent, don't buy, for the first café. Revisit buying once a site has proven itself.

5. Toll-roast from day one (own green coffee, own label, no roaster capex). Add an in-house roastery only when wholesale demand passes roughly 450-500 kg a month.

6. Not Sanand yet. Re-examine in 2028 as the semiconductor workforce arrives; it could then host a low-cost roastery.

Launch timeline (October 2026 to May 2027)

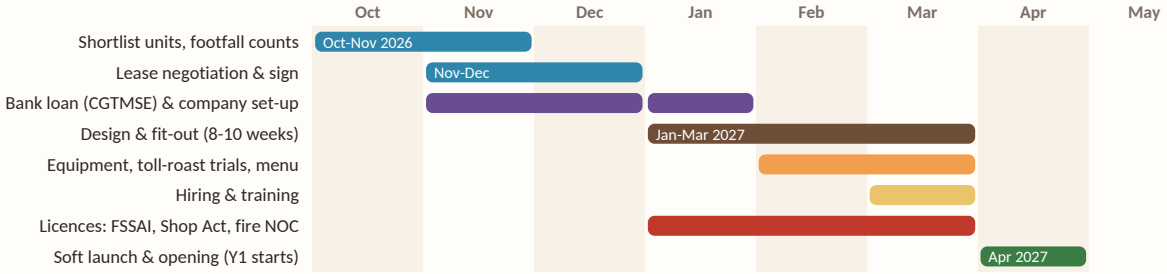


Figure 20.1. Six months from site search to opening, matching the model's Y0.

20.1 Before signing: what to verify

- **Footfall and bills**: count passers-by and competitors' queues at the unit on weekday mornings, evenings and weekends; the model needs about 110-170 bills a day.
- **Rent and terms**: carpet versus super area, CAM, escalation, lock-in, rent-free fit-out months, deposit (model: 4-6 months).
- **Fit-out reality**: power load (kW), drainage, exhaust, fire NOC and society rules for cafés; landlord contributions.
- **Toll roaster**: cup samples, minimum batch, lead time, price per kg, private-label terms; green-coffee contracts with two or three estates.
- **Team**: a head barista with specialty experience is the most important hire.

20.2 Funding options

Source	What it is	Typical terms	In the model
Promoter equity	Founder and family; angel investors	No repayment; dilution if outside investors	Model: ₹66-182 lakh depending on model
MSME term loan + CGTMSE guarantee	Banks and NBFCs; collateral-free credit guarantee (cover raised to ₹10 crore in Budget 2025)	~10-12%, 5-7 years	Model: 50% of capex at 11%, 5 years
PM MUDRA (Tarun / Tarun Plus)	Up to ₹20 lakh for micro enterprises	Small top-up	Useful for equipment or working capital
Equipment leasing / vendor finance	Espresso machines and grinders on lease	Preserves cash; higher effective cost	Alternative for ₹10-15 lakh of kit
Loan against property	For the owned-premises option	~9.5-10.5%, 10-15 years	Model D: 60% of capex + property at 10%
Cash credit / overdraft	Working-capital line	~12-13%	The model draws only if cash dips

20.3 The monthly dashboard to run the café by

KPI	Target (from the model)	Review
Bills per day (and by daypart)	Plan: Y1 65%, Y2 85%, Y3 100% of steady state (170 at SBR)	Weekly
Average bill (ex-GST)	₹367 at SBR in Y1, rising with menu prices	Weekly
Ingredient cost % of sales	~29-31%	Monthly stock count
Staff cost % of sales	~22-26% at steady state	Monthly
Rent + CAM % of sales	~11% (SBR), ~9% (Prahladnagar)	Monthly
EBITDA margin	Break-even in year 1 (SBR plan); 20%+ by year 3	Monthly
Cash in bank	Never below the minimum + one month of fixed costs	Weekly
Coffee freshness	Beans served 7-30 days from roast	Daily
Reviews and repeat rate	4.5+ ratings; loyalty-app repeat visits	Weekly

20.4 Phase two

- **Second outlet** (Model B at GIFT City if Model A opened first) once the first café beats plan for six consecutive months.
- **Own roastery** when wholesale demand passes roughly 450-500 kg a month: a 3-5 kg roaster with an afterburner in a light-industrial unit (Sanand or the city's edge) can then supply all outlets and wholesale accounts.
- **Retail and quick commerce**: own-label beans, drip bags and cold-brew bottles once the brand has local recognition.

20.5 How to use the workbook from here

Replace the judgement inputs with your own data as you learn it: real rent quotes on **Locations**, menu prices and mix on **Inputs**, quotes on **Capex**, and bean prices on **Sourcing**. The **Dashboard**, **Configs** ranking and the **Scenarios** tables update automatically, so the comparison in this report can be re-run in minutes as facts replace assumptions.

A

Appendix

Full Base-case statements for Model A, sources and glossary

Model A, Base, ₹ lakh	FY27	FY28	FY29	FY30	FY31	FY32	FY33	FY34	FY35	FY36	FY37
Revenue	–	146.1	201.6	250.2	270.7	292.8	316.7	340.8	366.8	394.8	424.9
Cost of goods sold	–	(44.0)	(60.4)	(74.6)	(80.2)	(86.3)	(92.8)	(99.4)	(106.4)	(113.9)	(122.0)
Gross profit	–	102.0	141.2	175.7	190.5	206.5	223.8	241.4	260.4	280.9	302.9
Staff	–	(46.9)	(50.6)	(54.9)	(60.2)	(66.0)	(72.4)	(79.3)	(86.7)	(94.9)	(103.9)
Occupancy	–	(25.9)	(27.2)	(28.6)	(30.0)	(31.5)	(33.1)	(34.7)	(36.5)	(38.3)	(40.2)
Other opex	(13.6)	(27.8)	(30.2)	(34.9)	(37.3)	(39.8)	(42.5)	(45.3)	(48.3)	(51.4)	(54.8)
EBITDA	(13.6)	1.5	33.2	57.2	62.9	69.1	75.8	82.1	89.0	96.2	104.0
Depreciation	–	(9.6)	(10.1)	(10.7)	(11.4)	(11.8)	(12.7)	(16.2)	(16.6)	(12.0)	(8.2)
EBIT	(13.6)	(8.1)	23.1	46.5	51.5	57.3	63.1	65.9	72.3	84.2	95.8
Interest expense	–	(4.4)	(3.5)	(2.6)	(1.7)	(0.9)	–	–	–	–	–
Interest income	–	1.0	0.4	1.2	2.5	4.0	5.7	7.4	10.0	12.8	15.9
Tax	–	(0.1)	0.3	(10.2)	(13.1)	(15.2)	(17.3)	(18.5)	(20.7)	(24.4)	(28.1)
Profit after tax	(13.6)	(11.7)	20.3	34.9	39.1	45.2	51.5	54.9	61.6	72.6	83.6
Cash from operations	(13.6)	(3.5)	29.5	44.5	49.4	55.7	62.8	69.1	76.0	83.4	91.5
Cash from investing	(92.0)	(2.2)	(3.0)	(3.8)	(4.1)	(4.4)	(20.1)	(5.1)	(5.5)	(5.9)	(6.4)
Cash from financing	129.9	(7.9)	(7.9)	(7.9)	(7.9)	(7.9)	–	–	–	–	–
Closing cash	24.3	10.6	29.2	62.0	99.4	142.7	185.5	249.5	319.9	397.4	482.6
Cash	24.3	10.6	29.2	62.0	99.4	142.7	185.5	249.5	319.9	397.4	482.6
Receivables + inventory	–	3.4	4.6	5.7	6.1	6.6	7.1	7.5	8.1	8.6	9.2
Rent deposit	12.6	12.6	12.6	12.6	12.6	12.6	12.6	12.6	12.6	12.6	12.6
Fixed assets	79.4	72.0	64.9	58.0	50.6	43.2	50.7	39.5	28.4	22.3	20.5
Total assets	116.3	98.6	111.3	138.3	168.7	205.2	255.8	309.2	369.0	441.0	524.8
Payables	–	1.8	2.5	3.1	3.3	3.5	3.8	4.1	4.4	4.7	5.0
Debt	39.7	31.8	23.8	15.9	7.9	–	–	–	–	–	–
Deferred tax	–	0.1	(0.1)	(0.7)	(1.6)	(2.7)	(3.9)	(5.7)	(7.7)	(8.6)	(8.7)
Equity	76.6	64.9	85.2	120.1	159.1	204.4	255.9	310.7	372.3	444.9	528.5
Liabilities + equity	116.3	98.6	111.3	138.3	168.7	205.2	255.8	309.2	369.0	441.0	524.8
Check	–	–	–	–	–	–	–	–	–	–	–
FCFF	(105.6)	(2.3)	29.1	40.7	44.8	49.0	38.5	58.4	63.0	67.9	73.2
FCFE	(65.9)	(13.6)	18.5	32.8	37.4	43.4	42.8	64.0	70.5	77.5	85.1

Source: Cafe_Financial_Model.xlsx with Inputs set to Sindhu Bhavan Road | Rent | Toll roasting | House | Base. Rows are links to the Model sheet; the check row is zero in every year.

Model C, Base, ₹ lakh	FY27	FY28	FY29	FY30	FY31	FY32	FY33	FY34	FY35	FY36	FY37
Revenue	–	106.9	146.8	181.4	195.2	210.1	226.1	243.3	261.9	281.9	303.4
Cost of goods sold	–	(32.7)	(44.6)	(54.8)	(58.6)	(62.8)	(67.2)	(71.9)	(77.0)	(82.5)	(88.3)
Gross profit	–	74.3	102.2	126.6	136.6	147.3	158.9	171.4	184.9	199.4	215.0
Staff	–	(39.1)	(42.2)	(45.6)	(49.8)	(54.5)	(59.6)	(65.2)	(71.3)	(78.1)	(85.4)
Occupancy	–	(14.3)	(15.0)	(15.7)	(16.5)	(17.4)	(18.2)	(19.1)	(20.1)	(21.1)	(22.2)
Other opex	(11.1)	(23.8)	(26.1)	(30.0)	(31.9)	(33.9)	(36.1)	(38.4)	(40.9)	(43.5)	(46.4)
EBITDA	(11.1)	(2.8)	19.0	35.3	38.3	41.6	45.0	48.7	52.6	56.7	61.1
Depreciation	–	(8.2)	(8.5)	(8.9)	(9.5)	(9.6)	(10.2)	(12.9)	(13.2)	(8.7)	(6.0)
EBIT	(11.1)	(11.0)	10.5	26.4	28.9	32.0	34.8	35.8	39.4	48.0	55.1
Net interest	–	(2.9)	(2.8)	(1.8)	(0.2)	1.4	3.1	4.0	5.5	7.1	8.9
Tax	–	(0.2)	0.2	(1.9)	(7.2)	(8.4)	(9.5)	(10.0)	(11.3)	(13.9)	(16.1)
Profit after tax	(11.1)	(14.0)	7.9	22.7	21.5	25.0	28.3	29.7	33.6	41.2	47.9
Cash from operations	(11.1)	(7.0)	15.8	30.8	30.0	33.5	37.5	41.0	44.9	49.1	53.7
Cash from investing	(71.1)	(1.6)	(2.2)	(2.7)	(2.9)	(3.2)	(14.9)	(3.7)	(3.9)	(4.2)	(4.6)
Cash from financing	101.2	(6.6)	(6.6)	(6.6)	(6.6)	(6.6)	–	–	–	–	–
Closing cash	19.1	3.9	10.8	32.2	52.7	76.4	99.0	136.3	177.3	222.2	271.4

Model C, Base, ₹ lakh	FY27	FY28	FY29	FY30	FY31	FY32	FY33	FY34	FY35	FY36	FY37
Fixed assets	66.5	59.9	53.6	47.4	40.9	34.4	39.1	29.9	20.7	16.1	14.7
Total assets	90.2	71.0	72.6	88.6	102.8	120.4	148.0	176.5	208.6	249.4	297.6
Debt	33.2	26.6	19.9	13.3	6.6	–	–	–	–	–	–
Equity	57.0	42.9	50.9	73.6	95.0	120.0	148.3	178.1	211.6	252.8	300.7
Check	–	–	–	–	–	–	–	–	–	–	–
FCFF	(82.1)	(5.7)	16.3	27.9	27.2	29.3	20.3	34.4	36.9	39.6	42.5

Source: Model C: Prahladnagar | Rent | Toll roasting | House | Base.

Model B, Base, ₹ lakh	FY27	FY28	FY29	FY30	FY31	FY32	FY33	FY34	FY35	FY36	FY37
Revenue	–	95.2	135.9	174.6	195.4	218.7	244.8	263.5	283.6	305.2	328.5
Cost of goods sold	–	(28.6)	(40.7)	(51.9)	(57.8)	(64.4)	(71.7)	(76.7)	(82.1)	(87.9)	(94.1)
Gross profit	–	66.5	95.2	122.7	137.6	154.4	173.2	186.8	201.5	217.3	234.4
Staff	–	(37.1)	(40.1)	(45.0)	(50.3)	(56.3)	(63.0)	(69.0)	(75.6)	(82.8)	(90.7)
Occupancy	–	(13.5)	(14.2)	(14.9)	(15.6)	(16.4)	(17.2)	(18.1)	(19.0)	(19.9)	(20.9)
Other opex	(10.6)	(19.7)	(21.4)	(24.9)	(27.0)	(29.3)	(31.9)	(33.9)	(36.1)	(38.4)	(40.8)
EBITDA	(10.6)	(3.8)	19.6	37.9	44.7	52.4	61.1	65.8	70.9	76.2	81.9
Depreciation	–	(7.7)	(8.0)	(8.4)	(9.0)	(9.1)	(9.7)	(12.3)	(12.7)	(8.4)	(6.0)
EBIT	(10.6)	(11.5)	11.6	29.4	35.7	43.3	51.4	53.6	58.2	67.8	75.8
Net interest	–	(2.7)	(2.6)	(1.6)	–	1.7	3.7	5.1	7.2	9.4	11.8
Tax	–	(0.2)	0.2	(3.0)	(9.0)	(11.3)	(13.9)	(14.8)	(16.5)	(19.4)	(22.1)
Profit after tax	(10.6)	(14.4)	9.1	24.8	26.7	33.7	41.2	43.9	48.9	57.7	65.6
Cash from operations	(10.6)	(7.5)	16.6	32.4	34.8	41.7	49.9	54.7	59.9	65.4	71.5
Cash from investing	(68.7)	(1.4)	(2.0)	(2.6)	(2.9)	(3.3)	(14.1)	(4.0)	(4.3)	(4.6)	(4.9)
Cash from financing	97.6	(6.3)	(6.3)	(6.3)	(6.3)	(6.3)	–	–	–	–	–
Closing cash	18.3	3.1	11.3	34.8	60.4	92.6	128.3	179.0	234.7	295.5	362.1
Fixed assets	62.8	56.4	50.5	44.6	38.6	32.8	37.2	28.9	20.5	16.7	15.6
Total assets	87.0	67.6	70.8	89.3	109.3	136.1	176.8	219.5	267.1	324.5	390.4
Debt	31.4	25.1	18.8	12.6	6.3	–	–	–	–	–	–
Equity	55.6	41.2	50.3	75.1	101.8	135.5	176.7	220.7	269.6	327.3	392.9
Check	–	–	–	–	–	–	–	–	–	–	–
FCFF	(79.3)	(6.2)	17.2	29.6	31.9	37.2	32.9	46.9	50.2	53.8	57.7

Source: Model B: GIFT City | Rent | Toll roasting | House | Base.

Model D, Base, ₹ lakh	FY27	FY28	FY29	FY30	FY31	FY32	FY33	FY34	FY35	FY36	FY37
Revenue	–	146.1	201.6	250.2	270.7	292.8	316.7	340.8	366.8	394.8	424.9
Cost of goods sold	–	(44.0)	(60.4)	(74.6)	(80.2)	(86.3)	(92.8)	(99.4)	(106.4)	(113.9)	(122.0)
Gross profit	–	102.0	141.2	175.7	190.5	206.5	223.8	241.4	260.4	280.9	302.9
Staff	–	(46.9)	(50.6)	(54.9)	(60.2)	(66.0)	(72.4)	(79.3)	(86.7)	(94.9)	(103.9)
Occupancy	–	(1.2)	(1.3)	(1.3)	(1.4)	(1.5)	(1.5)	(1.6)	(1.7)	(1.8)	(1.9)
Other opex	(9.4)	(27.8)	(30.2)	(34.9)	(37.3)	(39.8)	(42.5)	(45.3)	(48.3)	(51.4)	(54.8)
EBITDA	(9.4)	26.2	59.1	84.5	91.5	99.1	107.3	115.3	123.7	132.8	142.3
Depreciation	–	(12.7)	(13.1)	(13.7)	(14.5)	(14.8)	(15.7)	(19.3)	(19.7)	(15.1)	(11.3)
EBIT	(9.4)	13.5	46.0	70.7	77.0	84.3	91.6	96.0	104.0	117.7	131.1
Net interest	–	(22.4)	(21.7)	(18.3)	(14.6)	(10.9)	(6.9)	(3.1)	1.6	6.7	12.2
Tax	–	(4.0)	(3.1)	(7.6)	(15.7)	(18.5)	(21.3)	(23.4)	(26.6)	(31.3)	(36.1)
Profit after tax	(9.4)	(12.9)	21.1	44.9	46.7	54.9	63.4	69.5	79.1	93.1	107.2
Cash from operations	(9.4)	2.2	36.8	60.5	62.7	70.7	79.8	88.5	98.0	108.2	119.2
Cash from investing	(385.9)	(2.2)	(3.0)	(3.8)	(4.1)	(4.4)	(20.1)	(5.1)	(5.5)	(5.9)	(6.4)
Cash from financing	413.3	(15.1)	(31.2)	(23.1)	(23.1)	(23.1)	(23.1)	(23.1)	(23.1)	(23.1)	(23.1)
Closing cash	18.1	3.0	5.6	39.2	74.6	117.8	154.4	214.6	283.9	363.0	452.7
Fixed assets	385.9	375.3	365.2	355.2	344.8	334.3	338.7	324.5	310.3	301.2	296.3
Total assets	403.9	381.7	375.4	400.1	425.6	458.8	500.2	546.7	602.3	672.8	758.2
Debt	231.5	216.4	185.2	162.1	138.9	115.8	92.6	69.5	46.3	23.1	–
Equity	172.4	159.5	180.6	225.5	272.2	327.1	390.5	460.1	539.1	632.2	739.4
Check	–	–	–	–	–	–	–	–	–	–	–
FCFF	(395.3)	22.4	50.0	64.8	69.6	74.5	64.8	85.7	91.2	97.2	103.7

Source: Model D: Sindhu Bhavan Road | Own | Toll roasting | House | Base. Property is carried at cost (land not depreciated); its market value enters the terminal value.

Glossary of financial terms

Term	Meaning
NPV	Net present value: today's value of all future cash flows minus the investment
IRR	Internal rate of return: the discount rate at which NPV is zero
Project IRR / FCFF	Return on all capital, using free cash flow to the firm (before debt)
Equity IRR / FCFE	Return on the promoter's money, after debt flows
WACC	Weighted average cost of capital: blended required return of equity and debt
Cost of equity	Risk-free rate + beta x equity risk premium + specific premium
Terminal value	Value of cash flows after Y10, here a perpetuity growing at 4%
EBITDA	Earnings before interest, tax, depreciation and amortisation
Payback	Years until cumulative free cash flow turns positive
DSCR	Debt service cover: cash available / (interest + principal)
Break-even bills	Bills per day at which EBITDA is zero
WDV depreciation	Tax depreciation at a fixed % of the declining written-down value
Deferred tax	Tax effect of the gap between book and tax depreciation
Data table	Excel what-if tool that re-runs the model for many input values

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20. Modelling conventions: CFI Financial Modeling Guidelines; CFI 3-Statement Model template (supplied by the client)

How to read the numbers

Research facts are as published by the sources above; some (market sizes, chain revenues) come from secondary summaries and are flagged as indicative.

Café demand inputs (bills per day, average bill, ramp-up) and the location scores are the author's judgement, triangulated with chain economics and local listings. They are the inputs to validate first.

Illustrations are drawn in code; they are diagrams, not photographs.